Business Finance Midterm Cheat Sheet: Ch.2–7

Formulas · BA II Plus keystrokes · Key concepts | Prof. Gemma Lee, Kyung Hee Univ. | Lecture-slide scope only

0. BA II Plus Setup

0.1 Reset (before every problem)

- `2nd CLR TVM` clears N, I/Y, PV, PMT, FV (CE|C, ON/OFF, 2nd QUIT do **not**)
- Cash-flow (NPV) problems: `2nd CLR WORK`
- P/Y = 1: `2nd P/Y` `1 ENTER` `↓` `1 ENTER` `2nd QUIT`
- 4 decimals: `2nd FORMAT` `4 ENTER` `2nd QUIT`
- BGN on screen → switch to END (coupons and ordinary annuities pay at period end)
- Full reset (`2nd RESET ENTER`) sets P/Y back to 12 → reset P/Y = 1

0.2 Input rules

- Enter 4 of N, I/Y, PV, PMT, FV; `CPT` + the 5th
- Cash out = **−**, cash in = **+**; same sign on PV and FV → Error 5
- I/Y = rate per period as a % number (1% → 1)
- Single sum → PMT = 0
- Negative answer = amount to pay now, not an error
- Answer off by exactly (1 + r) → BGN mode is on

Ch.2 Financial Statements, Taxes and Cash Flow

2.1 Balance Sheet

- Snapshot at a point in time; **Assets = Liabilities + Equity**
- Assets in decreasing liquidity; liabilities by when due
- NWC = current assets − current liabilities
- Liquidity = speed and ease of conversion to cash; avoids financial distress

2.2 Book vs. Market Value; Equity Accounts

- Book = accounting value (historical cost); market = price in actual trade → use market value for decisions
- Common stock = shares × par; paid-in surplus = (issue price − par) × shares
- Equity = paid-in capital + retained earnings; par = arbitrary nominal amount
- Ex: 100 sh, par $1, issued at $6 → $100 + $500 = **$600**

2.3 Income Statement

- Performance over a period; sales first, then subtract costs
- Sales − COGS − Depreciation = **EBIT**; EBIT − Interest = taxable income; − Taxes = **NI**
- NI = dividends + addition to retained earnings
- Ex: 1,509 − 750 − 89 = 670; 670 − 70 = 600; tax 21% = 126; NI = **474**
- Trap: interest is deducted **after** EBIT

2.4 GAAP vs. IFRS

- GAAP: FASB, rules-based | IFRS: IASB, principles-based | Korea: K-IFRS
- Same purpose; presentation order flipped (GAAP current items first, IFRS non-current first)

2.5 Taxes

- Average = total tax ÷ taxable income | Marginal = rate on the next $1 (use for decisions)
- Flat tax → marginal = average; progressive → marginal > average
- US flat 21% (2017 reform; before: progressive up to 35%) | Korea progressive 10/20/22/25%
- OECD Pillar Two: 15% minimum effective tax for large MNEs (≥ €750M)
- Ex: Korea ₩500M: 10% × 200M + 20% × 300M = ₩80M → average **16%**, marginal **20%**

2.6 Cash Flow (CFFA)

- **CFFA = OCF − NCS − ΔNWC = CF to creditors + CF to stockholders**
- OCF = EBIT + Depreciation − Taxes

- NCS = End net fixed assets − Beg net fixed assets + Depreciation
- ΔNWC = End NWC − Beg NWC
- CF/CR = Interest − Net new borrowing | CF/SH = Dividends − Net new equity
- Ex: OCF = 670 + 89 − 126 = 633 | NCS = 1,709 − 1,644 + 89 = 154 | ΔNWC = (1,403 − 389) − (1,112 − 428) = 330
- Ex: CFFA = 633 − 154 − 330 = **149** | check: CF/CR 70 − 46 = 24, CF/SH 165 − 40 = 125, sum 149
- Ex (Zephyr): 3,180 − 1,400 − 600 = **1,180**
- Traps: depreciation is added back (OCF, NCS) | current-liability increase lowers NWC | negative CF/CR = borrowing > interest paid | omit any component → wrong
- Cash flow ≠ accounting statement of cash flows

Ch.3 Working with Financial Statements

3.1 Common-Size Statements; Users

- Balance sheet ÷ total assets | income statement ÷ sales; compare across years and across different-size firms
- Ex: inventory 84,600 ÷ 470,000 = **18.0%**
- Internal users: evaluation, planning, control | External: lenders, suppliers, customers, investors, competitors / acquirers / regulators

3.2 Ratio Categories

- Liquidity (short-term solvency) | Leverage (long-term solvency) | Turnover (asset management) | Profitability | Market value
- Slide firm: Sales 2,361; NI 474; TA 3,630; TE 2,625; price $115; 33M shares

3.3 Liquidity

- Current = CA ÷ CL → **1.29**
- Quick = (CA − Inventory) ÷ CL → **0.52** (excludes inventory)
- Cash = Cash ÷ CL → **0.18**

3.4 Leverage and Coverage

- Total debt ratio = (TA − TE) ÷ TA → **0.28**
- D/E = TD ÷ TE → **0.38** | EM = TA ÷ TE = 1 + D/E → **1.38**
- TIE = EBIT ÷ Interest = 741 ÷ 141 → **5.26**
- Cash coverage = (EBIT + Dep.) ÷ Interest = 1,017 ÷ 141 → **7.21** (always ≥ TIE)

3.5 Turnover

- Inventory = **COGS** ÷ Inventory → 3.18 (numerator is COGS, not sales)
- Receivables = Sales ÷ AR → 12.56 | Payables = COGS ÷ AP → 3.91
- Total asset turnover (TAT) = Sales ÷ TA → 0.65

3.6 Profitability

- PM = NI ÷ Sales → 20.08% | ROA = NI ÷ TA → 13.06% | ROE = NI ÷ TE → 18.06%
- Same numerator; only the denominators differ

3.7 Market Value

- EPS = NI ÷ shares → 14.36 | P/E = price ÷ EPS → 8.01
- P/S = price ÷ (Sales/share) → 1.61 | M/B = price ÷ (TE/share) → 1.45 (book value per share)
- EV = market cap + debt − cash; EBITDA = EBIT + Dep.
- Ex: EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 1,017; **EV/EBITDA = 4.62**

3.8 DuPont

- **ROE = PM × TAT × EM** (a product, not a sum) | PM = profitability / cost control, TAT = asset efficiency, EM = leverage
- Ex: Coca-Cola 27.34% × 0.46 × 3.26 ≈ **40.7%** (slide value; inputs rounded)
- KO vs PEP: both ROE ≈ 40%, different paths
  - KO: sells concentrate, asset-light, high margin, low turnover
  - PEP: finished goods and snacks, ~2× sales, thin margin, high turnover and leverage
  - PEP current ratio < 1 is not a red flag (cash sales, long payables terms, strong credit)

3.9 Benchmarks and Limits

- Benchmarks: time-trend analysis; peer group (SIC / NAICS)
- Limits: no underlying theory, conglomerates, global (IFRS vs GAAP), accounting methods (FIFO / LIFO, depreciation), different year-ends, capital structure, seasonality / one-offs, window dressing

Ch.4 Time Value of Money

4.1 Concepts

- A dollar today > a dollar tomorrow (it can be invested)
- Compounding (interest on interest) ↔ discounting
- Simple = interest on principal only | compound = on principal + past interest; gap grows with time
- Ex: $1,000, 10%, 3 yrs: simple 1,000 + 3 × 100 = 1,300 vs compound 1,000 × 1.10^3 = **1,331** (interest-on-interest $31)

4.2 Formulas

- $FV = PV(1 + r)^t$ | $PV = FV \div (1 + r)^t$
- $r = (FV \div PV)^{1/t} - 1$ | $t = \ln(FV \div PV) \div \ln(1 + r)$
- Slides favor `CPT I/Y` and `CPT N` over memorizing r and t formulas

4.3 Keystrokes

- FV: $2,000, 8%, 5 yrs → `5 N` `8 I/Y` `0 PMT` `2000 +/− PV` `CPT FV` = **2,938.66**
- PV: $10,000 in 6 yrs, 7% → `6 N` `7 I/Y` `0 PMT` `10000 FV` `CPT PV` = **−6,663.42**
- r: $5,000 → $10,000 in 9 yrs → `9 N` `5000 +/− PV` `0 PMT` `10000 FV` `CPT I/Y` = **8.01%**
- t: $1,000 → $2,500 at 9% → `9 I/Y` `1000 +/− PV` `0 PMT` `2500 FV` `CPT N` = **10.63 yrs**
- Practice (a) FV of $8,000, 6.5%, 10 yrs → 15,017.10
- Practice (b) PV of $20,000 in 10 yrs at 6.5% → −10,654.52
- Practice (c) $8,000 → $20,000 in 10 yrs, r → 9.60%
- Practice (d) $8,000 → $20,000 at 6.5%, t → 14.55 yrs
- Trap: same sign on PV and FV → Error 5

Ch.5 Discounted Cash Flow Valuation

5.1 Multiple Cash Flows

- Discount / compound each cash flow at its own t, then **add** (never average); including CF0 → NPV
- FV: $4,000 now + $7,000 in 1 yr, 8%, value at year 3
  - `3 N` `8 I/Y` `0 PMT` `4000 +/− PV` `CPT FV` = 5,038.85
  - `2 N` `7000 +/− PV` `CPT FV` = 8,164.80 → sum = **13,203.65**
- PV of uneven CFs 200 / 400 / 600 / 800 at 12%
  - `CF` `2nd CLR WORK` `0 ENTER ↓` (CF0) `200 ENTER ↓↓` `400 ENTER ↓↓` `600 ENTER ↓↓` `800 ENTER ↓↓`
  - `NPV` `12 ENTER ↓` `CPT` = **1,432.93**

5.2 Annuity Types and Formulas

- Annuity: level CFs, finite (car / home loans) | Annuity due: paid at start (rent, lease, insurance) → × (1 + r)
- Perpetuity: C ÷ r (preferred stock, consol) | Growing perpetuity: $C_1 \div (r - g)$, r > g
- $PV = C \times [1 - 1/(1 + r)^t] \div r$ | $FV = C \times [(1 + r)^t - 1] \div r$
- r = rate **per payment period** (not APR); t = **number of payments** (not years)

5.3 Annuity Keystrokes

- PV (loan amount): $632 / mo, 1% / mo, 48 mo → `48 N` `1 I/Y` `632 +/− PMT` `0 FV` `CPT PV` = **23,999.54**
- Practice: $450, 0.75%, 36 mo → 14,151.06
- PMT: $20,000, 0.5% / mo, 60 mo → `60 N` `0.5 I/Y` `20000 PV` `0 FV` `CPT PMT` = **−386.66**
- Practice: $15,000, APR 9% monthly, 4 yrs → r = 0.75%, N = 48 → −373.28
- r: cost $1,000, $200 / yr × 8 → `8 N` `1000 +/− PV` `200 PMT` `0 FV` `CPT I/Y` = **11.81%**
- Practice: $8,000, $1,200 / yr × 12 → 10.45%
- N: save $2,000 / yr at 7% to reach $50,000 → `7 I/Y` `2000 +/− PMT` `0 PV` `50000 FV` `CPT N` = **14.95 yrs**
- Practice: $400 / mo, 0.5% / mo, $30,000 → 63.85 mo (5.32 yrs)
- FV: $300 / mo, 0.75% / mo, 30 yrs → `360 N` `0.75 I/Y` `300 +/− PMT` `0 PV` `CPT FV` = **549,223.04**
- Practice: $2,500 / yr, 7%, 25 yrs → 158,122.59

5.4 Annuity Due

- Method 1: ordinary value × (1 + r) | Method 2: `2nd BGN` `2nd SET` `2nd QUIT`, enter as usual, then **restore END**
- Ex: $1,000 / mo, 1% / mo, 36 mo → ordinary PV 30,107.51 × 1.01 = **30,408.58**
- Practice: $1,800, 0.5%, 24 mo → 40,613.16 × 1.005 = 40,816.22
- Lease PMT (BGN): $48,000, 12% APR monthly, 4 yrs → `48 N` `1 I/Y` `48000 +/− PV` `0 FV` `CPT PMT` = **1,251.51** (= ordinary 1,264.02 ÷ 1.01)

5.5 Perpetuities

- PV = C ÷ r | Ex: preferred $1.50 / qtr at 2% / qtr → **75.00**
- Growing: $C_1 \div (r - g)$ | Ex: $2.00 next year, g 3%, r 10% → 2.00 ÷ 0.07 = **28.57**
- If $2.00 was just paid ($D_0$): $C_1 = D_0(1 + g)$ = 2.06 → 2.06 ÷ 0.07 = **29.43**

5.6 APR vs. EAR

- APR = quoted rate (periodic rate × m) | EAR = actual annual rate | **always compare in EAR**
- EAR = $(1 + APR/m)^m - 1$ | APR = $m \times [(1 + EAR)^{1/m} - 1]$ | continuous: EAR = $e^r - 1$
- More frequent compounding / higher rate → larger APR–EAR gap
- `2nd ICONV`: NOM → C/Y → `CPT EFF`; reverse: EFF → C/Y → `CPT NOM` (standard BA II Plus operation)
- Ex: 15% APR → annual 15.00%, semiannual 15.56%, quarterly 15.87%, monthly 16.08%, daily 16.18%
- Practice: 9% APR → quarterly 9.31%, monthly 9.38%, daily 9.42%
- Bank A 15% daily (EAR 16.18%) vs Bank B 15.5% annual (15.50%) → A for deposits
- Card A 22.9% monthly (EAR 25.46%) vs Card B 23.5% annual (23.50%) → B is cheaper
- EAR 12% as monthly APR → **11.39%** | EAR 9% as quarterly APR → **8.71%**
- Calculator returns a monthly rate: $250 / mo × 15 yrs → $75,000 → `180 N` `250 +/− PMT` `0 PV` `75000 FV` `CPT I/Y` = 0.5303% / mo → APR **6.36%**, EAR **6.55%**

5.7 Loan Types and Amortization

- Pure discount (T-bill, zero) | interest-only (corporate bond) | amortized (home / car loan)
- Loan balance = PV of remaining payments
- Payment constant; interest falls, principal rises
- Interest = beginning balance × r | principal = PMT − interest | ending = beginning − principal
- Ex: $5,000, 9%, 5 yrs → `5 N` `9 I/Y` `5000 PV` `0 FV` `CPT PMT` = **−1,285.46**
  - Year 1: interest 450.00, principal 835.46, ending balance 4,164.54
  - Total paid 6,427.30 = principal 5,000 + interest 1,427.30
- Practice: $10,000, 8%, 4 yrs → PMT 3,019.21 | yr-1 ending balance 7,780.79 | yr-2 interest 622.46 | balance after 2 yrs 5,384.05

Ch.6 Interest Rates and Bond Valuation

6.1 Bond Anatomy and Keys

- Face / par = repaid at maturity | coupon = fixed interest | coupon rate = fixed at issue | maturity
- YTM = market required return, changes daily | after issue only the **price** changes
- Price = PV(coupon annuity) + PV(face)
- Keys: N = periods to maturity | I/Y = YTM per period | PMT = coupon per period | FV = face | PV = price
- PMT and FV positive → PV comes out negative (cash you pay); solving YTM → enter PV as negative

6.2 Par, Discount, Premium

- Coupon rate = YTM → **par**
- Coupon rate < YTM → **discount** (capital gain at maturity)
- Coupon rate > YTM → **premium** (capital loss at maturity)

6.3 Pricing Keystrokes

- Discount: $1,000, 8% coupon, 10 yrs, YTM 10% → `10 N` `10 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−877.11**
- Premium: same bond, YTM 6% → `10 N` `6 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−1,147.20**
- Semiannual: 9% coupon, 7 yrs, YTM 8% → N = 2 × yrs, I/Y = YTM ÷ 2, PMT = coupon ÷ 2
  - `14 N` `4 I/Y` `45 PMT` `1000 FV` `CPT PV` = **−1,052.82**
- Zero coupon: 5 yrs, YTM 6% semiannual → `10 N` `3 I/Y` `0 PMT` `1000 FV` `CPT PV` = **−744.09**

6.4 Yield to Maturity

- YTM = single discount rate that sets PV of promised CFs = price (the bond's IRR)
- Assumptions: hold to maturity | no default | coupons reinvested at YTM
- Annual: 8% coupon, 8 yrs, price 918.50 → `8 N` `918.50 +/− PV` `80 PMT` `1000 FV` `CPT I/Y` = **9.50%**
- Semiannual: 7% coupon, 10 yrs, price 900.29 → `20 N` `900.29 +/− PV` `35 PMT` `1000 FV` `CPT I/Y` = 4.25 → × 2 = **8.50%** (EAY 8.68%)
- Three yields (8% coupon, price 918.50): coupon rate 8.00% < current yield 80 ÷ 918.50 = 8.71% < YTM 9.50%

6.5 Interest Rate Risk

- Price and yield move **inversely** (promised CFs are fixed); price curve is convex
- Longer maturity and lower coupon → more sensitive; zero coupon is most sensitive
- Interest rate risk = price risk (not default risk)
- 30-yr bond, yield 10% → 12%: coupon 5% → −17.50% | 10% → −16.11% | 15% → −15.61%

6.6 Clean vs. Dirty Price

- Dirty (invoice) = clean (quoted) + accrued interest
- Accrued = coupon × (elapsed ÷ coupon period)
- Ex: semiannual $40 coupon, 2 months elapsed, quote $985 → 40 × 2/6 = 13.33 → dirty **998.33**

6.7 Fisher Effect and Yield Components

- **1 + R = (1 + r)(1 + h)** | Ex: nominal 15.5%, inflation 5% → 1.155 ÷ 1.05 − 1 = **10.00%** real (R − h = 10.5% is only approximate)
- Nominal yield = real rate + inflation premium + interest-rate-risk premium (rises with maturity) + bond-specific: default, taxability, liquidity premiums
- Higher expected inflation → higher nominal yields, lower bond prices

6.8 Term Structure

- Relation between maturity and yield (pure-discount Treasuries); yield curve = its graph
- Upward sloping → expected inflation up | downward → expected inflation down

6.9 Treasuries and Municipals

- Treasury: no default risk, state / local tax-exempt, federally taxable
- Municipal: default risk, federally tax-exempt → lower yield, held by high-bracket investors
- Tax-equivalent yield = muni yield ÷ (1 − t) | 6 ÷ 0.65 = **9.23%** (t = 35%) | 6 ÷ 0.80 = **7.50%** (t = 20%)

6.10 Debt vs. Equity

- Creditors: no ownership, no vote | interest is a pre-tax (deductible) expense | dividends are paid after tax
- Unpaid interest → default; unpaid dividend → not default
- Liquidation: equity is the residual claim (last)

6.11 Indenture, Call, Covenants

- Indenture = contract between issuer and bondholders; trustee represents holders
- Terms: face, coupon, maturity | security | repayment (sinking fund) | seniority | call | covenants
- Call: issuer may repurchase before maturity | call premium = call price − face | deferred call = protected period
  - Exercised when rates fall → bond value lower; investors demand a higher yield
- Negative covenants: limit dividends, pledging assets, mergers, asset sales, extra long-term debt
- Positive covenants: maintain NWC, supply audited statements, maintain collateral, keep ratios
- Firms accept covenants because the yield reduction exceeds the flexibility cost

6.12 Ratings and Bond Types

- Investment grade AAA / Aaa to BBB / Baa | junk (speculative) BB / Ba and below
- BBB− / Baa3 = last investment grade (institutional-investment cutoff)
- Zero coupon: only CF at maturity | Floating rate: collar = floor + cap | Convertible: lower coupon
- Put provision: opposite of call, adds value | CoCo | Sukuk: asset-based return instead of interest

6.13 Bond Market

- Large volume but mostly **OTC** (no central exchange); dealer quotes; thin liquidity per issue; institution-dominated
- Transparency problem → TRACE (2002) reporting and disclosure

Ch.7 Equity Markets and Stock Valuation

7.1 Stock vs. Bond Valuation

- Dividends at board discretion (coupons are contractual)
- No maturity → infinite cash flows
- Must estimate R and g; price is highly sensitive to both

7.2 Dividend Discount Model

- P_0 = PV of all future dividends
- "Sell later at a higher price" is already inside the model
- Non-dividend firms still have value only if they eventually pay
- No slide keystrokes for Ch.7 → solve with formulas

7.3 Four Growth Cases

- **Zero growth** (preferred): $P_0 = D \div R$
- **Constant growth**: $D_1 = D_0(1 + g)$; $P_0 = D_1 \div (R - g)$, requires $R > g$; $P_t = D_{t+1} \div (R - g)$
- **Nonconstant**: list $D_1 \ldots D_t$, find P_t, discount all; P_t is discounted t periods
- **Two-stage** (g_1 for t yrs, then g_2): $P_0 = D_1/(R - g_1) \times [1 - ((1 + g_1)/(1 + R))^t] + P_t/(1 + R)^t$, with $P_t = D_{t+1}/(R - g_2)$ (standard form; slide formula is an image)
- Sustainable growth is low single digits
- Zero: D \$3.00, R 10% → **30.00**
- Constant: D_0 \$2.00, g 5%, R 12% → D_1 = 2.10; 2.10 ÷ 0.07 = **30.00**
- Future price: $D_6 = 2.00 \times 1.05^6$; $D_6 \div 0.07$ = **38.29**
- Nonconstant: D_0 \$1.00, 20% growth for 3 yrs then 5% forever, R 15%
  - $D_{1\ldots3}$ = 1.20, 1.44, 1.728 | D_4 = 1.8144 | P_3 = 1.8144 ÷ 0.10 = 18.144
  - $P_0 = 1.20/1.15 + 1.44/1.15^2 + 1.728/1.15^3 + 18.144/1.15^3$ = **15.20**
- No dividend for 4 yrs, \$0.50 at end of yr 5, g 4%, R 20% → $P_4 = 0.50 \div (0.20 - 0.04) = 3.125$; $\div 1.20^4$ = **1.51**

7.4 Return Decomposition

- **R = dividend yield ($D_1 \div P_0$) + capital gains yield (= g under constant growth)**
- Dividend yield is like a bond's current yield
- Ex: P \$20, D_1 \$1, g 10% → 5% + 10% = **15%**

7.5 Multiples

- P/E, price-sales: fallback when there is no dividend
- Benchmark = comparable firms or own history; "comparable" drives the answer
- Losses → P/E meaningless; P/S still works
- Price = EPS × benchmark P/E | Ex: 3.00 × 18 = **54.00**
- Price = sales per share × benchmark P/S | Ex: 10.00 × 2.5 = **25.00**

7.6 Common Stock Rights

- Elect directors; vote (mergers, charter amendments, new shares)
- Dividends only after board declaration | residual claim | limited liability (loss up to paid-in)
- Preemptive right: first claim on new shares, prevents dilution
- Proxy: delegated vote (management usually favored); proxy fight: rival solicits proxies
- Dual-class: Class B 10 votes vs Class A 1 (founder control); one-share-one-vote varies by law / listing rules

7.7 Dividends

- Discretionary (non-payment ≠ default)
- Paid from after-tax income (asymmetric to deductible interest)
- US corporate dividends-received deduction (DRD) → corporations prefer preferred stock

7.8 Preferred Stock

- Fixed dividend on par; paid before common, after creditors
- Cumulative: arrearages paid first | normally no vote (may gain vote in arrears)
- Looks like debt but is not: missed dividend ≠ bankruptcy trigger; dividends not tax-deductible; equity legally and in accounting

7.9 Stock Markets

- Primary (firm receives funds) vs secondary (firm receives nothing)
- Dealer: holds inventory, bears risk | broker: agent, earns commission
- Bid = dealer buys | ask = dealer sells | spread = dealer's compensation
- NYSE: hybrid (DMM, mostly electronic) | NASDAQ: dealer market, competing market makers, ECNs narrow spreads
- Quote fields: Last / Close, Net change, Bid / Ask, Volume, Dividend, Yield = dividend ÷ price, P/E = price ÷ EPS

Exam Trap Checklist

T.1 Calculator

- Same sign on PV and FV → Error 5; cash out −, cash in +
- `2nd CLR TVM` / `2nd CLR WORK` before each problem; P/Y = 1
- Match units of N and I/Y (semiannual: N = yrs × 2, I/Y = rate ÷ 2)
- First CF now → annuity due = ordinary × (1 + r); turn BGN off afterwards

T.2 Statements and Ratios

- Interest deducted after EBIT | NCS adds depreciation | OCF adds depreciation back
- Inventory turnover uses COGS | quick ratio excludes inventory | EM = 1 + D/E | M/B uses book value per share
- DuPont is a product, not a sum

T.3 Rates and Bonds

- APR given → convert to the periodic rate first; EAR = $(1 + APR/m)^m - 1$; compare in EAR
- After-tax interest = interest × (1 − T); dividends paid from after-tax income
- Price and YTM move inversely; coupon rate < YTM → discount; clean + accrued = dirty
- Tax-equivalent yield: **divide** by (1 − t) (6% × 0.65 = 3.90% is wrong)

T.4 Stocks and Written Answers

- Constant growth numerator is D_1 ($D_0 \times (1 + g)$); valid only if R > g
- P_3 is discounted 3 periods, not 4; losses → use P/S
- Written answers: definition + direction (up / down) + reason in one line

BUSINESS FINANCE — FINANCIAL RATIOS

Chapter 3 | One-page formula sheet · Use figures from the same fiscal period where possible

1. COMMON-SIZE STATEMENTS

Ratio / purpose	Formula
Balance sheet account (%)	Account ÷ Total Assets × 100
Income statement item (%)	Item ÷ Sales × 100

2. LIQUIDITY — short-term solvency

Ratio / purpose	Formula
Current ratio	Current Assets (CA) ÷ Current Liabilities (CL)
Quick ratio	(CA – Inventory) ÷ CL
Cash ratio	Cash ÷ CL

3. LEVERAGE / LONG-TERM SOLVENCY

Ratio / purpose	Formula
Total debt ratio	(Total Assets – Total Equity) ÷ Total Assets = Total Debt ÷ TA
Debt-to-equity (D/E)	Total Debt ÷ Total Equity
Equity multiplier (EM)	Total Assets ÷ Total Equity = 1 + D/E
Times interest earned (TIE)	EBIT ÷ Interest
Cash coverage	(EBIT + Depreciation) ÷ Interest

4. ASSET MANAGEMENT / TURNOVER

Ratio / purpose	Formula
Inventory turnover	COGS ÷ Inventory
Receivables turnover	Sales ÷ Accounts Receivable (AR)
Payables turnover	COGS ÷ Accounts Payable (AP)
Total asset turnover (TAT)	Sales ÷ Total Assets

5. PROFITABILITY

Ratio / purpose	Formula
Profit margin (PM)	Net Income (NI) ÷ Sales
Return on assets (ROA)	NI ÷ Total Assets
Return on equity (ROE)	NI ÷ Total Equity

6. MARKET VALUE

Ratio / purpose	Formula
Earnings per share (EPS)	Net Income ÷ Shares Outstanding
P/E ratio	Price per Share ÷ EPS
Price/Sales (P/S)	Price per Share ÷ Sales per Share
Market-to-book (M/B)	Price per Share ÷ Book Value per Share
Enterprise value (EV)	Market Value of Equity + Book Value of Liabilities – Cash
EBITDA	EBIT + Depreciation & Amortization
EV/EBITDA	Enterprise Value ÷ EBITDA

7. DUPONT IDENTITY — decompose ROE

Ratio / purpose	Formula
Core identity	ROE = PM × TAT × EM
Expanded	(NI ÷ Sales) × (Sales ÷ TA) × (TA ÷ Equity) = NI ÷ Equity

QUICK INTERPRETATION & EXAM CHECKS

Liquidity: higher ratios usually mean more short-term coverage, but excess idle assets may reduce efficiency. • **Leverage:** more debt can magnify shareholder returns and risk; TIE / cash coverage measure ability to cover interest. • **Turnover:** higher turnover generally means more efficient asset use; interpret inventory turnover in industry context. • **Profitability:** PM = profit per sales dollar; ROA = profit relative to assets; ROE = profit relative to equity. • **Common-size:** balance sheet uses total assets as denominator; income statement uses sales. • **DuPont:** ROE can come from high margin, efficient asset turnover, or high leverage. • **Calculation:** convert percentages to % only at the end; ratios/turnover are often shown as “x” (e.g., 1.5x). • **Comparison cautions:** compare across time and similar firms; accounting methods, seasonality, fiscal year-end, one-offs, and window dressing can mislead.

Investment Analysis Sub-Note (Ch.2–6, to §6.4)

Scope: Ch.2–Ch.6 (to §6.4), lecture-PPT order. Target: 80-point core. Tags: **DEF** = answer keywords · **CALC** = formula + solving order · **TRAP** = common error · ※ = SA/MC bank item not in PPT (do not use as model answer).

Ch.2 Asset Classes & Financial Instruments

2.1 Money Market

DEF

- Short-term (≤ 1 yr), highly marketable, liquid, low-risk debt = "cash equivalents"
- Large denominations → individuals access via **money-market mutual funds (MMF)**
- Dealer-driven, highly competitive market

Instrument	Keywords
T-bill	4/8/13/26/52 wks; sold at discount, face at maturity; most marketable; ~risk-free; exempt from state & local tax
CD	Time deposit; interest + principal at maturity; negotiable CD ≥ $100,000 marketable
Commercial paper (CP)	Unsecured; large sound firms; ≤ 270 days (usually 1–2 mo); bank line of credit; asset-backed CP = receivables collateral
Bankers' acceptance (BA)	Order to bank to pay at future date; international trade; once accepted = bank liability; trades at discount
Eurodollars	USD deposits at foreign banks / foreign branches of US banks; outside US jurisdiction → less regulation, higher yield; less liquid
Repo / reverse repo	Repo = sell securities + agree to buy back next day at higher price = short-term collateralized (overnight) loan; reverse repo = you lend cash, hold collateral
Fed funds	Reserve balances at Fed; overnight rate between banks; benchmark / monetary policy tool
Broker's call loans	Margin buyers borrow from broker; broker borrows at call money rate; callable on demand

- MMF: stable $1.00 NAV, check-writing, "parking place" for cash
- **TRAP** 2008 "break the buck": Reserve Primary Fund held Lehman CP → NAV < $1 → run → Treasury guarantee ⇒ low-risk ≠ risk-free
- ※ LIBOR (excluded from exam)

CALC — yield conventions (F = face, P = price, n = days)

Yield	Formula	Basis
BDY	$r_{BD} = (F - P)/F \times 360/n$	face value, 360 days; quoting convention only
BEY	$r_{BEY} = (F - P)/P \times 365/n$	purchase price, 365, simple
EAY	$r_{EAY} = (F/P)^{(365/n)} - 1$	purchase price, 365, compound; true annual yield

- Example F = 10,000, P = 9,600, n = 180: BDY = 400/10,000 × 2 = **8.00%**; BEY = 400/9,600 × 365/180 = **8.45%**; EAY = (10,000/9,600)^(365/180) − 1 = **8.63%**
- Order: **BDY < BEY < EAY**
- Discount-quoted (BDY): T-bills, CP, BA. Add-on yield: CDs, repos, fed funds, Eurodollars
- To compare → convert all to BEY or EAY

2.2 Bond Market (fixed-income capital market)

DEF — longer-term borrowing; fixed (or formula) income; segmented by issuer & tax treatment

Type	Keywords
Treasury notes / bonds	Notes ≤ 10 yr, bonds > 10 yr; semiannual coupons; par $1,000; quoted in **32nds**; ~default-free, liquid
TIPS	Principal indexed to inflation (CPI); coupon on adjusted principal; guaranteed **real** return; yield spread vs nominal = expected inflation
Federal agency	GSEs: Fannie Mae, Freddie Mac, FHLB; housing & agriculture credit; implicit guarantee; conservatorship 2008
International	Eurobond (currency ≠ issuer's country), Yankee (USD, sold in US by foreign issuer), Samurai (yen, sold in Japan by non-Japanese issuer)
Municipal	State & local; interest federal-tax-exempt; **GO** = full taxing power; **revenue** = project revenues; lower stated yield

Type	Keywords
Corporate	Semiannual coupon; default risk; rated (investment grade vs high yield); spread over Treasuries; secured (collateral) vs unsecured (debenture, lower priority); callable / convertible / puttable
MBS	Claim on mortgage pool; pass-through of P&I; agencies package conforming loans; securitization; subprime MBS = 2008 crisis

CALC

- Treasury quote: 100-08 = 100 + 8/32 = 100.25% × $1,000 = **$1,002.50** (digits after point = 32nds, not cents; "+" = ½ tick)
- WSJ quote 100.0320: 100 + 3/32 + 0.25/32 = 100.1016% = **$1,001.02**; ≠ 100.032%
- WSJ columns: coupon, bid (dealer buys), asked (you buy), chg (32nds), asked YTM
- TIPS: adjusted principal = P × (1 + inflation); coupon = coupon rate × adjusted principal. Ex: 2% coupon, 3% inflation → 1,030 → coupon **$20.60**
- Muni equivalent taxable yield: r_taxable = r_muni / (1 − t). Ex: 4%, t = 35% → 4/0.65 = **6.15%**; muni wins if taxable yield < cutoff

2.3 Equity Securities

Item	Keywords
Common stock	Residual claim (last in line); **limited liability** (max loss = investment); voting → elects board; dividends discretionary
Preferred stock	Debt + equity hybrid; fixed dividend, perpetuity; cumulative; priority over common, below debt; dividends NOT tax-deductible for issuer; callable / convertible
Priority	Debt → preferred → common (residual)
ADR / GDR	Negotiable certificate from depositary bank for foreign shares held by custodian; ratio sets shares per receipt; economic rights; USD-listed (ADR), London / Luxembourg (GDR); sponsored / unsponsored

- Stock quote fields: Last, Chg, Volume, 52-wk range, P/E = price ÷ EPS, Yield = annual dividend ÷ price
- **CALC** ADR price = underlying home price (in USD) × ratio. Ex: SK hynix 1 ADR = 1/10 share

2.4 Market Indexes

DEF — track market direction; benchmark; basis for index funds & derivatives; design: which stocks, how weighted, how averaged

Index	Weight	Examples
Price-weighted	Share price (high price dominates)	DJIA (30 blue chips)
Value-weighted	Market cap (large cap dominates)	S&P 500 (~80% of US equity value)
Equal-weighted	Equal dollars; avg of returns	—

CALC — setup: A $100 → $110 (20M sh), B $25 → $20 (100M sh)

- Price-weighted: (110 + 20)/(100 + 25) − 1 = **+4.0%**
- Value-weighted: cap A 2,000 → 2,200; B 2,500 → 2,000; 4,500 → 4,200 = **−6.7%**
- Equal-weighted: (10% + (−20%))/2 = **−5.0%**
- DJIA divisor: index = Σprice / divisor. $90 + $30, d = 2 → 60. Split 2-for-1: $45 + $30 = 75; naive 75/2 = 37.5 ×; new divisor = 75/60 = **1.25** (real divisor < 1)
- Others: Nasdaq Composite (tech-heavy), NYSE Composite, Wilshire 5000 (broadest), Nikkei, FTSE, DAX, Hang Seng
- Bond indexes: harder (more bonds, stale prices, matrix pricing, maturity drift, frequent rebalancing); Bloomberg, Merrill Lynch, Salomon

2.5 Derivatives

DEF

- Value derived from underlying asset
- **Call** = right to BUY at strike K; **Put** = right to SELL at K; buyer pays **premium**
- **Futures** = obligation to trade at set price on future date; no premium; exchange-traded, standardized; **marked to market** daily; zero-sum

Position	Payoff at expiry	Profit	Break-even
Long call	max(S_T − K, 0)	payoff − premium	K + premium
Long put	max(K − S_T, 0)	payoff − premium	K − premium

Position	Payoff at expiry	Profit	Break-even
Long futures	$S_T - F_0$	same (no premium)	F_0
Short futures	$F_0 - S_T$	same	F_0

	Options	Futures
Nature	Right	Obligation
Up-front	Premium	Margin only
Payoff	Asymmetric (capped loss)	Symmetric (linear)
Adverse move	Let expire	Must settle
Analogy	Insurance	Locked-in price

- **TRAP** Option writer (seller) has the obligation, not the buyer; call loss capped at premium only for the buyer

Ch.3 Securities Markets

3.1 Issuing Securities

	Primary market	Secondary market
Trade	New issues sold to investors	Existing securities among investors
Issuer	Receives proceeds	No new money
Role	Investment bankers assist	Liquidity, price discovery

- **IPO process**: ① file prospectus with SEC → ② underwriters market, **book-building** → ③ offer price set; **firm commitment** (underwriter bears placement risk) → ④ list & trade in secondary market
- **Underpricing**: IPO priced below first-day trading price → large first-day return = cost to issuer, benefit to allocated investors
- Firm commitment: syndicate buys entire issue, resells, spread = pay, underwriter bears unsold risk. Best efforts: agent only, unsold shares return to issuer (※ best efforts not in PPT)
- **CALC** IPO money left on table = (P_first-day close − P_offer) × shares. Ex: 20 → 26, 10M sh → **$60M**; first-day return = 26/20 − 1 = **30%**
- ※ Winner's curse, long-run IPO underperformance: not in PPT

3.2 How Securities Trade

Market type	Keywords
Direct search	Find each other; sporadic, low-value; lowest organization
Brokered	Broker searches for a fee; block trades, real estate
Dealer	Bid/ask quotes from inventory; profit = spread
Auction	All traders converge, one place & price; most integrated

CALC — bid / ask

- You **buy at ask** (higher), **sell at bid** (lower)
- Spread = Ask − Bid = dealer gross profit = investor round-trip cost. Ex: 50.10 − 49.90 = **$0.20**

Electronic trading

- **ECN**: matches orders directly, no dealer
- **Algorithmic**: timing, price, size → cut cost & market impact
- **HFT**: ultra-fast, tiny fleeting price gaps, milliseconds
- Benefits: speed, lower cost, transparency, anonymity for large orders, after-hours trading

Venue	Keywords
NYSE	Auction / order-driven; central limit-order book; **price–time priority**; hybrid (electronic + human oversight at open / close); blue chips
Nasdaq	Dealer / quote-driven; competing **market makers**; no floor; screen-based since 1971; tech / growth
KRX	KOSPI (large cap), KOSDAQ (growth / tech); auction; opening & closing call auctions; also clearing, derivatives

Venue	Keywords
Nextrade (NXT)	Korea's 1st ATS; Mar 2025; 34-firm consortium; 08:00–20:00; ~800 stocks; maker–taker fees; mid-price & stop-limit orders
Smart order routing	Broker auto-routes to better price (KRX vs NXT) → duty of **best execution**
Cboe	1973; 1st standardized options; largest US options exchange; **VIX** = fear gauge, expected 30-day S&P 500 volatility
CME Group	Largest futures exchange; CME + CBOT 2007 (+NYMEX); Globex; 1982 first equity-index futures; clearinghouse = central counterparty; CFTC

- Options / futures: **options** = right, premium, loss capped; **futures** = binding, standardized, marked-to-market, centrally cleared
- Regulator: securities & security options → **SEC**; futures & commodities → **CFTC**

3.3 Order Types

	Market	Limit	Stop
Execution	Immediate, best available price	Buy ≤ / sell ≥ set price	Dormant until trigger
Certain	Execution	Price	—
Uncertain	Price (**slippage**)	Execution (may never fill)	Price (becomes market)
Use	Liquid stock, fill matters	Price-sensitive, patient	Limit loss / protect gain

- **Stop-loss** → market order after trigger; **stop-limit** → limit order after trigger (price guaranteed, fill not)
- **Marketable limit** (crosses spread) fills now; **non-marketable** rests in book, adds liquidity
- Trade-off: market = certainty of fill; limit = certainty of price

CALC

- Fill rule: buys hit lowest asks; sells hit highest bids; average price = Σ(p × q)/Σq
- Slippage: ask book 100@50.10, 200@50.15, 500@50.25; market buy 500 → (5,010 + 10,030 + 10,050)/500 = **$50.18** vs last trade 50.10
- Limit orders at bid 49.90 / ask 50.10: buy @49.95 rests; buy @50.10 fills 50.10; sell @50.05 rests; sell @49.90 fills 49.90
- Book (bids 50.10×500, 50.05×300, 50.00×800; asks 50.20×400, 50.25×600, 50.30×700): market buy 600 → (400 × 50.20 + 200 × 50.25)/600 = **$50.217**; market sell 1,000 → (500 × 50.10 + 300 × 50.05 + 200 × 50.00)/1,000 = **$50.065**; limit buy 500 @50.15 → no fill (ask 50.20 > 50.15), rests as best bid
- Gap-down: own @80, stop 72, opens 68 → stop-loss fills ≈ **68**; stop-limit (72/72) **no fill**
- Scenarios: cap loss = stop-loss; buy only ≤ 45 = limit buy; must own now = market; lock in gain = stop-loss (sell stop)

3.4 Margin Trading

DEF

- Buy with own **equity** + **broker loan**; shares = collateral; pay interest
- **Initial margin** (Fed Reg T) ≥ 50%; **maintenance margin** lower (25–40%); below → **margin call** (add cash / securities or position sold)
- Loan fixed → equity absorbs all gains & losses → **leverage magnifies both**; profitable only if stock return > borrowing rate

CALC

1. Balance sheet: Assets = Loan + Equity
2. Margin % = Equity / Asset value = (V − Loan)/V
3. Buying power = Cash / initial margin = 1 / m_init (50% → 2×, 40% → 2.5×, 33% → 3×, 25% → 4×). Ex: cash 10,000 @50% → buy **20,000**, borrow 10,000
4. 400 sh @50 (loan 10,000): at $45 → equity 18,000 − 10,000 = 8,000; margin = 8,000/18,000 = **44.4%**
5. **Margin-call price (long)**: (N·P − Loan)/(N·P) = m → **P = Loan / [N(1 − m)]**. Ex: 10,000/(400 × 0.7) = **$35.71** (call when price falls below)
6. Return on equity = [V_1 − Loan(1 + r_b)]/Equity_0 − 1. Ex: price ±20% → ±**40%** (ignoring interest)

- **TRAP** Margin denominator = total asset value; call-price formula uses (1 − m)

3.5 Short Selling

DEF

- Borrow shares → sell high → buy back low (**cover**) → return; profit if price **falls**; motive: bearish view or hedge
- Owe shares, not cash; must reimburse dividends (payment in lieu); borrow fee (higher if hard-to-borrow); forced buy-in on recall; no fixed maturity
- Done in margin account; proceeds held as **collateral** (not withdrawn) + initial margin deposit (Reg T 50% of short value)
- Payoff: max gain **limited** (price → 0); loss **unlimited**
- Risks: unlimited loss, margin call on **price rise**, **short squeeze**, forced buy-in, fees & dividends, bans (uptick rules, halts)

CALC (short 1,000 sh @ $100)

1. Collateral = proceeds 100,000 + margin 50,000 = **150,000**
2. Equity = Collateral − Value owed = 150,000 − 1,000·P
3. Margin = Equity / Value owed. Start 50,000/100,000 = 50%; at $110 → 40,000/110,000 = **36.4%**
4. **Margin-call price (short)**: (C − N·P)/(N·P) = m → **P = C / [N(1 + m)]**. Ex m = 30%: 150,000/1,300 = **$115.38** (call when price rises above)
5. Return = (P_sell − P_cover − dividends/share)/(initial margin per share). Ex: 100 → 90 → 10,000/50,000 = **+20%**

- **TRAP** Long: (1 − m) in denominator, call on fall. Short: (1 + m), call on rise

3.6 Regulation

Pillar	Keywords
Securities Acts 1933 & 1934	Full disclosure at issue; created **SEC**
Sarbanes-Oxley 2002	Corporate governance, auditing, financial reporting after scandals
Self-regulation (FINRA)	Industry polices member conduct & trading practices
Insider trading rules	Prohibit trading on **material non-public information**

Ch.4 Mutual Funds & Other Investment Companies

4.1 Investment Companies

DEF

- Financial intermediary: pools funds of many investors → invests in wide range of securities
- **Pro-rata claim**: own 1% of shares = 1% of every asset + 1% of income; no specific securities owned; no guaranteed return

#	Four core services
1	Record keeping & administration (reports, distributions, reinvestment, tax reporting, switching within family)
2	**Diversification & divisibility** (fractional share of many securities)
3	**Professional management** (analysts, PMs) — beating market = empirical question, not guaranteed
4	**Lower transaction costs** (large blocks)

CALC — NAV

- **NAV = (Market value of assets − Liabilities) / Shares outstanding**
- Assets marked to market at close; liabilities = accrued fees, payables, borrowing
- Ex: assets 35.8M, liabilities 0.8M, shares 3.5M → 35.0/3.5 = **$10.00**
- Ex: 412M assets, 37M liabilities, 15M shares → 375/15 = **$25.00**; portfolio +4%, liabilities same → (412 × 1.04 − 37)/15 = 391.48/15 = **$26.10**
- New shares issued at NAV → assets & shares rise proportionally → NAV unchanged, no dilution
- Open-end: transact at NAV (struck once daily at 4:00 p.m. ET); closed-end & ETF: market price

4.2 Types of Investment Companies

Type	Keywords
UIT	Fixed, **unmanaged** portfolio in trust by sponsor; redeemable units; **self-liquidating** (set termination date); uniform assets (e.g., munis); very low fees; sold at NAV + 1–4% markup

Type	Keywords
Open-end (mutual fund)	Shares issued / redeemed; buy from / redeem with fund; price = **NAV** (once daily 4pm); all orders same price; no intraday trading
Closed-end	Fixed shares after IPO; no issuance / redemption; trade on exchange with other investors; price by supply & demand → **premium / discount** to NAV; intraday, limit orders, short sales
Commingled funds	Pooled vehicles for small trust / retirement accounts, managed collectively
REIT	Pools funds in income-producing real estate; diversified exposure without managing property
Hedge fund	Private pool; **accredited / institutional** only; not registered under Investment Company Act; short sales, derivatives, heavy leverage

	Mutual fund	Hedge fund
Investors	General public	Accredited / institutional
Regulation	Investment Company Act 1940, full disclosure	Largely unregulated, limited disclosure
Liquidity	Daily redemption at NAV	**Lock-up**, quarterly redemption with notice
Leverage / shorts	Tightly restricted	Essentially unrestricted
Fees	Expense ratio ~0.03–1.5%	**"2 and 20"**: ~2% of assets + performance fee

4.3 Mutual Funds

Policy	Keywords
Money market	Short-term, high-quality debt; liquidity, capital preservation
Equity	Income funds (dividends) vs growth funds (capital appreciation)
Sector	Single industry; high concentration risk
Bond	By issuer (gov, corp, muni, MBS), maturity, credit quality
International	Global (incl. US), international (ex-US), regional, emerging
Balanced	Stocks + bonds, relatively **stable mix** (※ not in PPT as a named type)
Asset allocation / flexible	Multiple asset classes; adjusts mix strategically / tactically
Index	Tracks benchmark; low turnover, research cost, fees

- Policy (prospectus) drives risk exposure more than manager skill
- Indexed assets overtook active for the first time (ICI): ~$21.8tn indexed vs $18.8tn active (May 2026)
- Sold: **direct-sold** (fund company, no broker, lower distribution cost); **broker-sold** (loads, conflict of interest); **fund supermarkets** (many families, no-transaction-fee)

4.4 Costs of Investing

Fee	Keywords
Expense ratio	Admin + advisory; deducted **daily** from assets, never billed → investors underestimate; reported return already **net**
Front-end load	Commission at purchase; legal cap **8.5%**, rarely > 6%; reduces amount invested
Back-end load (CDSC)	Redemption fee; ~5% declining 1 pt/yr (5, 4, 3, 2, 1, 0); **locks in** investor
12b-1 fee	SEC Rule 12b-1; annual, max **1%** of assets; marketing / adviser pay; inside expense ratio; economically = load paid in instalments
Soft dollars	In roadmap only; no detail in PPT

- Loads go to selling broker, not fund; published (NAV-based) returns **overstate** load-fund investor's return

CALC

- Front-end load: invested = P × (1 − load). 5.75% on 10,000 → **$9,425**; break-even = 10,000/9,425 − 1 = **6.10%** (> load %)
- Offer price = NAV/(1 − load)
- **Net return ≈ Gross − Expense ratio (− 12b-1 per PPT)**; loads applied separately to initial & final cash flows
- **Fund HPR = (NAV_1 − NAV_0 + income dist. + cap-gain dist.)/NAV_0**. Ex: (30.10 − 28.50 + 0.60 + 1.20)/28.50 = **11.93%** (net); gross = 11.93 + 1.20 = **13.13%**; with 5% front load: buy price 28.50/0.95 = 30.00 → 31.90/30.00 − 1 = **6.33%**
- Distributions must be added back (NAV drops by distribution on ex-date)

- **Share class**: A = 4% front + 1.00% exp → net 11.00%; C = no load, 1.75% exp, 1% CDSC yr1 → net 10.25% (12% gross)
  - A_n = 10,000(0.96)(1.11)^n; C_n = 10,000(1.1025)^n (yr1 × 0.99 for CDSC)
  - 1 yr: A 10,656 < C 10,915; 3 yr: 13,129 < 13,401; 6 yr ≈ break-even (17,956 vs 17,959); 10 yr: 27,258 > 26,533; 20 yr: 77,399 > 70,400
  - **Short horizon → Class C; long horizon → Class A** (load one-time; annual fee compounds)

4.5 Exchange-Traded Funds

DEF

- ETF: trades on exchange all day (like closed-end) near NAV (like open-end); **investors transact with each other, not the fund**
- **Creation / redemption in kind** by **Authorized Participant (AP)**: swaps basket of underlying shares ↔ ETF shares, no cash
- **Arbitrage**: ETF > basket → AP buys basket, delivers for new ETF shares, sells ETF → price falls back; ETF < basket → reverse

Gains	Costs
Trade intraday: limit, stop, short, options	Spread + commission on every trade
No cost of others' exits → **tax efficiency** (in-kind, no forced selling → fewer cap-gain distributions)	Arbitrage not instant / free (e.g., Korea-listed US-index ETFs trade while NY closed → no live NAV)
No idle cash buffer (mutual fund holds 1–5%)	Easy trading → overtrading → lower returns
	Label = how it trades, not what it holds (single-stock 2× ETF)

- Concentration: BlackRock, Vanguard, State Street = 59% of global ETF assets
- Sorting ETFs: what it holds (equity / bond), how run (passive / active), how built (plain / covered call / leveraged / inverse)

CALC — Leveraged ETF (2× resets **daily**; ±10% days)

- Index: 100 → 110 → 99 → 108.9 → 98.01 = **−1.99%**
- 2× ETF: 100 → 120 → 96 → 115.2 → 92.16 = **−7.84%**
- Volatility decay: compounding on changing base; multi-period ≠ 2× index return

CALC — Covered call (own stock @100, sell call K = 105, premium 2)

- Payoff = (S_T − 100) + 2 − max(S_T − 105, 0)
- S_T: 80 → −18; 90 → −8; 100 → +2; 105 → +7; 120 → +7; 140 → +7
- Max gain = (K − S_0) + premium = **+7**; **upside capped, downside open**; premium constant = not protection
- Payout = option premium, **not dividend**; best when flat, worst in strong rally

4.6 Investment Performance

Topic	Keywords
Active vs passive	SPIVA: 79% US large-cap active underperformed S&P 500 (2025); 92% of US domestic active underperformed over 20 yrs; worse with longer horizon
Arithmetic of active mgmt	Passive holds market → active as a group holds market → value-weighted avg gross active return = market return; **average active return = R_market − costs**
Limits	Says nothing about any single fund or dispersion; winners paid for by losers
Survivorship bias	Failed / merged funds drop from list (16% within 10 yr, 47% within 15 yr); visible funds are filtered; SPIVA counts them as underperformers
Persistence	Top-quartile 2021 (164 funds): 20.1% still top in 2022, **0.0%** in 2023; chance = 0.25^2 = **6.25%** → past winners don't repeat; fee is the lasting signal
Coin analogy	1,000 coins × 5 flips → $1{,}000/2^5 \approx$ **31** all heads → luck, not skill
Korea	77% of Korea-domiciled US equity funds underperformed S&P 500 (won terms)

- **CALC** Expected number of repeaters by chance = N × 0.25^k (k years top-quartile). Ex: 164 × 6.25% ≈ **10**

Ch.5 Risk, Return & the Historical Record

5.1 Rates of Return

DEF — HPR

- HPR = (P_1 − P_0 + Div)/P_0 = **capital gain yield + dividend yield**
- Denominator = **beginning (purchase) price**, never ending price
- Ex: 70,000 → 76,000, div 1,500 → gain 8.57% + div 2.14% = **10.71%**

Average	Formula	Keywords
Arithmetic	r̄ = Σr_t / n	Simple mean; **forecasts** next-period E(r); overstates compound growth
Geometric (time-weighted)	g = [Π(1 + r_t)]^(1/n) − 1	Realized compound growth; **past performance**; fund disclosure standard; g ≤ r̄, gap grows with volatility
Dollar-weighted (IRR)	PV(inflows) = PV(outflows)	Reflects investor's cash-flow timing; grades **investor**; differs from TW when flows added / withdrawn

- Rough link: **g ≈ r̄ − ½σ²** (volatility drag)
- Time-weighted → judge **manager**; dollar-weighted → judge **investor** (contaminated by client timing)

CALC

- Returns +25, −15, +30, −5: arithmetic = 40/4 = **8.75%**; wealth = 1.25 × 0.85 × 1.30 × 0.95 = 1.3122 → geometric = 1.3122^(1/4) − 1 = **7.03%**; check 8.75 − ½(0.1916)² ≈ 6.9%
- IRR ex: CF0 = −50,000; CF1 = −53,000 + 2,000 = −51,000; CF2 = 108,000 + 4,000 = +112,000 → solve 0 = −50,000 − 51,000/(1 + r) + 112,000/(1 + r)² → **r ≈ 7.1%**; time-weighted: sub-period 1 = (53,000 + 2,000 − 50,000)/50,000 = 10%, sub-period 2 = (2×54,000 + 4,000 − 2×53,000)/106,000 = 5.66% → √(1.10 × 1.0566) − 1 = **7.81%**

APR vs EAR

- APR = simple scaling, ignores compounding; **EAR = (1 + APR/n)^n − 1**
- Continuous: **EAR = e^APR − 1**; APR = ln(1 + EAR)
- EAR rises with n, converges as n → ∞; compare products on EAR; continuously compounded returns **add** over time
- APR 12%: n = 1 → 12.000%; 2 → 12.360%; 4 → 12.551%; 12 → 12.683%; 365 → 12.747%; ∞ → **12.750%**
- Ex: card APR 19.9% monthly → (1 + 0.199/12)^12 − 1 = **21.82%**

5.2 Inflation & Real Rate

DEF — Fisher equation: (1 + R) = (1 + r)(1 + i)

CALC

- Exact: **r = (1 + R)/(1 + i) − 1**; approx: r ≈ R − i
- Approximation **overstates** real return; error = r × i; negligible at low inflation; never for long-horizon / cumulative
- Ex: R 8%, i 3.5% → 4.35% (approx 4.50); R 20%, i 15% → 4.35% (approx 5.00) ⇒ same real return, different nominal; R 3.5%, i 5.1% → **−1.52%**
- Equilibrium nominal rate: **R = r_real + E(i)**; expected inflation passes 1-for-1 into R; real rate set by saving & investment; keeps lenders' real return

5.3 Risk & Risk Premiums

CALC — scenario analysis (list states s, p(s), r(s))

- **E(r) = Σ p·r**
- **σ² = Σ p·[r − E(r)]²**; **σ = √σ²**
- Ex: Boom .25/+40, Normal .35/+15, Slowdown .25/−5, Recession .15/−25 → E = 10 + 5.25 − 1.25 − 3.75 = **10.25%**; σ² = 221.27 + 7.90 + 58.14 + 186.38 = 473.69 → **σ = 21.76%**
- **Excess return** = r − r_f (realized, can be negative); **risk premium** = E(r) − r_f (expected, positive). Ex: 10.25 − 3 = **7.25%**
- Risk averse: needs positive premium, rejects fair game (standard assumption) · risk neutral: E(r) only · risk loving: accepts fair game

Normality

- 68-95-99.7: E ± 1σ / 2σ / 3σ (ex: −11.5–32.0; −33.3–53.8; −55.0–75.5)
- Convenient: 2 parameters define distribution; symmetric σ = full risk; sum of normals is normal
- Ex under N(10.25, 21.76²): P(r < 0) = Φ(−0.47) = **31.9%**; P(r > 32) = z = 1 → **15.9%**; P(r < −33.3) = z = −2 → **2.3%**; 95% CI = 10.25 ± 1.96σ = **−32.4 to 52.9%**

Departure	Formula	Keywords
Skewness	$E[(r - E)^3]/\sigma^3$	Asymmetry; negative = long left tail, mean < median < mode; equities left-skewed → σ understates risk
Kurtosis	$E[(r - E)^4]/\sigma^4 - 3$	Normal = 0; **fat tails**, extremes more frequent → tail risk

- Evidence: Black Monday 1987 (≈ −20.4%, −20σ), 2008 / 2020 KOSPI crashes → normal understates extreme losses

VaR (Value at Risk)

- **DEF**: loss exceeded with probability α over horizon T (α-quantile of loss); needs **horizon T, confidence c, base value V_0**
- 1-day 99% VaR ₩40M → exceeded ~1 day in 100; **not maximum loss**; silent on size beyond cut-off
- **Parametric VaR = $V_0 \times z_c \times \sigma \times \sqrt{T}$** (mean ignored: σ ∝ √T, μ ∝ T)
- z: 95% = **1.645**; 99% = **2.326**; 99.5% = 2.576
- √T rule assumes independence & constant σ (fails in crisis); **TRAP** do not multiply by T
- Ex: V_0 = ₩1bn, σ = 2%/day: 95% 1-day = 1.645 × 0.02 × 1bn = **₩32.9M**; 99% 1-day = **₩46.5M**; 99% 10-day = 46.5 × √10 = **₩147.1M**
- Limits: normality understates tails; no loss size past cut-off
- **Historical simulation**: sort actual past returns, read quantile; no distribution assumed, fat tails free, needs long sample
- **Monte Carlo**: simulate paths from assumed model; handles options / non-linear; model risk
- **Expected shortfall (ES / CVaR)**: average loss beyond VaR; **coherent (subadditive)**; Basel FRTB: 99% VaR → **97.5% ES**
- Others: max drawdown, Sortino, stress tests

Sharpe ratio

- **$S = [E(r_P) - r_f]/\sigma_P$** = reward per unit of volatility = slope of CAL
- Unchanged by mixing risk-free → measures quality of risky portfolio
- Fails: single security (denominator includes diversifiable risk), assumes normality (flatters fat left tails), interval-dependent, estimation error
- Ex: 7.25/21.76 = **0.33**

5.4 Historical Record (US, 1928–2021)

Asset	Avg return (arith.)	σ
Small caps (bottom decile)	18.5%	38.5%
Large stocks (S&P 500)	11.8%	19.5%
Baa corporates	7.2%	7.4%
10-yr T-bonds	5.1%	7.7%
3-mo T-bills	3.4%	3.1%

- Lessons: higher return ⇔ higher risk; **equity risk premium over T-bills = 8.44 pp** (11.82 − 3.38); T-bills 3.38% vs ~3% inflation → real ≈ 0
- S&P 500: arithmetic **11.82%** > geometric **9.98%**; $100 → $761,748
- Baa vs T-bond gap 2.1 pp = credit + term + liquidity premium (σ not comparable)

5.5 Asset Allocation

DEF

- Complete portfolio C: fraction **y** in risky P, **(1 − y)** in risk-free
- Risk-free: σ = 0, Cov = 0 → **$\sigma_C = y \cdot \sigma_P$**; y > 1 = borrowing
- Sharpe ratio **invariant to y** → choose P and y separately = **separation property**

CALC — complete portfolio

- **$E(r_C) = r_f + y[E(r_P) - r_f]$**; **$\sigma_C = y\sigma_P$**; premium_C = y·premium_P
- P: 14%, 20%, r_f 4%, y = 0.7 → E = **11%**, σ = **14%**, premium **7%**, Sharpe **0.50**; wealth 10m → 3.0m r_f + 7.0m P (60/40 → 4.2m equity, 2.8m bond); y = 0.5 → E 9%, σ 10%

CAL (capital allocation line)

- **$E(r) = r_f + S \cdot \sigma$**; intercept = r_f; slope = S = (14 − 4)/20 = **0.50**; left of P lending, right borrowing
- **Kink** at P: borrowing rate > r_f → slope falls beyond P

Utility

- **$U = E(r) - \frac{1}{2} \cdot A \cdot \sigma^2$** (decimals); A > 0 averse (typical 2–4), A = 0 neutral, A < 0 loving
- ½ A σ^2 = risk penalty (grows with σ^2); risk-free U = r_f
- **Certainty equivalent** = risk-free rate giving the same U

- Ex: P3 (13%, 20%), A = 8: 0.13 − ½·8·0.04 = **−3%**; table (U): P1 (7/5): 6.75 / 6.50 / 6.00; P2 (9/10): 8.00 / 7.00 / 5.00; P3 (13/20): 9.00 / 5.00 / −3.00 for A = 2 / 4 / 8 → choose P3 / P2 / P1

Optimal y*

1. U = r_f + y[E(r_P) − r_f] − ½ A y² σ_P²
2. dU/dy = 0 → **y* = [E(r_P) − r_f] / (A·σ_P²)**

- y* ↑ with risk premium; ↓ with A; ↓ with σ²; y* > 1 → borrow
- Ex: 0.10/(A × 0.04): A = 2 → **1.25** (E 16.5%, σ 25%, U 10.25%); A = 4 → **0.625** (E 10.25%, σ 12.5%, U 7.13%); A = 6 → **0.417** (E 8.17%, σ 8.33%, U 6.08%)
- Borrowing rate above r_f → y* capped below 1.25 (kink)

5.6 Passive Strategies & CML

- **Passive** = hold broad market index, no security analysis / timing
- **CML** = CAL when P = broad market index
- Rationale: **low cost** (0.03–0.2% vs 1%+ active); **free-rider** (others' analysis already in price); low turnover; automatic diversification
- Open: **Grossman–Stiglitz paradox** (if all passive, who sets prices?); index may not be well diversified; passive ≠ no risk

Ch.6 Efficient Diversification (§6.1–6.4)

6.1 Diversification & Portfolio Risk

	Systematic (market)	Firm-specific (idiosyncratic)
Source	Economy-wide: business cycle, rates, inflation, FX, oil, war, pandemic	One company / industry: CEO exit, failed drug trial, fire, lawsuit, fraud
Diversifiable?	**No** (non-diversifiable)	**Yes** (good / bad surprises offset)
Priced?	**Earns risk premium**	Not rewarded

- **Var(total) = Var(systematic) + Var(firm-specific)** (uncorrelated parts)
- **Insurance principle**: risk reduction by spreading exposure across many independent risk sources
- Enron 2001: ~60% of 401(k) in Enron stock; $90 → < $1; job + savings on same firm → doubling down
- Prices set by diversified investors → only systematic risk priced

CALC — n equally weighted stocks (w = 1/n)

- **σ²_P = (1/n)·avg variance + (1 − 1/n)·avg covariance**
- n → ∞: first term → 0; floor = **average covariance** (σ_P = √ avg Cov) = market risk
- Ex: σ = 40% (var 0.1600), ρ = 0.30 → avg Cov = 0.30 × 0.4 × 0.4 = **0.0480**
  - n = 1: 40.00%; n = 10: 0.0592 → **24.33%**; n = 50: 0.0502 → 22.41%; n = 100: 0.0491 → 22.16%; n = ∞: 0.0480 → **21.91%** (floor)
  - 1 → 10 stocks removes most diversifiable risk; 100 → ∞ almost nothing

6.2 Two Risky Assets

Inputs (running case): S: E 12%, σ 20%; B: E 6%, σ 10%; ρ = 0.20 → Cov = 0.20 × 20 × 10 = **40** (%²); r_f = 3%

Concept	Formula	Keywords
Portfolio return	E(r_P) = w_S E(r_S) + w_B E(r_B)	**Linear** in weights
Covariance	Cov = Σ p_i [r_S(i) − E(r_S)][r_B(i) − E(r_B)]	Move together (+) / offset (−); units hard to read
Correlation	ρ = Cov/(σ_S σ_B); Cov = ρσ_Sσ_B	Between **−1 and +1**; unit-free
Portfolio variance	σ²_P = $w_S^2\sigma_S^2 + w_B^2\sigma_B^2 + 2\, w_S\, w_B$ Cov	Interaction term = only term that can be **negative** = only source of diversification

- ρ = +1: no benefit; ρ = −1: perfect hedge (risk-free mix possible); 0 < ρ < 1: usual case (partial benefit)
- **Cost-free**: E(r) = weighted avg (no return sacrificed) but σ_P < weighted-avg σ → benefit exists only if ρ < +1

CALC — 60/40 step order

1. E = 0.6(12) + 0.4(6) = **9.60%**
2. Stock term = 0.36 × 400 = 144.0
3. Bond term = 0.16 × 100 = 16.0

4. Interaction = 2(0.6)(0.4)(40) = 19.2
5. Variance = 179.2; σ = **13.39%** (vs weighted-avg 16.00% → saved 2.61)
6. Sharpe = (9.60 − 3)/13.39 = **0.493**

Effect of ρ (60/40; fixed terms 144 + 16)

ρ	Cov	Interaction	Variance	σ_P	Sharpe
+1.0	200	+96	256	16.00%	0.413
+0.5	100	+48	208	14.42%	0.458
+0.2	40	+19.2	179.2	13.39%	0.493
0.0	0	0	160	12.65%	0.522
−0.5	−100	−48	112	10.58%	0.624
−1.0	−200	−96	64	8.00%	0.825

- ρ = +1: **$\sigma_P = w_S\sigma_S + w_B\sigma_B$**; ρ = −1: **$\sigma_P = |w_S\sigma_S - w_B\sigma_B|$**
- **Zero-risk mix (ρ = −1)**: 20w = 10(1 − w) → **$w_S = \sigma_B/(\sigma_S + \sigma_B) = 1/3$**; E = 1/3(12) + 2/3(6) = **8%**

Scenario inputs (Example 3) — p = 0.25 each: stock −22 / +6 / +18 / +38 → E = **10%**; bond +12 / +4 / +3 / +5 → E = **6%**; deviation products −192, +8, −24, −28 → **Cov = −59**

- Steps: (1) E = Σp·r; (2) variances Σp·dev²; (3) Cov = Σp·dev_S·dev_B, ρ = Cov/(σ_Sσ_B)
- 50/50 check: σ_P = **9.57%** (formula = state-by-state); weighted avg 0.5(21.73) + 0.5(3.54) = 12.64% → gap **3.07 pts** = diversification benefit; E = 8.00%

Opportunity set & efficiency

- Curve of portfolios over w; lower ρ → bows further **left**; ρ = +1 straight line; ρ = −1 touches vertical axis
- **Efficient** = upward-sloping part from **minimum-variance portfolio**; points below = **dominated** (same σ, lower E)

Minimum-variance portfolio

- **$w_{S,min} = [\sigma_B^2 - Cov]/[\sigma_S^2 + \sigma_B^2 - 2Cov]$**
- Ex: (100 − 40)/(400 + 100 − 80) = 60/420 = **14.29%** stocks; E = **6.86%**; σ = **9.56%** (< bond's 10%)
- Adding riskier asset reduces risk if **$\rho < \sigma_B/\sigma_S$** = 10/20 = **0.50** (negative ρ not required)
- **TRAP** Min-variance ignores expected return → not the best portfolio

6.3 Optimal Risky Portfolio with a Risk-Free Asset

DEF

- Complete portfolio: **$E(r_C) = r_f + y[E(r_P) - r_f]$**; **$\sigma_C = y\sigma_P$**
- **CAL**: E(r_C) = r_f + S_P·σ_C; **slope = Sharpe ratio**
- Each risky portfolio has its own CAL; **steeper = strictly better** (more E at every σ) → maximize Sharpe ratio
- **Optimal risky portfolio P*** = **tangency** point of CAL with opportunity set; highest Sharpe; lies on efficient frontier

Candidate (r_f = 3%)	w_S	E	σ	Sharpe
All bonds	0%	6.00%	10.00%	0.300
Min-variance	14.29%	6.86%	9.56%	0.403
60/40	60%	9.60%	13.39%	0.493
All stocks	100%	12.00%	20.00%	0.450
Optimal P*	**48.15%**	**8.89%**	**11.82%**	**0.498**

CALC — optimal weights (excess returns R_S = E(r_S) − r_f, R_B)

- **$w_S^* = [R_S\sigma_B^2 - R_B Cov] / [R_S\sigma_B^2 + R_B\sigma_S^2 - (R_S + R_B)Cov]$**; $w_B^* = 1 - w_S^*$
- Ex (R_S = 9, R_B = 3): numerator 9(100) − 3(40) = **780**; denominator 900 + 1,200 − 480 = **1,620**; w_S^* = **0.4815**, w_B^* = 0.5185
- E(r_P) = 0.4815(12) + 0.5185(6) = **8.89%**; variance = 92.73 + 26.89 + 19.97 = 139.59; σ = **11.82%**; Sharpe = 5.89/11.82 = **0.498**
- Reading: higher premium ↑ weight; higher own variance ↓; high Cov ↓ diversification value; if $w^* \notin [0, 1]$ → short sale needed

CALC — complete portfolio (A = 6, P*: 8.89%, 11.82%; **decimals**)

1. **$y^* = [E(r_P) - r_f]/(A\sigma_P^2)$** = 0.05889/(6 × 0.013959) = **0.7031**
2. T-bills = 1 − y* = **29.69%**
3. Look-through stocks = 0.7031 × 0.4815 = **33.86%**; bonds = 0.7031 × 0.5185 = **36.45%** (sum 100%)
4. E(r_C) = 3 + 0.7031(5.89) = **7.14%**; σ_C = 0.7031 × 11.82 = **8.31%**

5. U = 0.0714 − ½(6)(0.0831)² = **5.07%** (vs 3% T-bills)

- A = 3 → y* = **1.41** → borrow at r_f (leveraged P*)

Four-step recipe

1. Inputs: E(r), σ, Cov / ρ
2. Trace E(r_P), σ_P over weights; efficient (upper) branch
3. Maximize Sharpe → P*
4. y* from A; split P* vs r_f

- **Only step 4 depends on investor**; steps 1–3 identical for all = **separation property**

6.4 Many Risky Assets

Efficient frontier (Markowitz)

- Highest E(r) for each σ; found by **minimizing variance for each target E(r)**
- Only branch **above global minimum-variance portfolio** is efficient
- Individual assets lie **inside** the frontier → all dominated
- **P*** = unique Sharpe-maximizing tangency portfolio; same math as 2-asset, more bookkeeping

Separation property

- (1) Determine optimal risky portfolio → technical, same for all clients
- (2) Decide how much in it (y) → personal, from risk aversion
- Implication: one research team serves many clients; questionnaire sets y not composition; target-date funds glide y down; bridge to CAPM (everyone holds P* → market portfolio)
- **Breaks down**: different inputs (disagree), borrowing rate > lending rate (kinked CAL), taxes, human capital, constraints (no short sale, mandates)

CALC — input list

- n returns + n variances + n(n − 1)/2 covariances = **n(n + 3)/2**
- n = 10 → 65; n = 50 → 1,325; n = 500 → **125,750**; n = 800 → 321,200; n = 3,000 → 4,504,500
- Problems: computation (solved) + **estimation error** ("error maximization")

Home bias

- Investors overweight home country vs world share; causes: information costs, currency risk, tax / regulation, familiarity
- Foreign assets lower the **covariance term** (different industrial structure → different systematic shocks)
- Caveats: correlations risen, currency risk; benefit ≠ 0 while ρ < +1
- Covariance argument, not forecasting

Formula sheet (PPT)

- $E(r_P) = w_S E(r_S) + w_B E(r_B)$
- $\sigma^2_P = w_S^2\sigma_S^2 + w_B^2\sigma_B^2 + 2 w_S w_B Cov$
- $Cov = \rho\sigma_S\sigma_B$
- $w_{S,min} = (\sigma_B^2 - Cov)/(\sigma_S^2 + \sigma_B^2 - 2Cov)$
- $S = (E(r_P) - r_f)/\sigma_P$
- $w_S^* = (R_S\sigma_B^2 - R_B Cov)/(R_S\sigma_B^2 + R_B\sigma_S^2 - (R_S + R_B)Cov)$
- CAL: $E(r_C) = r_f + S\cdot\sigma_C$
- $y^* = (E(r_P) - r_f)/(A\sigma_P^2)$
- **Units warning**: y* must use decimals

Out of exam scope: §6.5 single-index model, §6.6 time diversification.

Final Check: Confusable Pairs

Pair	Distinction
Primary vs secondary market	Issuer gets proceeds vs no new money
Firm commitment vs best efforts	Underwriter bears risk vs agent only
Bid vs ask	Sell at bid, buy at ask; spread = round-trip cost
Market vs limit order	Certain fill / uncertain price vs certain price / uncertain fill
Stop-loss vs stop-limit	Becomes market (price not guaranteed) vs becomes limit (fill not guaranteed)

Pair	Distinction
Long vs short margin call	Price **falls**, P = Loan/[N(1 − m)] vs price **rises**, P = C/[N(1 + m)]
BDY vs BEY vs EAY	Face / 360 simple < price / 365 simple < price / 365 compound
Discount vs add-on	T-bill, CP, BA vs CD, repo, fed funds, Eurodollar
Price vs value vs equal weight	DJIA (price) vs S&P 500 (cap) vs average of returns
Option vs futures	Right, premium, asymmetric vs obligation, margin, linear
Open-end vs closed-end vs ETF	NAV once daily vs exchange price (premium / discount) vs exchange + AP arbitrage
Mutual fund vs hedge fund	Public, 1940 Act, daily NAV vs accredited, lock-up, 2 and 20, leverage
Expense ratio vs load vs 12b-1	Annual, in NAV vs one-time (front / CDSC) vs annual distribution fee inside expense ratio
Class A vs Class C	Load + low fee (long horizon) vs no load + high fee (short horizon)
Leveraged ETF vs covered-call ETF	Daily reset decay vs capped upside, open downside
Arithmetic vs geometric vs dollar-weighted	Forecast vs past compound / manager vs IRR / investor
APR vs EAR	Scaled rate vs compounded: (1 + APR/n)^n − 1
Exact vs approximate real return	(1 + R)/(1 + i) − 1 vs R − i (overstates by r·i)
Excess return vs risk premium	Realized r − r_f vs expected E(r) − r_f
VaR vs ES	Cut-off loss vs average loss beyond cut-off
Parametric vs historical vs Monte Carlo	Normal formula vs actual past sorted vs simulated model paths
Sharpe ratio vs utility	Slope of CAL (quality of P) vs U = E − ½Aσ² (investor-specific)
CAL vs CML	Any risky P vs broad market index as P
Min-variance vs optimal (tangency)	Lowest σ, ignores E(r) vs highest Sharpe
Systematic vs firm-specific	Not diversifiable, priced vs diversifiable, unpriced
y* vs weights inside P*	Investor-specific (A) vs same for everyone (separation)

Formula one-liners

- HPR = (P_1 − P_0 + D)/P_0 · NAV = (A − L)/shares · Buying power = 1/m
- Margin call: long P = L/[N(1 − m)]; short P = C/[N(1 + m)]
- EAR = (1 + APR/n)^n − 1 · r = (1 + R)/(1 + i) − 1
- E = Σp·r · σ^2 = Σp(r − E)² · VaR = V_0 z σ √T (z: 1.645 / 2.326)
- Sharpe = (E − r_f)/σ · U = E − ½Aσ^2 · y* = (E − r_f)/(Aσ^2)
- σ^2_P = w_S²σ_S² + w_B²σ_B² + 2w_S w_B$\rho\sigma$_Sσ_B · w_min = (σ_B² − Cov)/(σ_S² + σ_B² − 2Cov)

Basis

- Scope: lecture PPT Ch.2–6 (to §6.4); Ch.1, LIBOR, §6.5, §6.6 excluded
- ※ Not in PPT (SA / MC bank only): best efforts, winner's curse, IPO long-run underperformance, 2010 flash crash, balanced fund as a named type, soft-dollar detail
- PPT graphics lost in text extraction (blank cells) were rebuilt from the formulas shown on the slides

Econometrics — Exam-Eve Summary (Key Points Only)

Wooldridge Ch.1–6 · Econometrics 1–2 · course R code | **[S]** not in material, standard form · **[?]** needs check · **[D]** derived from material · ✔ recomputed in code

Ch.1 · Nature of Econometrics & Data

1.1 Core definitions

- Econometrics = statistical methods to analyze economic data | mostly **nonexperimental** data
- Goals: estimate relationships | test theories | evaluate policy
- Most of econometrics = specification of the error u
- **Ceteris paribus** = "other relevant factors being equal" | causal questions hold other factors fixed
- Population = entire set to study | sample = subset | **external validity** = conclusions valid outside the study

1.2 Data types (→ method choice)

Type	Key feature
Cross-sectional	one point in time | ≈ independent (random sampling)
Time series	serially correlated | order matters | trends, seasonality
Pooled cross sections	independent samples in different years | policy before/after
Panel	**same units** over time | controls time-invariant unobservables | lags

Ch.2 · Simple Regression

Slide equations are images (not extracted) → formulas are **[S]** unless marked.

2.1 Model & OLS

$y = \beta_0 + \beta_1 x + u$ | $E(u|x) = 0$ → PRF $E(y|x) = \beta_0 + \beta_1 x$ **[S]**

$\hat{\beta}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$ (num = covariation, den = $SST_x > 0$)
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ **[S]**

- Procedure: $\bar{x}$, $\bar{y}$ → deviations → Σ cross products → Σ(x dev)² → $\hat{\beta}_1$ → $\hat{\beta}_0$
- Residual $\hat{u}_i = y_i - \hat{y}_i$ (actual − fitted) | $\Sigma\hat{u}_i = 0$ | $\Sigma x_i\hat{u}_i = 0$ | line through $(\bar{x}, \bar{y})$ **[S]**

2.2 CEO salary example (slide table)

- Fitted line from table: **salaryhat = 963.191 + 18.501·roe** **[D]**
- Slope = (2004.808 − 1072.348)/(56.3 − 5.9) = **18.501** **[D]**
- Salary in $1,000s → CEO 12: 933 − 1459.023 = **−526.023** → over-predicted by $526,023
- CEO 5: 1368 − 1218.508 = **+149.493** → under-predicted
- Slide typo: CEO 7 residual = 1078 − 1266.611 = **−188.611** (slide: +) ✔

2.3 Goodness of fit

$SST = \Sigma(y_i - \bar{y})^2$ | $SSE = \Sigma(\hat{y}_i - \bar{y})^2$ | $SSR = \Sigma\hat{u}_i^2$
$SST = SSE + SSR$ → $R^2 = SSE/SST = 1 - SSR/SST$ **[S]**

- Simple regression: $R^2 = corr(y, x)^2$ ✔ (regression.R)

2.4 Log forms (interpretation)

Model	β_1 means
y on x	$\Delta y = \beta_1 \Delta x$
y on log x	$\Delta y \approx (\beta_1/100)\cdot\%\Delta x$ **[S]**
log y on x	$\%\Delta y \approx (100\cdot\beta_1)\Delta x$ → **semi-elasticity**
log y on log x	$\%\Delta y \approx \beta_1\cdot\%\Delta x$ → **constant elasticity**

2.5 Assumptions & theorems

- SLR.1 linear in parameters | SLR.2 random sampling | SLR.3 sample variation in x | SLR.4 $E(u|x)=0$ | SLR.5 $Var(u|x)=\sigma^2$
- **Thm 2.1** (SLR.1–4): $E(\hat{\beta}_j) = \beta_j$ → unbiased = correct **on average** over repeated samples

Thm 2.2 (SLR.1–5): $Var(\hat{\beta}_1) = \sigma^2/SST_x$ **[S]**
Thm 2.3: $\hat{\sigma}^2 = SSR/(n-2)$ | $se(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$ **[S]**

- Var ↑ with σ^2 (noise) | Var ↓ with more variation in x

2.6 Binary x & treatment effects

- $x \in \{0,1\}$: β_0 = mean of y if x=0 | β_1 = **difference in means** **[S]**
- ATE $\tau = E[y_i(1) - y_i(0)]$ | never observe both outcomes | **random assignment** → OLS unbiased for τ
- Job training: +$1,790 → 1,790/4,550 = **39.3%** ✔

Ch.3 · Multiple Regression: Estimation

3.1 Model & interpretation

$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$ **[S]** | "linear" = linear in **parameters**

- β_j = effect of x_j **holding other x fixed** (ceteris paribus)
- College GPA: hsGPA +1 (ACT fixed) → colGPA **+0.453**
- Econometrics 2: in multiple regression β_1 is the **partial slope**, not the slope of y on x_1
- Partialling out: regress x_1 on other x → residual $\hat{r}_1$ → regress y on $\hat{r}_1$ → slope = $\hat{\beta}_1$

3.2 Arrest example (implied coefficients [D])

- Prop. prior arrests +0.5 → −7.5 per 100 men → coef. **−0.150**
- Months in prison +12 → −0.408 per man → **−0.034**/month
- Quarters employed +1 → −10.4 per 100 men → **−0.104**
- Small R^2 can still give good ceteris paribus estimates

3.3 MLR assumptions

- MLR.1 linear | MLR.2 random sample | MLR.3 **no perfect collinearity** | MLR.4 $E(u|x_1...x_k)=0$ | MLR.5 homoskedasticity
- Unbiased: MLR.1–4 | **Gauss–Markov**: MLR.1–5 → OLS = **BLUE**

3.4 Omitted variable bias

True $y = \beta_0+\beta_1x_1+\beta_2x_2+u$, omit x_2:
$E(\tilde{\beta}_1) = \beta_1 + \beta_2\cdot\tilde{\delta}_1$, $\tilde{\delta}_1$ = slope of x_2 on x_1 [S]

Bias [S]	corr(x_1,x_2) > 0	corr < 0
$\beta_2 > 0$	**upward (+)**	downward (−)
$\beta_2 < 0$	downward (−)	**upward (+)**

- No bias if $\beta_2 = 0$ (irrelevant) or $\tilde{\delta}_1 = 0$ (uncorrelated)
- Ability omitted in wage eq. → return to educ biased upward [S]
- Irrelevant variable included → no bias, variance may ↑

3.5 Variance & multicollinearity

$\mathrm{Var}(\hat{\beta}_j) = \sigma^2 / [SST_j(1 - R_j^2)]$, R_j^2 = R^2 of x_j on other x
$\hat{\sigma}^2$ = SSR/(**n − k − 1**) | $se(\hat{\beta}_j) = \hat{\sigma}/\sqrt{[SST_j(1-R_j^2)]}$ | VIF = $1/(1-R_j^2)$ [S]

- σ^2 ↑ → Var ↑ | SST_j ↑ → Var ↓ | R_j^2 ↑ → Var ↑ | R_j^2 → 1 → Var → ∞
- Multicollinearity inflates only the involved variables' variances | **not** an MLR.3 violation
- Dropping a variable reduces variance but may cause OVB → bias–variance trade-off

Ch.4 · Inference

4.1 Distribution

- MLR.6: u ~ Normal(0, σ^2), independent of x | MLR.1–6 = CLM [S]
- Normality impossible: wages (≥ 0), arrests (few integers), unemployment (0/1) | log(y) may help

$t = (\hat{\beta}_j - a_j) / se(\hat{\beta}_j) \sim t_{(n-k-1)}$ (a_j = 0 usually) [S]

4.2 Critical values & rules

H_1	Reject if	Slide (5%)
$\beta_j > 0$	t > c	df 28: **1.701** ✓
$\beta_j < 0$	t < −c	df 18: **−1.734** ✓
$\beta_j \neq 0$	|t| > c	**2.06** (df ≈ 25 [?])

- Large df (two-sided): 10% **1.645** | 5% **1.96** | 1% **2.576** [S] ✓
- Steps: H_0/H_1 → t → df = n−k−1 → c (α, one/two-sided) → decide → interpret

4.3 p-value

- p = smallest significance level at which H_0 is rejected | reject ⇔ **p < α**
- Two-sided: P(|t| > 1.85) = 2 × 0.0359 = **0.0718** ✓

4.4 Confidence interval

$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$, c = 97.5th pct of $t_{(n-k-1)}$ (95%) [S]

- Random bounds: covers β_j in 95% of repeated samples
- a_j outside 95% CI ⇔ reject H_0: $\beta_j = a_j$ at 5% (two-sided)
- Statistical ≠ economic significance → always discuss magnitude | "wrong" sign → misspecification?

4.5 Linear combination ($\beta_1 = \beta_2$)

- $t = (\hat{\beta}_1 - \hat{\beta}_2)/se(\hat{\beta}_1 - \hat{\beta}_2)$ | se needs Cov → not in standard output [S]
- Trick: $\theta = \beta_1 - \beta_2$ → regress y on x_1 and $(x_1 + x_2)$ → $se(\hat{\theta})$ directly [S]

4.6 F test

$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-k-1)] \sim F(q, n-k-1)$
R^2 form: $F = [(R^2_{ur} - R^2_r)/q] / [(1-R^2_{ur})/(n-k-1)]$
Overall: $F = (R^2/k) / [(1-R^2)/(n-k-1)]$ [S]

- q = number of restrictions | F ≥ 0 | reject if F > c
- regression.R: (0.07449/2)/((1−0.07449)/82) = **3.30** ✓
- Restricted model with different y (e.g. rational house assessments) → use **SSR form** only [S]

Ch.5 · OLS Asymptotics

Property	Needs
Unbiased	MLR.1–4
Variance formulas, BLUE	MLR.1–5
Exact t / F	MLR.1–6
Consistent	MLR.1–4 (weaker MLR.4′ suffices)
Asymptotic t / F / CI	MLR.1–5 (no normality)

- Consistency: P(estimate close to true value) → 1 as n ↑ | minimum requirement
- MLR.4′: $E(u) = 0$, $\mathrm{Cov}(x_j, u) = 0$ [S]
- plim $\hat{\beta}_1 = \beta_1 + \mathrm{Cov}(x_1,u)/\mathrm{Var}(x_1)$ | inconsistency doesn't vanish with n [S]
- Large n: t ≈ N(0,1) → t, F, CI valid **without MLR.6** | homoskedasticity still needed
- $se(\hat{\beta}_j) \approx c_j/\sqrt{n}$ → why large samples are better [S]

Ch.6 · Further Issues

6.1 Logs

- Pros: % / elasticity reading | invariant to rescaling | less outliers | helps normality, homoskedasticity
- Don't log: years | percentage points | zero or negative values

6.2 Quadratic

$y = \beta_0 + \beta_1x + \beta_2x^2 + u$ → **$\Delta\hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2x)\Delta x$**
Turnaround **$x^* = |\hat{\beta}_1 / (2\hat{\beta}_2)|$** [S]

- Wage peaks at **24.4 years** of experience (coefficients in image [?])

6.3 Interaction & APE

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u \rightarrow \partial y/\partial x_2 = \beta_2 + \beta_3 x_1$ [S]

- β_2 alone = effect at $x_1 = 0$ | center ($x_1 - \mu_1$) → coefficient = effect at the mean
- APE = average of each unit's partial effect over the sample

6.4 Adjusted R^2

$R^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ [S]

- regression.R: $1 - (0.92551)(84/82) =$ **0.05192** ✓
- $R^2 \uparrow \Leftrightarrow$ new regressor's $|t| > 1$ | use for nonnested models
- Never compare R^2 / R^2 across **different dependent variables**
- High $R^2 \neq$ causal | low $R^2 \neq$ bad partial effects

6.5 Too many controls · prediction

- Don't control for mechanisms: beer consumption (beer tax → fatalities) | doctor visits (pesticides → health spending)
- Add regressors uncorrelated with x → $\sigma^2 \downarrow$ → more precise
- log(y) model: exp(predicted log y) underestimates → $\hat{y} = \hat{\alpha}_0 \cdot \exp(\ldots)$, $\hat{\alpha}_0 = n^{-1}\Sigma\exp(\hat{u}_i)$ [S]
- CI for mean $E(y|x)$ narrow | prediction interval for one new y wider ($+\hat{\sigma}^2$) [S]

Ch.R · R Code — Reading the Output

R.1 regression.R (Weidmann & Deming 2021)

- n = **85** groups | y = cryptography score | x: RMET (eye), skill, Ravens | all standardized → intercept ≈ 0
- `lm(perform ~ eye)` (outcome first) | `summary()` | `confint()` | `predict()`

Model	$\hat{\beta}$ (se)	p	R^2
~ eye	0.192 (0.108)	0.079	0.037
~ skill	0.214 (0.107)	0.049	0.046
~ ravens	0.217 (0.107)	0.046	0.047
~ skill + ravens	0.171 / 0.175 (0.110 each)	0.122 / 0.115	0.074

- All recomputed in R ✓
- Overall F(2, 82) = **3.3**, **p = 0.04184** → reject H_0: both = 0
- Each significant alone (p ≈ 0.05) but not when both included (corr 0.246, computed) → partial effects, larger se
- Skill **99%** CI [−0.069, 0.497] contains 0 while p = 0.049 → consistent (not significant at 1%)
- Simple regression: $F = t^2$ ($1.997^2 = 3.989$) ✓
- `predict(skill=0.5, ravens=1.5)` = **0.3475** | (1.5, 0.5) = **0.3441** ✓
- Trap: `eye_regr` overwritten by skill model → later CI-band plot wrong

R.2 Simulation (simple regression demonstration.R)

- True $\beta_0 = 3$, $\beta_1 = 0.2$ | n = 10 | `rnorm(n, 10, 2)` → 3rd arg = **SD**
- 10,000 samples: mean $\hat{\beta}_1$ = **0.2027 ≈ 0.2** → unbiasedness ✓
- Reject 1,908 / 10,000 (19.1%) at 5% → H_0 is false → this is **power** ✓
- Larger slope or n → power ↑ | true null → reject ≈ 5% [S]
- 20 CIs (slope 2): **19/20** contain 2 ≈ 95% coverage ✓

R.3 Lab 2 & Lab 4 essentials

- $\Sigma(x - \bar{x}) = 0$ always | `var()` = SS/(**n−1**) | `sd()` = √var
- competence: SS 124.07 → var **4.772**, sd **2.185** ✓
- IQR = Q3 − Q1: competence 6.5 − 4 = **2.5** | breaks 4.5 − 1 = **3.5** ✓
- Skewed data → IQR better than SD | boxplot outlier beyond 1.5×IQR (grade 36) ✓
- `to12("12")` → "12 AM" (noon bug) | `"13" - 12` → error (character) ✓

Exam-Eve Trap Checklist

C.1 Signs & units

- Residual = **actual − fitted** ($\hat{u} < 0$ → over-predicted)
- OVB sign = sign(β_2) × sign(corr(x_1, x_2))
- log–level: **×100** for % | log–log: β = elasticity | level–log: ÷100
- Check units (CEO salary in $1,000s)
- Quadratic $\beta_2 < 0$ → maximum at x*

C.2 Degrees of freedom

- Simple $\hat{\sigma}^2$: **n − 2** | multiple: **n − k − 1** | sample variance: **n − 1**
- F: numerator q | denominator n − k − 1

C.3 Commonly confused

- R_j^2 (x_j on other x) ≠ model R^2
- Two-sided p = 2 × one tail
- One-sided vs two-sided critical value (1.701 vs 2.06)
- Multicollinearity ≠ perfect collinearity
- Unbiased ≠ consistent | BLUE needs homoskedasticity
- R^2 form of F invalid if y differs

C.4 R pitfalls

- `set.seed()` first | reused object names overwrite
- `as.integer()` before arithmetic on strings
- `==` on doubles → `round()`

C.5 Written answers

- "Holding [other x] fixed, a one-unit increase in x is associated with a $\hat{\beta}$ [unit] change in y"
- Causal claim → cite MLR.4 / random assignment
- Test: H_0/H_1 → stat → df → c or p → decision → economic meaning
- Name the assumption (SLR/MLR number) behind each property

Chapter 2: Financial Statements, Taxes and Cash Flow

Multiple Choice

1. A firm has net sales of $2,400, cost of goods sold of $1,300, depreciation of $200 and interest paid of $150. The tax rate is 21%. What is net income?
a. $900.00
b. $711.00
c. $592.50
d. $750.00

2. A Korean firm has taxable income of ₩500 million. Using the national corporate tax schedule in the table, what are its average and marginal tax rates?

Taxable income	Rate
Up to ₩200 million	10%
₩200 million to ₩20 billion	20%
₩20 billion to ₩300 billion	22%
Over ₩300 billion	25%

a. Average 15%; marginal 20%
b. Average 16%; marginal 20%
c. Average 16%; marginal 10%
d. Average 20%; marginal 20%

3. A firm issues 2,000 shares of common stock with a par value of $0.50 at an issue price of $12 per share. How much goes to additional paid-in capital (paid-in surplus)?
a. $1,000
b. $24,000
c. $22,000
d. $23,000

4. Zephyr Logistics, 2025: EBIT $2,600, depreciation $1,000, taxes $420. Net fixed assets rose from $9,000 to $9,400. Current assets rose from $5,000 to $5,800 and current liabilities rose from $2,500 to $2,700. What is the cash flow from assets (CFFA)?
a. $1,180
b. $1,780
c. $2,380
d. $3,180

5. A firm paid interest of $100 and dividends of $885. Its long-term debt rose from $3,000 to $3,200 and its common stock and paid-in surplus rose from $2,500 to $2,600 because of a new share issue. What are its cash flow to creditors and cash flow to stockholders?

a. −$100 and $785

b. $300 and $985

c. $100 and $885

d. $785 and −$100

Chapter 2: Financial Statements, Taxes and Cash Flow — Answers

1. c
EBIT = 2,400 − 1,300 − 200 = 900; taxable income = 900 − 150 = 750; taxes = 21% × 750 = 157.50; net income = 750 − 157.50 = $592.50.

2. b
Tax = 10% × 200M + 20% × 300M = ₩80M, so the average rate = 80 / 500 = 16%; the marginal rate is the 20% bracket that applies to the next won of income.

3. d
Paid-in surplus = (issue price − par) × shares = (12 − 0.50) × 2,000 = $23,000; the $1,000 of par goes to the common stock line.

4. a
OCF = 2,600 + 1,000 − 420 = 3,180; NCS = 9,400 − 9,000 + 1,000 = 1,400; ΔNWC = (5,800 − 2,700) − (5,000 − 2,500) = 600; CFFA = OCF − NCS − ΔNWC = 3,180 − 1,400 − 600 = $1,180.

5. a
CF to creditors = interest paid − net new borrowing = 100 − 200 = −$100; CF to stockholders = dividends paid − net new equity raised = 885 − 100 = $785; their sum, $685, equals CFFA.

Chapter 3: Working with Financial Statements

Multiple Choice

1. Coastal Furniture has inventory of $84,600, equity of $351,000, total assets of $470,000 and sales of $612,300. What is the common-size percentage for inventory?
a. 74.7%
b. 24.1%
c. 18.0%
d. 13.8%

2. A firm has cash of $98, current assets of $708 (including inventory of $422) and current liabilities of $548. What is its quick ratio?
a. 0.18
b. 0.52
c. 1.29
d. 0.77

3. A firm has total assets of $3,630 and total equity of $2,625. What are its debt/equity ratio and its equity multiplier?
a. D/E = 1.38; EM = 0.38
b. D/E = 0.28; EM = 1.28
c. D/E = 0.28; EM = 0.38
d. D/E = 0.38; EM = 1.38

4. A firm has EBIT of $741, depreciation of $276 and interest of $141. What are its times interest earned (TIE) and cash coverage ratios?
a. TIE = 3.30; cash coverage = 5.26
b. TIE = 7.21; cash coverage = 5.26
c. TIE = 5.26; cash coverage = 3.30
d. TIE = 5.26; cash coverage = 7.21

5. A firm has sales of $2,361, cost of goods sold of $1,344, inventory of $422 and accounts receivable of $188. What is its inventory turnover?
a. 3.18
b. 0.31
c. 7.15
d. 5.59

6. A firm has net income of $474, sales of $2,361, total assets of $3,630 and total equity of $2,625. What are its profit margin, ROA and ROE?
a. 18.06%, 13.06%, 20.08%
b. 13.06%, 20.08%, 18.06%
c. 20.08%, 13.06%, 18.06%
d. 20.08%, 18.06%, 13.06%

7. A firm has net income of $474 million, total equity of $2,625 million and 33 million shares outstanding. The market price is $115 per share. What are its P/E ratio and market-to-book ratio?
a. P/E = 14.36; M/B = 79.55
b. P/E = 8.01; M/B = 1.45
c. P/E = 1.45; M/B = 8.01
d. P/E = 8.01; M/B = 0.69

8. A firm has 33 million shares at $115 each, total liabilities (book value) of $1,005 million, cash of $98 million, EBIT of $741 million and depreciation of $276 million. What is its enterprise value / EBITDA ratio?
a. 4.72
b. 4.62
c. 3.73
d. 6.35

9. Northwind: sales $5,000M, net income $1,050M, total assets $5,000M and total equity $2,500M. Using the DuPont identity, what is its ROE?
a. 21.0%
b. 10.5%
c. 31.5%
d. 42.0%

Chapter 3: Working with Financial Statements — Answers

1. c
Common-size balance sheet accounts are divided by total assets: 84,600 / 470,000 = 18.0%.

2. b
Quick ratio = (current assets − inventory) / current liabilities = (708 − 422) / 548 = 0.52.

3. d
Total debt = 3,630 − 2,625 = 1,005; D/E = 1,005 / 2,625 = 0.38; EM = TA / TE = 3,630 / 2,625 = 1.38 = 1 + D/E.

4. d
TIE = EBIT / interest = 741 / 141 = 5.26; cash coverage = (EBIT + depreciation) / interest = 1017 / 141 = 7.21.

5. a
Inventory turnover = COGS / inventory = 1,344 / 422 = 3.18 times; a higher turnover generally indicates better performance.

6. c
Profit margin = NI / sales = 474 / 2,361 = 20.08%; ROA = NI / TA = 474 / 3,630 = 13.06%; ROE = NI / TE = 474 / 2,625 = 18.06%.

7. b
EPS = 474 / 33 = 14.36, so P/E = 115 / 14.36 = 8.01; book value per share = 2,625 / 33 = 79.55, so M/B = 115 / 79.55 = 1.45.

8. b
EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 741 + 276 = 1,017; EV/EBITDA = 4,702 / 1,017 = 4.62.

9. d
PM = 1,050 / 5,000 = 21%; TAT = 5,000 / 5,000 = 1.00; EM = 5,000 / 2,500 = 2.00; ROE = PM × TAT × EM = 0.21 × 1.00 × 2.00 = 42.0%.

Chapter 4: Introduction to Valuation — Time Value of Money

Multiple Choice

1. You invest $1,000 for 3 years at 10% per year. How much more do you earn with compound interest than with simple interest?
a. $31
b. $331
c. $100
d. $0

2. You invest $2,000 today at 8% per year, compounded annually. How much will you have after 5 years?
a. $1,361.17
b. $2,800.00
c. $2,938.66
d. $2,720.98

3. You want $10,000 in 6 years and can earn 7% per year, compounded annually. How much must you invest today?
a. $7,042.25
b. $15,007.30
c. $6,663.42
d. $9,345.79

4. An investment of $5,000 grows to $10,000 in 9 years. What annual rate did it earn?
a. 7.18%
b. 8.01%
c. 9.05%
d. 11.11%

5. How long does it take for $1,000 to grow to $2,500 at 9% per year, compounded annually?
a. 10.18 years
b. 16.67 years
c. 11.63 years
d. 10.63 years

6. An investment of $18,000 today will be worth $31,000 in 7 years. Assuming P/Y = 1 and cleared TVM registers, which BA II Plus sequence returns the annual rate of return?

a. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT I/Y

b. 7 N → 31,000 +/− PV → 18,000 FV → 0 PMT → CPT I/Y

c. 7 N → 18,000 PV → 31,000 FV → 0 PMT → CPT I/Y

d. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT N

Chapter 4: Introduction to Valuation — Time Value of Money — Answers

1. a
Simple: 1,000 + 3 × 100 = 1,300; compound: 1,000 × $(1.10)^3$ = 1,331; the extra $31 is interest on interest.

2. c
FV = 2,000 × $(1.08)^5$ = $2,938.66. BA II Plus: 2nd [CLR TVM], 5 N, 8 I/Y, 0 PMT, 2000 +/− PV, CPT FV.

3. c
PV = 10,000 / $(1.07)^6$ = $6,663.42. BA II Plus: 6 N, 7 I/Y, 0 PMT, 10000 FV, CPT PV gives −6,663.42 (the cash you pay in).

4. b
r = $(10{,}000 / 5{,}000)^{1/9} - 1$ = 8.01%. BA II Plus: 9 N, −5000 PV, 0 PMT, 10000 FV, CPT I/Y.

5. d
2,500 = 1,000 × $(1.09)^t$, so t = ln(2.5) / ln(1.09) = 10.63 years. BA II Plus: 9 I/Y, −1000 PV, 0 PMT, 2500 FV, CPT N.

6. a
Cash paid out is negative and cash received is positive, so PV = −18,000 and FV = +31,000; with N = 7 and PMT = 0, CPT I/Y returns 8.08 (% per year).

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice

1. You deposit $4,000 today and another $7,000 at the end of year 1. The account earns 8% per year. How much is in the account at the end of year 3?
a. $13,203.65
b. $11,000.00
c. $13,856.83
d. $12,225.60

2. An investment pays $200, $400, $600 and $800 at the end of years 1 through 4. The required return is 12%. What is it worth today?
a. $1,604.88
b. $1,271.04
c. $1,432.93
d. $2,000.00

3. You can afford a car payment of $632 per month for 48 months, and the bank quotes 1% per month. How large a loan can you take?
a. $24,239.54
b. $23,999.54
c. $2,466.04
d. $30,336.00

4. You finance $15,000 of a car over 4 years at 9% APR compounded monthly. What is the monthly payment?
a. $1,371.92
b. $385.84
c. $312.50
d. $373.28

5. An investment costs $1,000 today and pays $200 at the end of each year for 8 years. What annual return does it offer?
a. 20.00%
b. 7.50%
c. 6.05%
d. 11.81%

6. You deposit $400 at the end of every month into an account earning 0.5% per month. How many years until the balance reaches $30,000?
a. 5.32 years
b. 63.85 years
c. 75.00 years
d. 6.25 years

7. You put $2,500 into an index fund at the end of every year for 25 years, earning 7% per year. What is the balance at the end?
a. $169,191.18
b. $62,500.00
c. $158,122.59
d. $13,568.58

8. A two-year apartment lease requires $1,800 at the beginning of every month. The relevant rate is 0.5% per month, compounded monthly. What is the lease worth today?
a. $39,016.22
b. $40,816.22
c. $43,200.00
d. $40,613.16

9. A company has just paid a dividend of $2.00 per share. Dividends are expected to grow 3% per year forever, and investors require 10% per year. What is the share worth?
a. $20.00
b. $29.43
c. $28.57
d. $68.67

10. Bank A quotes 15% APR compounded daily; Bank B quotes 15.5% APR compounded annually. What are their EARs, and where would you rather deposit?
a. A = 16.18%; B = 16.76%; Bank B
b. A = 15.56%; B = 15.50%; Bank A
c. A = 15.00%; B = 15.50%; Bank B
d. A = 16.18%; B = 15.50%; Bank A

11. You borrow $10,000 for 4 years at 8%, repaid in equal annual payments of $3,019.21. What is the loan balance after two years?
a. $5,384.05
b. $7,780.79
c. $5,561.58
d. $3,961.58

Chapter 5: Discounted Cash Flow Valuation — Answers

1. a
FV = 4,000 × $(1.08)^3$ + 7,000 × $(1.08)^2$ = 5,038.85 + 8,164.80 = $13,203.65; the $4,000 is invested 3 years and the $7,000 only 2.

2. c
PV = 200/1.12 + 400/$(1.12)^2$ + 600/$(1.12)^3$ + 800/$(1.12)^4$ = $1,432.93; each cash flow is discounted by its own t and then added.

3. b
PV = 632 × [1 − 1/$(1.01)^{48}$] / 0.01 = $23,999.54. BA II Plus: 48 N, 1 I/Y, −632 PMT, 0 FV, CPT PV; N counts payments, not years.

4. d
r = 9% / 12 = 0.75% per month and t = 48: 15,000 = C × [1 − 1/$(1.0075)^{48}$] / 0.0075, so C = $373.28. BA II Plus: 48 N, 0.75 I/Y, 15000 PV, 0 FV, CPT PMT.

5. d
1,000 = 200 × [1 − 1/$(1 + r)^8$] / r cannot be solved by algebra; the calculator iterates: 8 N, −1000 PV, 200 PMT, 0 FV, CPT I/Y = 11.81%.

6. a
30,000 = 400 × [$(1.005)^t$ − 1] / 0.005 gives t = 63.85 months (CPT N); 63.85 / 12 = 5.32 years.

7. c
FV = 2,500 × [$(1.07)^{25}$ − 1] / 0.07 = $158,122.59. BA II Plus: 25 N, 7 I/Y, −2500 PMT, 0 PV, CPT FV.

8. b
PV = ordinary annuity × (1 + r) = 1,800 × [1 − 1/$(1.005)^{24}$] / 0.005 × 1.005 = 40,613.16 × 1.005 = $40,816.22.

9. b
C_1 is the cash flow one period from today: D_1 = 2.00 × 1.03 = 2.06; PV = D_1 / (r − g) = 2.06 / 0.07 = $29.43.

10. d
EAR(A) = $(1 + 0.15/365)^{365}$ − 1 = 16.18%; EAR(B) = 15.50% because compounding is annual; Bank A pays more. BA II Plus ICONV: NOM = 15, C/Y = 365, CPT EFF.

11. a
Balance = PV of the remaining 2 payments = 3,019.21 × [1 − 1/$(1.08)^2$] / 0.08 = $5,384.05.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice

1. A bond has a $1,000 face value, an 8% annual coupon and 10 years to maturity. Comparable bonds yield 10%. What is its price?
a. $1,000.00
b. $385.54
c. $877.11
d. $1,122.89

2. A $1,000 bond pays a 9% coupon semiannually and matures in 7 years. Comparable bonds yield 8% (quoted annual rate). What is its price?
a. $1,528.16
b. $1,000.00
c. $1,052.82
d. $1,052.06

3. A 5-year zero coupon bond with a $1,000 face value is priced to yield 6% compounded semiannually. What is its price?
a. $558.39
b. $744.09
c. $862.61
d. $747.26

4. A $1,000 bond with an 8% annual coupon and 8 years to maturity trades at $918.50. What is its yield to maturity?
a. 8.7%
b. 10.0%
c. 8.0%
d. 9.5%

5. A $1,000 bond with an 8% annual coupon and 8 years to maturity is priced at $918.50. What is its current yield?
a. 8.71%
b. 9.50%
c. 8.87%
d. 8.00%

6. A $1,000 bond pays an 8% coupon semiannually ($40 per coupon). Two months have passed since the last coupon, and the bond is quoted at $985. What is the invoice (dirty) price?

a. $998.33

b. $1,025.00

c. $985.00

d. $991.67

7. You invest $1,000 for one year at a nominal rate of 15.5%. Inflation is 5%. What is the real rate?

a. 16.28%

b. 10.50%

c. 10.00%

d. 14.76%

8. An investor in the 35% tax bracket compares a 6% municipal bond with a taxable bond. What is the equivalent taxable yield of the muni?

a. 7.50%

b. 9.23%

c. 6.00%

d. 3.90%

Chapter 6: Interest Rates and Bond Valuation — Answers

1. c
Price = $80 \times [1 - 1/(1.10)^{10}] / 0.10 + 1{,}000 / (1.10)^{10} = \$877.11$. BA II Plus: 10 N, 10 I/Y, 80 PMT, 1000 FV, CPT PV = −877.11.

2. c
Work in periods: N = 14, I/Y = 8% / 2 = 4%, PMT = 90 / 2 = 45, FV = 1,000; price = $1,052.82.

3. b
A zero pays only at maturity: price = $1{,}000 / (1.03)^{10} = \$744.09$ (N = 10, I/Y = 3).

4. d
$918.50 = 80 \times [1 - 1/(1 + r)^{8}] / r + 1{,}000 / (1 + r)^{8}$ gives r = 9.5% (9% prices at $944.65 and 10% at $893.30). BA II Plus: 8 N, −918.50 PV, 80 PMT, 1000 FV, CPT I/Y.

5. a
Current yield = annual coupon / market price = 80 / 918.50 = 8.71%; it counts the coupon but ignores the capital gain at maturity.

6. a
Accrued interest = 40 × (2 months / 6 months) = 13.33; dirty price = clean price + accrued interest = 985 + 13.33 = $998.33.

7. c
Fisher: $1 + R = (1 + r)(1 + h)$, so $r = 1.155 / 1.05 - 1 = 10.00\%$.

8. b
Equivalent taxable yield = muni yield / (1 − t) = 6% / 0.65 = 9.23%.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice

1. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the stock worth?
a. $17.50
b. $28.57
c. $42.00
d. $30.00

2. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the expected price five years from today, just after the fifth dividend has been paid?
a. $36.47
b. $30.00
c. $40.20
d. $38.29

3. A firm just paid a $1.00 dividend. Dividends will grow at 20% for the next three years and at 5% forever after that. The required return is 15%. What is the stock worth today?
a. $15.20
b. $14.63
c. $13.64
d. $3.27

4. A firm will pay no dividend for the next four years. It will pay $0.50 at the end of year 5, and that dividend will then grow at 4% forever. The required return is 20%. What is the stock worth today?
a. $3.12
b. $1.21
c. $1.51
d. $1.26

5. A stock sells for $20.00. Its next dividend will be $1.00, and dividends are expected to grow at 10% a year. What are its dividend yield, capital gains yield and required return?
a. 5.0%; 10.0%; 5.0%
b. 5.0%; 10.0%; 15.0%
c. 5.0%; 5.0%; 10.0%
d. 10.0%; 5.0%; 15.0%

6. A firm is expected to earn $3.00 per share next year. Comparable firms trade at 18 times earnings. What is the estimated share price?
a. $18.00
b. $54.00
c. $0.17
d. $21.00

7. A loss-making firm has sales of $10.00 per share. Comparable firms trade at 2.5 times sales. What is the estimated share price?
a. $4.00
b. $12.50
c. $0.00
d. $25.00

Chapter 7: Equity Markets and Stock Valuation — Answers

1. d
$D_1 = 2.00 \times 1.05 = 2.10$; $P_0 = D_1 / (R - g) = 2.10 / (0.12 - 0.05) = \$30.00$.

2. d
The model works from any date if you use the following period's dividend: $P_5 = D_6 / (R - g) = 2.00 \times (1.05)^6 / 0.07 = \$38.29$.

3. a
$D_1 = 1.20$, $D_2 = 1.44$, $D_3 = 1.728$, $D_4 = 1.728 \times 1.05 = 1.8144$; $P_3 = D_4 / (0.15 - 0.05) = 18.144$; PV $= 1.20/1.15 + 1.44/(1.15)^2 + 1.728/(1.15)^3 + 18.144/(1.15)^3 = \$15.20$.

4. c
Apply the model one period before the first dividend: $P_4 = 0.50 / (0.20 - 0.04) = 3.125$; $P_0 = 3.125 / (1.20)^4 = \$1.51$.

5. b
Dividend yield $= D_1 / P_0 = 1.00 / 20.00 = 5.0\%$; under constant growth the price grows at g, so the capital gains yield $= g = 10.0\%$; $R = 5.0\% + 10.0\% = 15.0\%$.

6. b
Price = EPS × benchmark P/E $= 3.00 \times 18 = \$54.00$; the answer is only as good as the word “comparable”.

7. d
Price = sales per share × benchmark price-sales ratio $= 10.00 \times 2.5 = \$25.00$; sales are always positive, so P/S still works when earnings are negative and a P/E ratio is meaningless.

Business Finance Midterm Short-Answer Quiz: Concepts and Keywords

1. **[Ch.2]** What does a balance sheet report, how are its two sides ordered, and how is net working capital defined?

2. **[Ch.2]** Why are the book value and the market value of a firm's assets often very different, and which one matters more for decisions?

3. **[Ch.2]** Compare US GAAP and IFRS in terms of who sets them, their style, and how the order of the balance sheet differs.

4. **[Ch.2]** Define the average tax rate and the marginal tax rate. Which one is used for financial and investment decisions, and how do the two rates compare under a flat tax and under a progressive tax?

5. **[Ch.2]** Define cash flow from assets (CFFA). What are its three components, and what other two cash flows must it equal in total?

6. **[Ch.3]** What are common-size balance sheets and income statements, and why are they useful?

7. **[Ch.3]** List the five categories of financial ratios and give the question each category answers.

8. **[Ch.3]** State the DuPont identity and explain what each of its three components measures.

9. **[Ch.3]** Coca-Cola and PepsiCo both earn an ROE of about 40%. Using the DuPont figures in the table, explain how each firm reaches that ROE.

Item	Coca-Cola (KO)	PepsiCo (PEP)
Profit margin	27.34%	8.77%
Total asset turnover	0.46	0.87
Equity multiplier	3.26	5.26
ROE	40.7%	40.4%

10. **[Ch.3]** How can a ratio be benchmarked, and what problems can distort a comparison of ratios across firms?

11. **[Ch.4]** Why is a dollar today worth more than a dollar tomorrow? Distinguish simple from compound interest, and compounding from discounting.

12. **[Ch.4]** State the sign convention and the clearing rule for the TI BA II Plus time-value keys, and what happens if the signs are wrong.

13. **[Ch.5]** Define an annuity, an annuity due, a perpetuity and a growing perpetuity. What condition does the growing perpetuity need, and which cash flow goes in the numerator?

14. **[Ch.5]** Distinguish the APR from the EAR. Why must quoted rates be converted before comparing them, and what happens to the EAR as compounding becomes more frequent?

15. **[Ch.5]** In a loan repaid with equal payments, how is each payment divided between interest and principal over time, and how do you find the loan balance at any date?

16. **[Ch.5]** List common mistakes when solving annuity problems on the calculator or with the formula.

17. **[Ch.6]** Name the defining terms of a bond and say which are fixed at issue and which changes. How is a bond valued?

18. **[Ch.6]** How does the relation between coupon rate and yield to maturity determine whether a bond sells at par, a premium or a discount? How do maturity and coupon size affect interest rate risk?

19. **[Ch.6]** Define yield to maturity. What three assumptions lie behind it, and how do the coupon rate, current yield and YTM rank for a discount bond?

20. **[Ch.6]** What is a bond indenture, who enforces it, and what are the basic terms it contains? Why does a call provision lower a bond's value, and why do firms accept protective covenants?

21. **[Ch.6]** What do bond ratings measure, which is the lowest investment-grade rating, and what are lower-rated bonds called?

22. **[Ch.6]** State the Fisher effect and list the components that make up a corporate bond's nominal yield.

23. **[Ch.7]** Why is a stock harder to value than a bond? What is the value of a share, and how can a firm that pays no dividend still have value?

24. **[Ch.7]** Name the four dividend growth cases and state what to watch for in each.

25. **[Ch.7]** Split a stock's required return into its two components and describe each. Under constant growth, what is the capital gains yield?

26. **[Ch.7]** When are the P/E and price-sales ratios used to value a share, and why does the word "comparable" matter?

27. **[Ch.7]** Is preferred stock debt or equity? Describe its main features.

28. **[Ch.7]** Distinguish primary from secondary markets and a dealer from a broker. What is the bid-ask spread, and how does NASDAQ differ from the NYSE?

Answers

1. [Ch.2] The balance sheet is a **snapshot** of the firm's assets and liabilities at a given **point in time**, and it satisfies assets = liabilities + stockholders' **equity**. Assets are listed in order of **decreasing liquidity**; liabilities are listed in ascending order of **when due** to be paid. **Net working capital** = current assets − current liabilities, which is usually positive for a healthy firm.
Keywords: snapshot · decreasing liquidity · when due · net working capital · equity

2. [Ch.2] Book value is the accounting value shown on the balance sheet, based on **historical cost**. **Market value** is the price at which assets, liabilities or equity can actually be bought or sold, so it reflects the **future** cash-generating ability of the assets. The two differ because historical cost does not track current values or growth prospects, and the **market value** is what matters for financial **decision** making.
Keywords: historical cost · market value · future · decision

3. [Ch.2] US GAAP is set by the **FASB** (under SEC oversight) and is more **rules-based**. IFRS (including K-IFRS in Korea) is set by the **IASB** and is more **principles-based**. Both report the same core statements; the labels and layout differ, e.g. GAAP lists current (liquid) items first while IFRS lists **non-current** items first, so the order flips.
Keywords: FASB · IASB · rules-based · principles-based · non-current

4. [Ch.2] The average tax rate = **total taxes** ÷ taxable income, the overall tax bite. The marginal tax rate is the rate paid on the **next dollar** of income, and it is the one used for **decision** making. Under a **flat** tax the marginal rate equals the average rate (U.S. 21%); under a **progressive** schedule (e.g. Korea) the marginal rate is higher than the average rate.
Keywords: total taxes · next dollar · decision · flat · progressive

5. [Ch.2] CFFA = **operating cash flow** (OCF) − **net capital spending** (NCS) − the change in **net working capital**. It shows how cash is generated by using assets and how it is paid to those who finance them, so it equals cash flow to **creditors** plus cash flow to **stockholders**. This differs from the accounting statement of cash flows.
Keywords: operating cash flow · net capital spending · net working capital · creditors · stockholders

6. [Ch.3] On a common-size balance sheet every account is a **percent of total assets**; on a common-size income statement every line item is a **percent of sales**. Standardized statements make it easy to compare **year-to-year** results and to compare companies of **different sizes**, particularly within the same industry.
Keywords: percent of total assets · percent of sales · year-to-year · different sizes

7. [Ch.3] Liquidity (short-term solvency) ratios ask whether the firm can pay its short-term bills; financial **leverage** (long-term solvency) ratios ask how much debt is used; asset management or **turnover** ratios ask how efficiently assets are used; **profitability** ratios ask how much profit is earned; and **market value** ratios (P/E, M/B, EBITDA ratio) compare the stock price with accounting numbers.
Keywords: liquidity · leverage · turnover · profitability · market value

8. [Ch.3] ROE = **profit margin** × total **asset turnover** × **equity multiplier**. Profit margin (NI/Sales) measures profitability and **cost control**; total asset turnover (Sales/TA) measures asset-use **efficiency**; the equity multiplier (TA/TE = 1 + D/E) measures financial **leverage**.
Keywords: profit margin · asset turnover · equity multiplier · leverage · efficiency · cost control

9. [Ch.3] Coca-Cola gets there through a very high **profit margin** (27.3%, an asset-light concentrate model) but low asset turnover (0.46). **PepsiCo** has a thin margin (8.8%) and relies on faster **asset turnover** (0.87) and greater financial **leverage** (equity multiplier 5.26 vs 3.26). Same destination, different routes.
Keywords: Coca-Cola · PepsiCo · profit margin · asset turnover · leverage

10. [Ch.3] A ratio means little alone. Benchmark it with **time-trend** analysis (the firm's own history) or **peer group** analysis (similar firms or industry averages). Comparisons can mislead because of different **accounting methods** (FIFO/LIFO, depreciation), different **fiscal year** ends or seasonality, **conglomerates** with no clean peer group, differences in capital structure, and **window dressing** around reporting dates; there is no underlying theory giving a single correct value.
Keywords: time-trend · peer group · accounting methods · fiscal year · conglomerate · window dressing

11. [Ch.4] A dollar today can be **invested** to **earn a return**. **Simple interest** is paid only on the original principal, while compound interest pays **interest on interest**, so value grows faster and faster. Compounding multiplies by $(1 + r)^t$ to get a **future value**; discounting divides by $(1 + r)^t$ to get a **present value**.
Keywords: earn a return · simple interest · interest on interest · present value · future value

12. [Ch.4] Clear the TVM registers before every problem (2nd CLR TVM), because old values stay in memory. Cash paid out is **negative** and cash received is positive, so PV and FV need **opposite** signs; otherwise **Error 5** appears when solving for I/Y or N. Enter four of N, I/Y, PV, **PMT**, FV and press **CPT** on the fifth; PMT = 0 for a single lump sum.
Keywords: clear · opposite · CPT · PMT · Error 5

13. [Ch.5] An annuity is a **level** cash flow for a fixed number of periods; an annuity due pays at the **beginning** of each period, so its value = ordinary annuity × (1 + r). A perpetuity is a cash flow that **never ends** (PV = C / r). A growing perpetuity (PV = C_1 / (r − g)) requires r **greater than g**, and C_1 is the cash flow **one period from today**, not the one just paid.
Keywords: level · beginning · never ends · greater than g · one period from today

14. [Ch.5] The APR is a **quoted** rate equal to the periodic rate × m; the **effective annual rate** (EAR = $(1 + APR/m)^m$ − 1) is the rate actually earned or paid over a year. Two loans with the same APR can cost different amounts, so convert both to EARs before you **compare** them. With **more frequent** compounding the EAR is **higher** for the same APR (15% APR: 15.00% annual, 16.08% monthly, 16.18% daily).
Keywords: quoted · effective annual rate · compare · more frequent · higher

15. [Ch.5] Each payment covers **interest and principal**. The payment is **constant**, but **interest falls** each period because the balance falls, so the **principal rises**. The loan balance at any date equals the **present value of the remaining payments**, discounted at the loan rate.
Keywords: interest and principal · interest falls · principal rises · constant · present value of the remaining payments

16. [Ch.5] Leaving the calculator in **BGN** mode corrupts later answers by a factor of (1 + r). The rate and N must be per payment period: use the **rate per period** (not the APR) and the **number of payments**, not the number of years. Always **clear** the TVM registers, and give PV and PMT the correct **sign**.
Keywords: BGN · rate per period · number of payments · clear · sign

17. [Ch.6] A bond's **face value** (repaid at **maturity**) and its **coupon** (coupon rate × face value) are fixed at issue. The **yield to maturity**, the market's required return, **changes** every day with the **market**. Bond value = the **present value** of the coupon annuity plus the present value of the face value, so everything that happens after issue happens to the price.
Keywords: face value · coupon · maturity · yield to maturity · present value · changes

18. [Ch.6] If the coupon rate equals the yield the bond sells at **par**; if the coupon rate exceeds the yield it sells at a **premium**; if it is below the yield it sells at a **discount**. Prices and yields move in the **opposite** direction. Interest rate risk is greater for a **longer maturity** and for a **lower coupon** (a zero coupon bond is the extreme).
Keywords: premium · discount · par · inverse · longer maturity · lower coupon

19. [Ch.6] YTM is the single discount rate that sets the present value of the promised cash flows equal to the **market price**: the bond's **internal rate of return**. It assumes the bond is **held to maturity**, there is **no default**, and coupons are **reinvested** at the YTM. For a discount bond, coupon rate < **current yield** < YTM.
Keywords: internal rate of return · market price · held to maturity · no default · reinvested · coupon rate < current yield

20. [Ch.6] The **indenture** is the written agreement between the issuer and its creditors (face value, coupon, security, seniority, repayment, call provisions, covenants); a **trustee**, usually a bank, enforces it. A **sinking fund** retires the issue gradually. A **call** provision gives the issuer the option to buy the bonds back, usually when rates have fallen, so investors demand a **higher yield** (a lower value). Protective **covenants** limit the borrower (negative and positive) and firms accept them because the reduction in yield is worth more than the lost flexibility.
Keywords: indenture · trustee · sinking fund · call · higher yield · covenant

21. [Ch.6] Ratings assess the likelihood that the promised payments are made (**default** risk) and the protection creditors have if they are not. **Investment grade** runs from AAA/Aaa down to **BBB** (Baa3 / BBB−, the lowest investment-grade rating); below that the bonds are **speculative** grade, or junk.
Keywords: default · investment grade · speculative · BBB

22. [Ch.6] The **Fisher** effect is 1 + R = (1 + r)(1 + h): nominal yields rise with expected inflation. The nominal yield = the **real rate** + **inflation premium** + **interest rate risk premium** (the term structure) plus bond-specific premiums: **default risk premium**, taxability premium and **liquidity premium**.
Keywords: Fisher · real rate · inflation premium · interest rate risk premium · default risk premium · liquidity premium

23. [Ch.7] A dividend is paid at the board's **discretion** (a bond coupon is contractual), a stock has **no maturity** date, and both the required return and the growth rate must be **estimated**. The price of a share is the **present value of all future dividends**; "I buy it to sell higher" is contained in the model. A firm that pays nothing now must **eventually** pay, or its shares are worthless.
Keywords: discretion · no maturity · estimate · present value of all future dividends · eventually

24. [Ch.7] **Zero growth** is a perpetuity (preferred stock; $D_1 = D_0$). **Constant growth** is the dividend growth model $P_0 = D_1 / (R - g)$: use the **next dividend** D_1 and require **R > g**. **Nonconstant** growth is irregular and then constant: find $P_t = D_{t+1} / (R - g)$ and discount it by t. **Two-stage** growth is g_1 for t years and then g_2.
Keywords: zero growth · constant growth · nonconstant · two-stage · next dividend · R > g

25. [Ch.7] R = **dividend yield** + **capital gains yield**. The dividend yield is **next year's dividend** divided by today's price (D_1 / P_0), the analogue of a bond's current yield. Under constant growth the price grows at exactly **g**, so the capital gains yield equals the **growth rate** g.
Keywords: dividend yield · capital gains yield · growth rate · next year's dividend

26. [Ch.7] When there is no dividend to discount, price = EPS × a **benchmark price-earnings** ratio, or sales per share × a **price-sales** ratio, taken from **comparable** firms (same industry, size and prospects) or the firm's own history. With **negative earnings** a P/E is meaningless but sales are always positive. The answer is only as good as the word comparable.
Keywords: price-earnings · price-sales · negative earnings · comparable · benchmark

27. [Ch.7] Preferred stock looks like debt (a **fixed dividend**, no voting rights, ranked **ahead of common** stock and often callable), but it is equity: skipping a dividend is **not default**, dividends are **not tax-deductible**, and holders rank **behind creditors**. Most issues are **cumulative**, so arrears must be paid before common dividends resume.
Keywords: fixed dividend · not default · cumulative · ahead of common · behind creditors · not tax-deductible

28. [Ch.7] In the **primary** market the issuer sells securities for the first time and raises money; in the **secondary** market investors trade existing shares and the firm receives nothing. A **dealer** trades for its own account and bears inventory risk; a **broker** only matches trades for a commission. The dealer's **bid** is the price it pays, the ask is the price it charges, and the **spread** is its compensation. NASDAQ is a dealer market (a computer network of competing market makers); the NYSE is a hybrid auction market with DMMs, now largely electronic.
Keywords: primary · secondary · dealer · broker · bid · spread

Chapter 2: Asset Classes and Financial Instruments

Multiple Choice

1. A Treasury bill with a face value of $10,000 and 120 days to maturity is selling for $9,780. What are its bank discount yield (BDY) and bond equivalent yield (BEY)?
a. BDY 6.60%; BEY 6.84%
b. BDY 6.84%; BEY 6.60%
c. BDY 6.69%; BEY 6.84%
d. BDY 6.60%; BEY 6.69%

2. For the same Treasury bill, which ordering of the three yield measures is correct?
a. Effective annual yield < bond equivalent yield < bank discount yield
b. Bond equivalent yield < bank discount yield < effective annual yield
c. Bank discount yield < bond equivalent yield < effective annual yield
d. All three are equal because they describe the same bill

3. Which of the following money market instruments is quoted on an interest-bearing (add-on) basis rather than on a discount basis?
a. Negotiable certificate of deposit
b. Treasury bill
c. Commercial paper
d. Banker's acceptance

4. In a repurchase agreement (repo), the dealer:
a. Buys securities from an investor with an agreement to sell them back later at a lower price
b. Sells securities to an investor with an agreement to buy them back the next day at a higher price
c. Lends securities to an investor in exchange for an unsecured promissory note
d. Exchanges dollar deposits for foreign-currency deposits at an agreed forward rate

5. A municipal bond yields 3.2% and a taxable corporate bond of comparable risk yields 4.5%. For an investor in the 32% marginal tax bracket, what is the equivalent taxable yield of the municipal bond, and which bond is better?
a. Equivalent taxable yield 4.71%; the corporate bond
b. Equivalent taxable yield 2.18%; the corporate bond
c. Equivalent taxable yield 4.71%; the municipal bond
d. Equivalent taxable yield 4.22%; the municipal bond

6. A taxable corporate bond yields 7.0% and a comparable-risk municipal bond yields 4.2%. At what marginal tax rate would an investor be indifferent between the two?

a. 28%

b. 35%

c. 40%

d. 60%

7. A Treasury bond with a par value of $1,000 is quoted at an asked price of 103-12. What is the dollar price of the bond?

a. $1,031.20

b. $1,031.88

c. $1,033.75

d. $1,037.50

8. A TIPS bond has a par value of $1,000 and a coupon rate of 2.0%. Inflation over the year is 3.5%, and the principal is adjusted accordingly. What coupon payment does the holder receive for the year?

a. $20.00

b. $20.70

c. $35.00

d. $55.70

9. Which pair of characteristics best describes common stock?

a. Fixed claim and unlimited liability

b. Residual claim and unlimited liability

c. Residual claim and limited liability

d. Fixed claim and limited liability

10. Preferred stock is often called a hybrid of debt and equity because:

a. It can be converted into the firm’s bonds at the holder’s discretion at any time and pays interest instead of dividends

b. It carries full voting rights like common stock and a guaranteed maturity date like a bond

c. It pays a fixed dividend like a perpetual bond with no maturity, yet it is equity that ranks below bondholders and above common stockholders

d. Its dividend is tax-deductible for the issuer, like interest, and rises automatically with the firm’s earnings

11. A foreign company's common share trades at KRW 540,000 in Seoul. One ADR represents 1/10 of a common share, and the exchange rate is KRW 1,350 per U.S. dollar. Ignoring costs, what is the ADR's approximate dollar price?

a. $4.00

b. $40.00

c. $400.00

d. $54,000

12. Consider the three stocks in the table, where Pt is the price and Qt the shares outstanding at time t. Stock C splits two-for-one in the last period (its price falls from 132 to 70 after the split). What is the rate of return on a price-weighted index for the first period (t = 0 to t = 1), and what must the divisor become for the second period so that the split does not change the index level?

Stock	P0	Q0	P1	Q1	P2	Q2
A	80	150	86	150	89	150
B	45	300	42	300	40	300
C	120	100	132	100	70	200

a. Return 3.20%; new divisor 2.24

b. Return 6.12%; new divisor 3.00

c. Return 6.12%; new divisor 2.24

d. Return 6.12%; new divisor 2.38

13. Using the three stocks in the table (stock C splits two-for-one in the last period), what are the first-period returns (t = 0 to t = 1) on an equally weighted index and on a value-weighted index?

Stock	P0	Q0	P1	Q1	P2	Q2
A	80	150	86	150	89	150
B	45	300	42	300	40	300
C	120	100	132	100	70	200

a. Equally weighted 3.61%; value-weighted 3.20%

b. Equally weighted 3.20%; value-weighted 3.61%

c. Equally weighted 3.61%; value-weighted 6.12%

d. Equally weighted 6.12%; value-weighted 3.20%

14. Which statement about an equally weighted index is correct?
a. It corresponds to a buy-and-hold strategy, so no trading is needed as prices change
b. It gives the greatest weight to the highest-priced stocks, as the Dow Jones Industrial Average does
c. It gives the greatest weight to the largest-capitalization firms, as the S&P 500 does
d. Maintaining it requires periodic rebalancing — selling stocks that have risen and buying stocks that have fallen

15. Which statement about a call option is correct?
a. The holder is obligated to buy the asset at expiration regardless of the asset's price
b. The holder has the right but not the obligation to buy the asset at the exercise price, and loses at most the premium paid
c. The holder's loss is unlimited, because the asset price can rise without bound and the holder must deliver at the strike
d. A call is exercised only when the market price falls below the strike price at or before expiration

16. You buy a call option with a strike price of $50 for a premium of $3 per share, and the stock trades at $57 at expiration. You also buy a put option with a strike price of $60 for a premium of $4. What is your profit per share on the call, and at what stock price does the put break even?
a. Call profit $7; put break-even $56
b. Call profit $4; put break-even $64
c. Call profit $10; put break-even $60
d. Call profit $4; put break-even $56

17. Which statement correctly distinguishes a long futures position from a long call position?
a. The futures buyer has the right but not the obligation to take delivery, while the call buyer is obligated to buy
b. Both positions require an up-front premium payment of the same size
c. The futures buyer's maximum loss is limited to the initial margin posted
d. The futures buyer is obligated to settle at the futures price, while the call buyer will exercise only if it is profitable

Chapter 2: Asset Classes and Financial Instruments — Answers

1. a
BDY = [(F − P) ÷ F] × (360 ÷ n) = (220 ÷ 10,000) × (360 ÷ 120) = 6.60%; BEY = [(F − P) ÷ P] × (365 ÷ n) = (220 ÷ 9,780) × (365 ÷ 120) = 6.84%.

2. c
BDY divides by face value over 360 days, BEY divides by the lower purchase price over 365 days, and EAY also compounds, so accuracy and size rise BDY → BEY → EAY.

3. a
T-bills, commercial paper and bankers' acceptances are sold below face value (discount basis), whereas CDs, repos, federal funds and Eurodollar deposits pay principal plus interest at maturity (add-on).

4. b
A repo is a short-term, often overnight, collateralized loan to the dealer; the higher repurchase price is the interest, and in a reverse repo the investor lends cash and holds the collateral.

5. c
Equivalent taxable yield = 3.2% ÷ (1 − 0.32) = 4.71%, which exceeds the corporate bond's 4.5%, so the municipal bond leaves more after tax (3.2% versus 4.5% × 0.68 = 3.06%).

6. c
Set after-tax yields equal: 7.0% × (1 − t) = 4.2%, so t = 1 − 4.2 ÷ 7.0 = 40%; investors above this bracket prefer the muni, those below prefer the taxable bond.

7. c
The digits after the dash are 32nds of a point: 103 + 12/32 = 103.375% of par, i.e. $1,033.75 per $1,000.

8. b
TIPS pay the coupon on the inflation-adjusted principal: $1,000 × 1.035 = $1,035, and $1,035 × 2.0% = $20.70.

9. c
Shareholders are last in line, receiving only what remains after creditors and preferred holders are paid, and they can lose no more than the amount invested because personal assets are shielded.

10. c
Preferred pays a fixed, often cumulative dividend and has priority over common but ranks below debt; unlike bond interest, its dividends are not tax-deductible for the issuer.

11. b
ADR price = share price × ratio ÷ exchange rate = 540,000 × 1/10 ÷ 1,350 = $40.00.

12. c
Price-weighted return = (86 + 42 + 132) ÷ (80 + 45 + 120) − 1 = 260 ÷ 245 − 1 = 6.12%; just before the split the index is (86 + 42 + 132) ÷ 3 = 86.67, and after it the sum is 86 + 42 + 66 = 194, so the divisor must be d = 194 ÷ 86.67 = 2.24.

13. a
Equally weighted = average of the three returns = (7.50% − 6.67% + 10.00%) ÷ 3 = 3.61%; value-weighted = total market value growth = (86×150 + 42×300 + 132×100) ÷ (80×150 + 45×300 + 120×100) − 1 = 38,700 ÷ 37,500 − 1 = 3.20%.

14. d
An equally weighted index averages stock returns, i.e. an equal dollar amount in each stock; prices drift the weights apart, so the investor must keep trading back to equal dollars.

15. b
The buyer pays a premium for a right: the loss is capped at the premium and the gain is open-ended, and the call is exercised only when the market price is above the strike.

16. d
Call profit = max(57 − 50, 0) − 3 = $4; a put pays K − S_T, so the premium is recovered when S_T = K − premium = 60 − 4 = $56.

17. d
An option is a right, bought for a premium, with a capped loss; a futures contract is an obligation with no premium (margin only), marked to market daily, so a move against you must still be settled.

Chapter 3: Securities Markets

Multiple Choice

1. Which of the following transactions takes place in the primary market?
a. An investor buys 100 shares of Samsung Electronics from another investor on the exchange
b. A mutual fund sells its holdings of Toyota stock to a pension fund
c. A firm sells newly issued shares to the public in an initial public offering
d. A dealer buys shares into inventory from a customer

2. In a firm commitment underwriting, the investment bank:
a. Acts only as an agent and returns any unsold shares to the issuer
b. Guarantees the issuing firm a minimum share price for one year after the offering
c. Buys the entire issue from the firm and bears the placement risk of reselling it to the public
d. Lends the issuing firm the offering proceeds, to be repaid out of future earnings

3. A firm sells 200,000 shares in an initial public offering at an offer price of $25. The underwriter’s explicit fees are $150,000, and the share price closes at $28 on the first day of trading. What is the best estimate of the total cost of the equity issue to the firm?
a. $150,000
b. $600,000
c. $750,000
d. $4,850,000

4. The sell side of a limit-order book for a stock shows the orders in the table. You submit a market order to buy 700 shares. What total amount will you pay, ignoring commissions?

Ask price	Shares offered
$31.40	300
$31.45	500
$31.55	600

a. $21,980
b. $22,000
c. $22,015
d. $22,085

5. Which statement correctly contrasts a market order with a limit order?
a. A market order guarantees a price but not execution; a limit order guarantees execution but not price
b. A market order executes immediately at the best available price; a limit order executes only at the specified price or better, and may never execute
c. Both orders execute immediately at the quoted price, but a limit order pays a lower commission
d. A limit order can only be used to buy, while a market order can be used to buy or sell and is the only order allowed during trading hours

6. You own a stock at $120 and place a stop-loss order at $108. Overnight, bad news makes the stock open the next morning at $102. What happens?
a. The shares are sold at exactly $108, because a stop-loss order guarantees the stop price
b. The order is never triggered, because the price skipped over $108 without trading at it
c. The stop is triggered and a market order sells the shares near the $102 opening price
d. The order becomes a limit order at $108, which cannot be filled at the lower opening price

7. Which statement correctly contrasts the NYSE and Nasdaq?
a. The NYSE is an auction, order-driven market with a central limit-order book, while Nasdaq is a dealer, quote-driven market with competing market makers
b. The NYSE is a dealer market with competing market makers, while Nasdaq is an auction market with a central limit-order book
c. Both are pure dealer markets with no limit-order book
d. Nasdaq has a physical trading floor, while the NYSE has no human oversight at all

8. You have $15,000 of cash and the initial margin requirement is 40%. What is the maximum value of stock you can buy, and how much do you borrow? (Ignore commissions.)
a. Buy $37,500; borrow $15,000
b. Buy $37,500; borrow $22,500
c. Buy $30,000; borrow $15,000
d. Buy $25,000; borrow $10,000

9. An investor buys 800 shares at $25 per share using the maximum allowable initial margin of 50%, so the margin loan is $10,000. The maintenance margin is 35%. At what share price will the investor receive a margin call?
a. $8.75
b. $12.50
c. $16.25
d. $19.23

10. An investor buys 300 shares at $60 per share, borrowing 40% of the purchase price at an interest rate of 7% per year. One year later the shares are sold at $66. The stock pays no dividends and commissions are ignored. What is the rate of return on the investor’s equity?

a. 9.67%

b. 10.00%

c. 12.00%

d. 16.67%

11. Which statement about buying stock on margin is correct?

a. Margin reduces the volatility of the investor’s return because the loan is a fixed obligation

b. Margin magnifies both gains and losses on the investor’s equity, and interest on the loan accrues whatever the stock does

c. The investor’s maximum loss on a margined position equals the loss on an unmargined position

d. The initial margin requirement is set by each individual broker and cannot be changed by regulators

12. Which statement about the cash flows and obligations in a short sale is correct?

a. The proceeds are held in the account as collateral, additional margin must be posted, and the short seller must reimburse any dividends the stock pays

b. The short seller receives the sale proceeds in cash and may withdraw them immediately to invest elsewhere, with no further obligation

c. The short seller is entitled to receive any dividends paid on the borrowed shares, because the seller is the owner of record

d. The position requires no margin because the shares were sold rather than bought, so no capital is at risk

13. An investor sells short 200 shares at $40 per share and deposits the required initial margin of 50% in cash. The maintenance margin is 35%. At what share price will the investor receive a margin call?

a. $39.00

b. $44.44

c. $54.00

d. $60.00

14. An investor sells short 500 shares at $60 per share and deposits an initial margin of 50% of the short value. One year later the investor covers at $54, after the stock paid a dividend of $1.50 per share. Treating the initial margin as the amount invested, what is the rate of return on the short position?

a. 7.50%

b. 15.00%

c. 20.00%

d. 25.00%

15. Which statement about U.S. securities regulation is correct?

a. The Securities Exchange Act of 1934 tightened governance after the 2002 scandals, and the CFTC regulates futures

b. The Sarbanes–Oxley Act of 2002 tightened auditing and governance standards after accounting scandals, and the CFTC regulates futures

c. The Sarbanes–Oxley Act of 2002 tightened governance, and FINRA is the government agency that regulates futures

d. The Securities Act of 1933 tightened governance after the 2002 scandals, and the SEC regulates futures

Chapter 3: Securities Markets — Answers

1. c
The primary market is where new securities are issued and the issuing firm receives the proceeds; every later trade among investors is a secondary-market trade that brings the firm no new money.

2. c
The underwriter sets an offer price from book-building demand and, under a firm commitment, bears the placement risk; its compensation is the spread between the price it pays and the price it charges.

3. c
Cost = explicit fee + money left on the table = \$150,000 + (28 − 25) × 200,000 = \$750,000.

4. b
A market buy fills against the lowest asks first: 300 × \$31.40 + 400 × \$31.45 = \$22,000 (an average of \$31.43).

5. b
A market order buys certainty of execution but not of price; a limit order buys certainty of price but not of execution, and you rarely get both.

6. c
A stop-loss guarantees execution, not price, so after a gap down it fills near the opening price; a stop-limit order, by contrast, would guarantee the price but might not fill at all.

7. a
The NYSE uses a central limit-order book with price–time priority (with human oversight at the open and close); Nasdaq has no trading floor and liquidity comes from multiple dealers posting quotes.

8. b
Buying power = cash ÷ initial margin = \$15,000 ÷ 0.40 = \$37,500; the loan is \$37,500 − \$15,000 = \$22,500.

9. d
Margin = (800P − 10,000) ÷ 800P = 0.35 → 800P × 0.65 = 10,000, so P = \$19.23; below this price a margin call is issued.

10. c
Position = 300 × \$60 = \$18,000; equity = \$10,800, loan = \$7,200; ending equity = 300 × \$66 − \$7,200 × 1.07 = \$12,096, so the return is 12,096 ÷ 10,800 − 1 = 12.00% (versus 10.00% unlevered).

11. b
The shares are collateral for a fixed loan, so all price variation falls on the investor's equity; leverage scales the return up and down, and Reg T sets initial margin at a minimum of 50% equity.

12. a
Short sales must be done in a margin account: the proceeds are not withdrawn, the investor also deposits initial margin, and dividends and borrow fees are costs of staying short.

13. b
Collateral = proceeds \$8,000 + margin \$4,000 = \$12,000; margin = (12,000 − 200P) ÷ 200P = 0.35, so P = 12,000 ÷ (200 × 1.35) = \$44.44; for a short position the call comes when the price rises.

14. b
Profit = (60 − 54 − 1.50) × 500 = \$2,250 (the dividend must be reimbursed); the investment is the margin of \$15,000, so the return is 2,250 ÷ 15,000 = 15.00%.

15. b
Sarbanes–Oxley (2002) responded to the accounting scandals; the SEC oversees securities and security options, the CFTC oversees futures, and FINRA is the industry self-regulatory body.

Chapter 4: Mutual Funds and Other Investment Companies

Multiple Choice

1. Which of the following is NOT one of the four core services an investment company provides to small investors?
a. Diversification and divisibility
b. Lower transaction costs from trading in large blocks
c. Record keeping and administration
d. A guaranteed return above the market index

2. A fund holds 120,000 shares of A at $52, 200,000 shares of B at $18, 90,000 shares of C at $64, and 350,000 shares of D at $9.50. Liabilities are $0.9 million and there are 2.2 million fund shares outstanding. What is the NAV per share?
a. $6.68
b. $8.19
c. $8.60
d. $9.01

3. An open-end fund has a NAV of $32.00. An investor buys $1.6 million of the fund. How many new shares are issued, and what is NAV afterward?
a. 50,000 shares; NAV rises to $32.80
b. 1,600,000 shares; NAV stays at $32.00
c. No new shares; NAV rises
d. 50,000 shares; NAV stays at $32.00

4. You place an order to buy shares of an open-end mutual fund at 10:30 a.m. At what price is the order filled?
a. At the last NAV published before 10:30 a.m.
b. At the market price quoted on the exchange at 10:30 a.m.
c. At any limit price you specify, if the market reaches it
d. At the NAV struck at the 4:00 p.m. close that day

5. Which statement about closed-end funds is correct?
a. Shares outstanding are fixed after the IPO, and the market price can differ from NAV
b. Investors redeem shares directly with the fund at NAV whenever they wish
c. Shares outstanding change daily as investors buy new shares from, and redeem them to, the fund
d. The market price is required by law to equal NAV at the close of each trading day

6. You buy a closed-end fund at $21.00 when its NAV is $24.00. One year later NAV is still $24.00, the price is $22.50, and the fund distributed $1.05 per share. What is your rate of return?
a. 4.38%
b. 7.14%
c. 11.33%
d. 12.14%

7. Which feature describes a unit investment trust (UIT)?
a. An actively managed portfolio overseen by a board of directors
b. A private pool open only to accredited investors
c. A fixed, unmanaged portfolio that terminates on a set date
d. A fund that trades continuously on an exchange at NAV

8. Compared with a mutual fund, a hedge fund typically:
a. Offers daily redemption at NAV to the general public, like an open-end mutual fund
b. Faces tighter limits on leverage and short sales because it is regulated more heavily
c. Is open only to accredited or institutional investors and is largely exempt from the Investment Company Act
d. Charges a lower fee than a mutual fund because it is less regulated and pays no incentive fee

9. Which statement about 12b-1 fees is correct?
a. They are one-time charges paid when shares are purchased, and are deducted from the amount invested
b. They are annual distribution fees of up to 1% of assets and work like a sales load spread over time
c. They are paid to portfolio managers as a performance bonus when the fund beats its benchmark
d. They are charged only on redemption within the first five years, and decline by one point annually

10. A fund reports a one-year return of 9%. For an investor who bought Class A shares with a 4% front-end load, the reported return:
a. Understates her return, because loads are refunded after one year
b. Equals her return, because loads are included in the expense ratio
c. Equals her return, because loads are deducted from NAV
d. Overstates her return, because published returns are computed before loads

11. You invest $30,000 in a fund with a 6% front-end load. How much is actually invested in the fund, and what return must the fund earn for you to get back to $30,000?
a. $28,200 invested; 6.00% return needed
b. $28,302 invested; 6.00% return needed
c. $31,800 invested; 5.66% return needed
d. $28,200 invested; 6.38% return needed

12. You invest $20,000 in Class A shares with a 4% front-end load. The fund’s reported return for the year is 12.0%. What is your return?

a. 7.52%

b. 8.00%

c. 11.54%

d. 12.00%

13. A fund charges a CDSC of 5%, 4%, 3%, 2% and 1% in years 1–5 and 0% thereafter. You redeem shares worth $22,000 during year 4. If the charge applies to the redemption value, what is it?

a. $220

b. $440

c. $660

d. $880

14. A fund’s NAV rises from $24.60 to $25.90 during the year. It pays income distributions of $0.45 and capital gain distributions of $0.95 per share, and its expense ratio is 1.25%. What are its net return and approximate gross return?

a. Net 10.98%; gross 12.23%

b. Net 5.28%; gross 6.53%

c. Net 10.42%; gross 11.67%

d. Net 10.98%; gross 9.73%

15. You invest $10,000 in a fund with a gross return of 8% a year. The two share classes are shown in the table. Which class leaves you with more after 1 year, and which after 10 years?

Feature	Class A	Class C
Front-end load	4.5%	none
Expense ratio	0.75%	1.45%
CDSC	none	1% if redeemed in year 1

a. 1 year: Class A ($10,242); 10 years: Class A ($19,230)

b. 1 year: Class C ($10,548); 10 years: Class C ($18,860)

c. 1 year: Class C ($10,548); 10 years: Class A ($19,230)

d. 1 year: Class A ($10,725); 10 years: Class C ($18,860)

16. Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?
a. ETFs are exempt from capital gains tax by law, so no gains are ever distributed
b. ETFs hold only bonds, which by their nature do not produce capital gains
c. ETF sellers trade with other investors on the exchange, so the fund does not have to sell stock to raise cash
d. ETF managers never sell securities for any reason, not even to rebalance the index

17. An ETF trades at $75.60 while its underlying basket is worth $75.00 per ETF share. Creation units are 20,000 shares. Ignoring trading costs, what is the AP's profit from one creation, and what does the AP do?
a. $12,000 — buy ETF shares, redeem them, sell the basket
b. $12,000 — buy the basket, create ETF shares, sell them
c. $1,200 — buy the basket, create ETF shares, sell them
d. $24,000 — buy ETF shares, redeem them, sell the basket

18. An index moves +8%, −8%, +8%, −8% on four consecutive days, starting at 100. A 2× daily leveraged ETF also starts at 100. What are the ending values?
a. Index 98.72; 2× ETF 94.95
b. Index 98.72; 2× ETF 97.45
c. Index 100.00; 2× ETF 100.00
d. Index 98.72; 2× ETF 98.72

19. You own a stock at $40 and sell a call with a $44 strike for a $1.20 premium. What is your profit per share on the covered call if the stock ends at $50, and if it ends at $35?
a. $10.00 at $50; −$5.00 at $35
b. $11.20 at $50; $1.20 at $35
c. $5.20 at $50; −$3.80 at $35
d. $5.20 at $50; −$6.20 at $35

20. Why must the average actively managed dollar underperform the market after costs?
a. Active managers are, on average, less skilled than the managers who build index funds
b. Before costs, active investors as a group hold the market, so their average gross return equals the market return
c. Active funds hold more cash than the market does, which drags returns in rising years
d. Regulations force active funds to hold riskier stocks than the index contains

21. Of 164 U.S. large-cap funds in the top quartile in 2021, 20.1% were still in the top quartile in 2022 and 0.0% in 2023, while pure chance would give 6.25%. What does this suggest?

a. Ranking funds by past returns is a good way to choose a fund

b. Past top performance is not a reliable signal of future top performance

c. Top funds deliberately take less risk after a good year

d. The result is caused by survivorship bias inflating the winners

Chapter 4: Mutual Funds and Other Investment Companies — Answers

1. d
The four services are record keeping and administration, diversification and divisibility, professional management and lower transaction costs; whether managers succeed on average is an empirical question.

2. b
Assets = 120,000×52 + 200,000×18 + 90,000×64 + 350,000×9.5 = $18,925,000; NAV = (18,925,000 − 900,000) ÷ 2,200,000 = $8.19.

3. d
New shares = $1,600,000 ÷ $32.00 = 50,000; assets and shares rise in proportion, so NAV stays at $32.00.

4. d
Open-end funds transact only at NAV, struck once daily at 4:00 p.m. ET, so every order that day gets the same price and you do not know it when you order; intraday prices and limit orders belong to ETFs and closed-end funds.

5. a
A closed-end fund neither issues nor redeems shares after its IPO; investors trade with each other on an exchange, so supply and demand set a price that may be at a premium or discount to NAV.

6. d
Return = (22.50 − 21.00 + 1.05) ÷ 21.00 = 12.14%, measured on the market price actually paid.

7. c
A sponsor deposits a fixed portfolio into a trust and sells redeemable units; with essentially no trading afterwards, management fees are very low and the trust is self-liquidating.

8. c
Hedge funds are private pools that disclose little, may use short sales, derivatives and heavy leverage, impose lock-up periods, and charge about “2 and 20”.

9. b
Named for SEC Rule 12b-1, this annual charge (maximum 1% of assets) pays for marketing and adviser compensation; it is included in the expense ratio but economically it is a load paid in instalments.

10. d
Reported returns are computed from NAV, which is net of expenses but not of loads, so a load-fund investor earns less than the published figure.

11. d
The load is a percentage of the gross amount paid: $30,000 × (1 − 0.06) = $28,200; to recover $30,000 the fund must earn 30,000 ÷ 28,200 − 1 = 6.38%, because the load is 6% of the gross but 6.38% of the net amount invested.

12. a
Amount invested = 20,000 × 0.96 = $19,200; end value = 19,200 × 1.12 = $21,504; return = 21,504 ÷ 20,000 − 1 = 7.52%.

13. b
The year-4 charge is 2%: 2% × $22,000 = $440.

14. a
Net return = (25.90 − 24.60 + 0.45 + 0.95) ÷ 24.60 = 10.98%, since distributions must be added back and the denominator is the beginning NAV; gross return ≈ net return + expense ratio = 10.98% + 1.25% = 12.23%.

15. c
Net growth: A = 8% − 0.75% = 7.25%, C = 8% − 1.45% = 6.55%; after 1 year A = 10,000 × 0.955 × 1.0725 = $10,242 and C = 10,000 × 0.99 × 1.0655 = $10,548 (the 1% CDSC applies in year 1); after 10 years A = 10,000 × 0.955 × 1.0725^{10} = $19,230 and C = 10,000 × 1.0655^{10} = $18,860.

16. c
In a mutual fund a redemption forces the fund to sell stock and realize gains that land on holders who sold nothing, whereas an ETF seller simply finds a buyer on the exchange.

17. b
The ETF is rich relative to the basket, so the AP buys the basket and creates ETF shares to sell; profit = (75.60 − 75.00) × 20,000 = $12,000.

18. a
Index: 100 × 1.08 × 0.92 × 1.08 × 0.92 = 98.72 (−1.28%); 2× ETF: 100 × 1.16 × 0.84 × 1.16 × 0.84 = 94.95 (−5.05%), about four times the index’s loss rather than twice.

19. c
At $50 the stock is called away at $44: profit = (44 − 40) + 1.20 = $5.20, so the upside above $44 was sold for the premium; at $35 the call expires worthless: profit = (35 − 40) + 1.20 = −$3.80, because the premium only softens the loss by $1.20 and is not protection.

20. b
Passive investors hold the market in proportion, so what remains for active investors is also the market; subtract their higher fees and the average must lag, though this says nothing about any single fund.

21. b
If skill drove rankings, winners would repeat more often than chance; the evidence shows no persistence, so past winners are not a dependable guide.

Chapter 5: Risk, Return, and the Historical Record

Multiple Choice

1. You buy a share at KRW 52,000. One year later it trades at KRW 57,200, and you received a cash dividend of KRW 1,300. What is the holding-period return?
a. 2.50%
b. 10.00%
c. 11.36%
d. 12.50%

2. A fund returns +30%, −25%, +15% and −10% in four consecutive years. What are its arithmetic and geometric average returns?
a. Arithmetic 2.50%, geometric 2.50%
b. Arithmetic 0.23%, geometric 2.50%
c. Arithmetic 2.50%, geometric 0.91%
d. Arithmetic 2.50%, geometric 0.23%

3. At t = 0 you buy one share at KRW 50,000. At t = 1 the share pays a KRW 1,500 dividend and you buy a second share at KRW 56,000. At t = 2 each share pays a KRW 1,500 dividend and you sell both shares at KRW 54,000. What are your dollar-weighted and time-weighted (geometric) returns?

Time	Share price	Dividend per share	Trade
t = 0	KRW 50,000	–	Buy 1 share
t = 1	KRW 56,000	KRW 1,500	Buy 1 more share
t = 2	KRW 54,000	KRW 1,500	Sell both shares

a. Dollar-weighted 4.15%; time-weighted 6.76%
b. Dollar-weighted 6.76%; time-weighted 4.15%
c. Dollar-weighted 4.15%; time-weighted 7.05%
d. Dollar-weighted 15.00%; time-weighted 6.76%

4. A deposit quotes 9% compounded monthly. What is the effective annual rate?
a. 0.75%
b. 9.00%
c. 9.31%
d. 9.38%

5. An investment has an effective annual rate of 12%. What is the equivalent continuously compounded annual rate?

a. 11.33%

b. 11.39%

c. 12.00%

d. 12.75%

6. A bond earns a nominal return of 7.5% while inflation is 3.2%. What is the exact real return?

a. 4.08%

b. 4.17%

c. 4.30%

d. 10.94%

7. A portfolio returns 40% in a boom (probability 0.25), 12% in a normal year (0.45), and −18% in a bust (0.30). What are its expected return and standard deviation?

State	Probability	Return
Boom	0.25	40%
Normal	0.45	12%
Bust	0.30	−18%

a. E(r) = 10.00%, σ = 21.49%

b. E(r) = 10.00%, σ = 4.62%

c. E(r) = 11.33%, σ = 23.68%

d. E(r) = 10.00%, σ = 41.09%

8. Two return distributions have the same mean and standard deviation. Distribution N is symmetric, while distribution L has negative skew. Which statement about L is correct?

a. It has a long right tail, so upside surprises are more frequent than in N

b. Its mean, median and mode coincide, as they do in the symmetric distribution N

c. It has a long left tail (mean < median < mode), so the standard deviation understates its downside risk

d. Its standard deviation overstates the risk of large losses, since the left tail is long

9. Compared with a normal distribution having the same mean and standard deviation, a return distribution with positive excess kurtosis has:

a. Thinner tails, so extreme returns are rarer than normality predicts

b. A long left tail only, so the mean lies below the median and mode

c. A taller peak and heavier tails on both sides, so extreme returns occur more often than normality predicts

d. A larger standard deviation by construction, since kurtosis is defined from the variance

10. A portfolio worth ₩5 billion has a daily volatility of 1.4%. Assuming normal returns, what are its one-day 95% VaR and its 10-day 99% VaR? (Use z = 1.645 for 95% and z = 2.326 for 99%.)
a. ₩115.15 million and ₩162.82 million
b. ₩115.15 million and ₩364.14 million
c. ₩115.15 million and ₩1,628.20 million
d. ₩115.15 million and ₩514.88 million

11. A bank reports a one-day 95% VaR of ₩30 million. Which statement is correct?
a. The largest loss the bank can suffer in one day is ₩30 million
b. The bank expects to lose ₩30 million per day on average
c. On 95 days out of 100, the bank will lose exactly ₩30 million
d. On about 5 trading days in 100, the loss is expected to exceed ₩30 million

12. Which statement correctly distinguishes value at risk (VaR) from expected shortfall (ES)?
a. VaR measures the average loss beyond the cut-off, while ES only marks the cut-off at a given confidence level
b. VaR marks the cut-off loss at a confidence level, while ES measures how large the loss is on average once that cut-off is passed
c. VaR and ES always give the same number when returns are normally distributed at any confidence level
d. ES ignores losses beyond the cut-off, which is why regulators prefer it to VaR for capital rules

13. Fund A has E(r) = 14% and σ = 24%; Fund B has E(r) = 9% and σ = 10%. The risk-free rate is 3%. Which fund has the higher Sharpe ratio?
a. Fund B: 0.90 versus 0.58
b. Fund A: 0.46 versus 0.60, because it earns more
c. Fund A: 2.18 versus 1.67
d. Fund B: 0.60 versus 0.46

14. As the investor changes y (the fraction of wealth in risky portfolio P, with the rest in the risk-free asset), what happens to the Sharpe ratio of the complete portfolio?
a. It is unchanged for every y
b. It rises as y rises
c. It falls as y rises
d. It rises until y = 1 and then falls

15. Over 1928–2021 the S&P 500 had an arithmetic average return of 11.82% and 3-month T-bills averaged 3.38%. What is the equity risk premium over T-bills?
a. 3.50%
b. 6.60%
c. 8.44%
d. 15.20%

16. Risky portfolio P has E(rP) = 15% and σP = 24%; the risk-free rate is 5%. An investor places y = 0.8 of her wealth in P and the rest in the risk-free asset. What are the expected return, standard deviation and Sharpe ratio of the complete portfolio?

a. E(rC) = 13.00%; σC = 24.00%; Sharpe 0.33

b. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.42

c. E(rC) = 12.00%; σC = 19.20%; Sharpe 0.36

d. E(rC) = 13.00%; σC = 19.20%; Sharpe 0.68

17. Wealth is KRW 30 million, y = 0.6, and risky portfolio P is 55% equities and 45% bonds. How is the money actually allocated?

a. Equities KRW 16.5m, bonds KRW 13.5m, T-bills KRW 0.0m

b. Equities KRW 9.9m, bonds KRW 8.1m, T-bills KRW 12.0m

c. Equities KRW 18.0m, bonds KRW 0.0m, T-bills KRW 12.0m

d. Equities KRW 6.6m, bonds KRW 5.4m, T-bills KRW 18.0m

18. The portfolios in the table are available to investors with utility $U = E(r) - \tfrac{1}{2} \cdot A \cdot \sigma^2$ and A = 2, 5 and 10. Which portfolio does each investor choose?

Portfolio	E(r)	σ
Conservative	6%	4%
Moderate	10%	12%
Aggressive	15%	22%

a. A = 2: Conservative; A = 5: Moderate; A = 10: Aggressive

b. All three investors choose Aggressive (highest E(r))

c. A = 2: Moderate; A = 5: Moderate; A = 10: Aggressive

d. A = 2: Aggressive; A = 5: Moderate; A = 10: Conservative

19. Portfolio Z has E(r) = 14% and σ = 24%. An investor with A = 4 can instead hold a 3% T-bill. What is Z’s certainty equivalent, and what does she hold?

a. Certainty equivalent 2.48%; she holds the T-bill

b. Certainty equivalent 2.48%; she holds Z because E(r) = 14% is higher

c. Certainty equivalent −9.04%; she holds the T-bill

d. Certainty equivalent 25.52%; she holds Z

20. E(rP) = 12%, σP = 20%, rf = 3% and the investor’s risk aversion is A = 4. What are the optimal risky weight y*, and the resulting E(rC), σC and utility $U = E(r) - \tfrac{1}{2}A\sigma^2$?

a. y* = 1.1250; E(rC) = 13.12%; σC = 22.50%; U = 3.00%

b. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 5.53%

c. y* = 0.1125; E(rC) = 4.01%; σC = 2.25%; U = 3.91%

d. y* = 0.5625; E(rC) = 8.06%; σC = 11.25%; U = 8.06%

21. Risky portfolio P has E(rP) = 12% and σP = 20%; the risk-free rate is 3%. An investor with A = 2 has optimal weight y* = 1.125 and wealth of KRW 16 million. Assuming she can borrow at the risk-free rate, what does she do?

a. Invests KRW 2.0m in P and holds the rest in T-bills

b. Borrows KRW 18.0m and invests KRW 34.0m in P

c. Borrows KRW 2.0m and invests KRW 18.0m in P

d. Holds KRW 18.0m in P and KRW 2.0m in T-bills

22. Holding everything else fixed, which change would make the optimal risky weight y* smaller?

a. A rise in the risk premium E(rP) − rf

b. A fall in the coefficient of risk aversion A

c. A fall in the variance of the risky portfolio

d. A rise in the variance of the risky portfolio

23. In $U = E(r) - \frac{1}{2} \cdot A \cdot \sigma^2$, which statement about A is correct?

a. A < 0 describes a risk-averse investor

b. A = 0 describes a risk-neutral investor who judges by E(r) alone

c. A larger A means the investor is more tolerant of risk

d. A has no effect when σ is below 20%

24. Risky portfolio P has E(rP) = 15% and σP = 25%; the risk-free rate is 5%. What is the slope of the capital allocation line, and where does it cross σ = 0?

a. Slope 0.60; intercept 5%

b. Slope 0.40; intercept 5%

c. Slope 0.40; intercept 0%

d. Slope 2.50; intercept 15%

25. What is the capital market line (CML)?

a. The line joining the lowest-risk and highest-return securities in the market

b. The capital allocation line of the best-performing active fund over the sample period

c. A line showing the mean and variance of every individual stock in the market

d. The capital allocation line that uses a broad market index as the risky portfolio P

Chapter 5: Risk, Return, and the Historical Record — Answers

1. d
HPR = (57,200 − 52,000 + 1,300) ÷ 52,000 = 12.50%, i.e. capital gain yield 10.00% plus dividend yield 2.50%.

2. d
Arithmetic = (30 − 25 + 15 − 10) ÷ 4 = 2.50%; geometric = (1.30 × 0.75 × 1.15 × 0.90)^(1/4) − 1 = 0.23%, always below the arithmetic mean.

3. a
Net cash flows: t0 = −50,000; t1 = −56,000 + 1,500 = −54,500; t2 = 2 × 1,500 + 2 × 54,000 = +111,000, whose IRR is the dollar-weighted return, 4.15%. Period returns: year 1 = (56,000 − 50,000 + 1,500) ÷ 50,000 = 15.00%, year 2 = (54,000 − 56,000 + 1,500) ÷ 56,000 = −0.89%, so the time-weighted return = (1.15 × 0.99107)^(1/2) − 1 = 6.76%; the dollar-weighted return is lower because more money was invested in the weaker second year.

4. d
EAR = $(1 + 0.09/12)^{12} - 1$ = 9.38%.

5. a
APR (continuous) = ln(1 + EAR) = ln(1.12) = 11.33%.

6. b
Fisher equation: 1 + R = (1 + r)(1 + i), so r = 1.075 ÷ 1.032 − 1 = 4.17%.

7. a
E(r) = 0.25(40) + 0.45(12) + 0.30(−18) = 10.00%; σ^2 = $0.25(30)^2 + 0.45(2)^2 + 0.30(-28)^2$ = 462, so σ = √462 = 21.49%.

8. c
With negative skew the left tail is long and the mean falls below the median and mode; equity returns are typically left-skewed, so σ alone understates their risk.

9. c
Fat tails mean extremes arrive far more often than a normal distribution implies (Black Monday 1987 was about a −20σ day), so normality understates the chance of crashes.

10. d
VaR = V0 × z × σ × √T. One-day 95%: ₩5,000 million × 1.645 × 0.014 × 1 = ₩115.15 million; 10-day 99%: ₩5,000 million × 2.326 × 0.014 × √10 = ₩514.88 million.

11. d
VaR is the loss exceeded with probability 5% over the stated horizon; it is not a maximum loss and says nothing about how large the loss is on those worst days.

12. b
VaR says nothing about the size of losses past the cut-off; ES (also CVaR) answers that and is coherent, and Basel FRTB replaced the 99% VaR charge with a 97.5% expected shortfall.

13. d
Sharpe = (E(r) − rf) ÷ σ: A = (14 − 3) ÷ 24 = 0.46; B = (9 − 3) ÷ 10 = 0.60, so B gives more reward per unit of volatility.

14. a
Both the risk premium and σC scale with y, so their ratio stays equal to the slope of the CAL; this is why choosing P and choosing y can be solved separately (the separation property).

15. c
The equity risk premium is 11.82% − 3.38% = 8.44 percentage points.

16. b
E(rC) = rf + y[E(rP) − rf] = 5% + 0.8 × 10% = 13.00%; σC = y·σP = 0.8 × 24% = 19.20%; Sharpe = (13.00 − 5) ÷ 19.20 = 0.42, identical to P's 10 ÷ 24, because the risk premium and σC both scale with y.

17. b
T-bills = 30m × 0.4 = KRW 12.0m; P = 30m × 0.6 = KRW 18.0m, of which equities = 18.0 × 0.55 = KRW 9.9m and bonds = 18.0 × 0.45 = KRW 8.1m.

18. d
Utilities (A = 2 / 5 / 10): Conservative 5.84 / 5.60 / 5.20; Moderate 8.56 / 6.40 / 2.80; Aggressive 10.16 / 2.90 / −9.20, so more risk aversion means a safer choice.

19. a
U = 0.14 − ½ × 4 × 0.24^2 = 2.48%, below the 3% certain return of the T-bill, so she holds the T-bill; a risk-free asset has σ = 0, so its certainty equivalent is its own return.

20. b
y* = [E(rP) − rf] ÷ (A·σP^2) = 0.09 ÷ (4 × 0.04) = 0.5625; E(rC) = 3% + 0.5625 × 9% = 8.06%; σC = 0.5625 × 20% = 11.25%; U = 0.0806 − ½ × 4 × 0.1125^2 = 5.53%.

21. c
y* = 1.125 > 1: she invests 1.125 × 16m = KRW 18.0m in P, financed by her own 16.0m plus KRW 2.0m borrowed at the risk-free rate.

22. d
y* = [E(rP) − rf] ÷ (A·σP^2): a bigger reward invites a bigger bet, while more risk aversion or more variance shrinks it.

23. b
A > 0 is risk averse (empirically about 2 to 4), A = 0 is risk neutral and A < 0 is risk loving; the penalty ½·A·σ^2 grows with both A and σ.

24. b
Slope = (E(rP) − rf) ÷ σP = (15 − 5) ÷ 25 = 0.40, the Sharpe ratio, and the line starts at rf = 5% where σ = 0.

25. d
A passive strategy holds a broad market index without security analysis or market timing; when that index is P, the resulting CAL is called the CML.

Chapter 6: Optimal Risky Portfolios

Multiple Choice

1. Each of n stocks has a standard deviation of 30% and every pair has a correlation of 0.25. An investor holds them with equal weights. What is the portfolio standard deviation when n = 20, and what does it converge to as n grows without limit?
a. 15.00% at n = 20; floor 16.09%
b. 6.71% at n = 20; floor 0.00%
c. 16.09% at n = 20; floor 7.50%
d. 16.09% at n = 20; floor 15.00%

2. Which of the following is an example of firm-specific (diversifiable) risk?
a. The central bank unexpectedly raises interest rates for the whole economy
b. A pandemic lowers the earnings of nearly all listed firms at once
c. A pharmaceutical company announces that its main drug failed its clinical trial
d. The won depreciates sharply against the dollar, hurting importers and exporters alike

3. A stock has a total return variance of 0.0400 (σ = 20%), of which 0.0256 (σ = 16%) is systematic. What is the standard deviation attributable to firm-specific risk?
a. 1.44%
b. 4.00%
c. 12.00%
d. 25.61%

4. Four equally likely states produce the returns in the table. What are the covariance (in %²) and correlation between the stock and bond returns?

State	Prob.	Stock return	Bond return
Severe recession	0.25	−20%	+11%
Mild recession	0.25	+4%	+5%
Normal growth	0.25	+16%	+2%
Boom	0.25	+30%	+2%

a. Cov = −64.5; ρ = −0.96
b. Cov = 64.5; ρ = 0.96
c. Cov = −27.0; ρ = −0.40
d. Cov = −64.5; ρ = −2.93

5. A stock fund has E(r) = 11% and σ = 22%; a bond fund has E(r) = 5% and σ = 8%; their correlation is 0.3. What are the expected return and standard deviation of a portfolio with 70% in stocks and 30% in bonds?

a. E(rP) = 9.20%, σP = 17.80%

b. E(rP) = 9.20%, σP = 15.59%

c. E(rP) = 8.00%, σP = 12.78%

d. E(rP) = 9.20%, σP = 16.28%

6. A stock fund (E(r) = 14%, σ = 30%) and a bond fund (E(r) = 6%, σ = 10%) have a correlation of −1. What weight in stocks creates a zero-risk portfolio, and what is its expected return?

a. 75.0% in stocks; E(r) = 12.0%

b. 25.0% in stocks; E(r) = 8.0%

c. 33.3% in stocks; E(r) = 8.7%

d. 50.0% in stocks; E(r) = 10.0%

7. Stock fund σS = 24%, bond fund σB = 8%, correlation 0.1. What is the weight in stocks of the minimum-variance portfolio, and its standard deviation?

a. 10.00% in stocks; σ = 7.81%

b. 92.55% in stocks; σ = 22.28%

c. 7.45% in stocks; σ = 8.00%

d. 7.45% in stocks; σ = 7.79%

8. A portfolio is 100% in a bond fund with σB = 12%. A stock fund has σS = 30%. For which correlation with the bond fund does adding a small amount of stock reduce the portfolio's risk?

a. ρ = 0.25

b. ρ = 0.40

c. ρ = 0.60

d. ρ = 0.90

9. As the correlation between two risky assets decreases while the expected returns and standard deviations stay the same, what happens to the portfolio opportunity set?

a. It becomes a straight line between the two assets

b. It shifts upward, so every mix earns a higher expected return

c. It bows further to the left, so a given expected return can be earned at lower risk

d. It is unchanged, because expected returns are linear in the weights

10. On the opportunity set of two risky assets, which portfolios are efficient?

a. Those on the upward-sloping part starting at the minimum-variance portfolio

b. Those on the downward-sloping part below the minimum-variance portfolio

c. Every portfolio on the curve

d. Only the two single-asset portfolios

11. Stock fund: E(r) = 13%, σ = 24%; bond fund: E(r) = 5%, σ = 9%; correlation 0.25; rf = 2%. What are the weights, expected return, standard deviation and Sharpe ratio of the optimal risky portfolio?

a. Stocks 39.13%, bonds 60.87%; E(rP) = 8.13%, σP = 12.00%, Sharpe 0.511

b. Stocks 4.92%, bonds 95.08%; E(rP) = 5.39%, σP = 8.93%, Sharpe 0.380

c. Stocks 34.02%, bonds 65.98%; E(rP) = 7.72%, σP = 11.23%, Sharpe 0.509

d. Stocks 60.87%, bonds 39.13%; E(rP) = 9.87%, σP = 15.86%, Sharpe 0.496

12. rf = 2%. Which of the candidate risky portfolios in the table gives the steepest capital allocation line?

Candidate portfolio	w in stocks	E(rP)	σP
All bonds	0.00%	5.00%	9.00%
Minimum-variance portfolio	4.92%	5.39%	8.93%
60 / 40 portfolio	60.00%	9.80%	15.69%
All stocks	100.00%	13.00%	24.00%
Optimal risky portfolio	39.13%	8.13%	12.00%

a. Minimum-variance portfolio (lowest risk)

b. All stocks (highest expected return)

c. All bonds (simplest low-risk choice)

d. Optimal risky portfolio (highest Sharpe ratio)

13. The optimal risky portfolio has E(rP) = 8.13%, σP = 12.00% with 39.13% in the stock fund; rf = 2%. What fraction of total wealth should an investor with A = 5 hold in the stock fund?

a. 14.81%

b. 33.33%

c. 39.13%

d. 85.19%

14. Two clients with risk aversion A = 3 and A = 8 face the same inputs and can lend and borrow at the same risk-free rate. According to the separation property:

a. They hold the same stock-to-bond ratio inside the risky portfolio but different amounts of T-bills

b. They hold different risky portfolios, with the more risk-averse client holding more bonds inside P

c. They hold the same total allocation to risky assets

d. The client with A = 8 holds no risky assets

15. Which of the following is a reason why the separation property breaks down in practice?

a. All investors face identical taxes and the same constraints

b. Investors agree on the input list and hold the same expectations

c. Human capital is the same for every investor in the economy

d. Borrowing rates exceed lending rates, which kinks the CAL

16. Which statement about the Markowitz efficient frontier of n risky assets is correct?
a. Each individual asset lies on the efficient frontier, since single securities are its building blocks
b. Individual assets lie inside the frontier, and only the branch above the minimum-variance portfolio is efficient
c. The entire frontier is efficient, including the part below the minimum-variance portfolio
d. The tangency portfolio is found by minimizing variance for a zero expected return

17. To run the Markowitz optimization on n = 200 securities, how many estimates are needed in total, and how many of them are covariances?
a. 19,900 in total; 19,900 covariances
b. 20,300 in total; 39,800 covariances
c. 20,300 in total; 19,900 covariances
d. 40,000 in total; 19,900 covariances

18. Korean households hold mostly domestic equities, although Korea is a small share of world market capitalization. Why would adding foreign equities usually reduce risk?
a. Foreign equities always earn higher expected returns than domestic equities
b. Correlations between countries are exactly zero, so foreign risk is fully independent
c. Adding foreign equities removes market risk entirely, because currencies offset equity moves
d. Foreign markets have different industrial structures, so their correlation with domestic stocks is below +1

Chapter 6: Optimal Risky Portfolios — Answers

1. d
Average covariance = $0.25 \times 30^2 = 225$; variance = $(1/20) \times 900 + (1 - 1/20) \times 225 = 258.75$, so $\sigma = 16.09\%$; as $n \to \infty$ the first term vanishes and $\sigma \to \sqrt{225} = 15.00\%$, the systematic risk that diversification cannot remove.

2. c
Firm-specific risk arises from events affecting one company or industry and cancels out in a large portfolio; interest rates, pandemics and exchange rates hit every firm at once, so they are market (systematic) risk.

3. c
Var(total) = Var(systematic) + Var(firm-specific) because the two parts are uncorrelated, so firm-specific variance = $0.0400 - 0.0256 = 0.0144$ and $\sigma = \sqrt{0.0144} = 12.00\%$.

4. a
E(S) = 8% and E(B) = 5%; Cov = Σ p × (dev S) × (dev B) = −64.5; $\sigma S = 18.35\%$ and $\sigma B = 3.67\%$, so $\rho = -64.5 \div (18.35 \times 3.67) = -0.96$; the stock does worst exactly when the bond does best.

5. d
$E(rP) = 0.7(11) + 0.3(5) = 9.20\%$; variance = $0.49(484) + 0.09(64) + 2(0.7)(0.3)(52.8) = 265.10$, so $\sigma P = 16.28\%$.

6. b
Set $wS\sigma S = wB\sigma B$: $30w = 10(1 - w)$, so $w = 10 \div (30 + 10) = 0.25$; $E(r) = 0.25(14) + 0.75(6) = 8.0\%$, with zero standard deviation.

7. d
$Cov = 0.1 \times 24 \times 8 = 19.2$; $wS = (\sigma B^2 - Cov) \div (\sigma S^2 + \sigma B^2 - 2Cov) = (64 - 19.2) \div (576 + 64 - 38.4) = 0.0745$; $\sigma = 7.79\%$, below the bond fund's own 8%.

8. a
Adding the riskier asset lowers risk whenever $\rho < \sigma B / \sigma S = 12/30 = 0.40$; $\rho = 0.25$ satisfies this, while $\rho = 0.40$ is exactly the break-even and higher correlations raise risk.

9. c
Expected return depends only on the weights, so diversification shows up as a leftward bow; at $\rho = +1$ the set is a straight line, and at $\rho = -1$ it touches the vertical axis.

10. a
Points below the minimum-variance portfolio are dominated by points above it with the same risk but a higher expected return, so only the upper branch is efficient.

11. a
Cov = 54.0; RS = 11, RB = 3; $wS^* = [11(81) - 3(54.0)] \div [11(81) + 3(576) - 14(54.0)] = 729 \div 1863 = 0.3913$; $E(rP) = 0.3913(13) + 0.6087(5) = 8.13\%$; $\sigma P^2 = 143.93$, so $\sigma P = 12.00\%$; Sharpe = $(8.13 - 2) \div 12.00 = 0.511$.

12. d
Sharpe ratios: All bonds 0.333; Minimum-variance portfolio 0.380; 60 / 40 portfolio 0.497; All stocks 0.458; Optimal risky portfolio 0.511; the CAL slope is the Sharpe ratio, and neither the lowest-risk nor the highest-return portfolio maximizes it.

13. b
$y^* = [E(rP) - rf] \div (A \cdot \sigma P^2) = 0.0613 \div (5 \times 0.014393) = 0.8519$; stocks = $y^* \times wS^* = 0.8519 \times 0.3913 = 33.33\%$ of total wealth (look-through).

14. a
The tangency portfolio is the same for everyone; only y^* differs, so the more risk-averse client simply puts more in T-bills while keeping the identical mix within P.

15. d
The other limits listed in the slides are tax differences, human capital (a semiconductor engineer should tilt away from semiconductors), disagreement over the input list, and binding constraints.

16. b
The frontier is found by minimizing variance for each target return; single assets are dominated, and the tangency portfolio P* remains the unique Sharpe-maximizing portfolio.

17. c
Total = n expected returns + n variances + $n(n - 1)/2$ covariances = 200 + 200 + 19,900 = 20,300 = $n(n + 3)/2$; covariances are symmetric, so only $n(n - 1)/2 = 19{,}900$ are distinct.

18. d
The benefit is a covariance argument, not a forecasting argument: as long as cross-border correlations are below +1 the interaction term helps, although correlations have risen over recent decades and currency risk must be handled separately.

Investment Analysis Midterm - Short Answer Quiz

1. **[Ch.2]** What defines a money market instrument? Explain why a Treasury note is not one, and name two money market instruments.

2. **[Ch.2]** In a repurchase agreement (repo), what does the dealer do, and why is a repo considered a very safe short-term loan?

3. **[Ch.2]** What are Eurodollars? Does the prefix "Euro" mean that the deposit must be held in Europe?

4. **[Ch.2]** What is a mortgage-backed pass-through security, and how does securitization shift risk?

5. **[Ch.2]** Which two characteristics define common stock as a claim on the firm, and what does each mean?

6. **[Ch.2]** Why is preferred stock called a hybrid security? What happens if the firm skips a preferred dividend?

7. **[Ch.2]** What is an American Depositary Receipt (ADR), and what is its advantage for a U.S. investor?

8. **[Ch.2]** A stock in a price-weighted index splits 2-for-1. Why must the divisor be changed, and what happens to that stock's weight in the index?

9. **[Ch.2]** What does an equally weighted index imply for an investor's portfolio, and why does maintaining it require trading?

10. **[Ch.2]** What rights does the holder of a call option have, what is the holder's maximum loss, and what happens if the asset price ends below the exercise price?

11. **[Ch.2]** When is a put option in the money, and why is being in the money not the same as being profitable for the buyer?

12. **[Ch.2]** Distinguish a long futures position from a long call position in terms of right versus obligation, up-front payment and potential loss.

13. **[Ch.2]** At what point is an investor indifferent between a taxable bond and a municipal bond, and why are municipal bonds mostly held by high-bracket investors?

14. **[Ch.2]** State the two flaws of the bank discount yield as a measure of the return on a Treasury bill, and how the bond equivalent yield corrects them.

15. **[Ch.2]** For the same Treasury bill, rank the bank discount yield, the bond equivalent yield and the effective annual yield from lowest to highest, and explain what drives each step.

16. **[Ch.2]** What does it mean for a money market fund to "break the buck," and what triggered it for the Reserve Primary Fund in 2008?

17. **[Ch.2]** How do TIPS protect an investor against inflation?

18. **[Ch.2]** Distinguish general obligation bonds from revenue bonds. Which is riskier in terms of default?

19. **[Ch.2]** Compare price-weighted, value-weighted and equally weighted indexes. Which kind of stock dominates the first two?

20. **[Ch.3]** In which market do the proceeds of a sale go to the issuing firm, and why does the secondary market still matter to firms?

21. **[Ch.3]** Distinguish firm commitment from best efforts underwriting. Who bears the risk of an unsold issue in each?

22. **[Ch.3]** What does the evidence say about average first-day and long-run IPO returns, and what explains the first-day pattern?

23. **[Ch.3]** A dealer quotes a bid of $34.75 and an asked price of $34.90. At which price does an investor buy, at which does she sell, and what does the spread represent?

24. **[Ch.3]** Compare a market order and a limit order in terms of certainty of execution and certainty of price.

25. **[Ch.3]** A stock trades at $50 and an investor places a limit buy order at $47. What happens to the order, and how does it relate to market liquidity?

26. **[Ch.3]** What does a stop-loss sell order at $55 do for an investor who holds a stock trading at $60, and what does it not guarantee?

27. **[Ch.3]** What is high-frequency trading, and what do proponents and critics argue about its effect on liquidity?

28. **[Ch.3]** Explain how buying on margin affects an investor's return and risk, and who sets the initial margin requirement.

29. **[Ch.3]** When does a margin call occur for a margin-purchased (long) position and for a short sale, and why do the two differ?

30. **[Ch.3]** Describe a short sale: what does the seller do, when does the position profit, and how do the maximum gain and loss compare?

31. **[Ch.3]** In a short sale, what happens to the sale proceeds, what margin is required, and who receives the dividends paid on the borrowed shares?

32. **[Ch.3]** How do ECNs and algorithmic (high-frequency) trading change the way securities are traded, and what are the benefits of electronic trading?

33. **[Ch.3]** Name the four pillars of securities regulation and state what each does.

34. **[Ch.4]** What exactly does an investor who owns 2% of a mutual fund's shares own, and what is not guaranteed?

35. **[Ch.4]** Name the four core services an investment company provides to small investors, and say whether a return above the market is one of them.

36. **[Ch.4]** How is a fund's net asset value (NAV) computed, which items are deducted, and why does issuing new shares at NAV leave NAV unchanged?

37. **[Ch.4]** At what price is an order to buy open-end mutual fund shares at 10:00 a.m. filled, and how do closed-end funds and ETFs differ?

38. **[Ch.4]** Describe a closed-end fund: how its shares are issued, who trades with whom, and how its market price relates to NAV.

39. **[Ch.4]** What is a unit investment trust (UIT), and why are its management fees very low?

40. **[Ch.4]** How does a hedge fund differ from a mutual fund in who may invest, regulation, strategies, fees and redemptions?

41. **[Ch.4]** Distinguish a balanced fund from an asset allocation fund.

42. **[Ch.4]** Why do investors tend to underestimate a fund's expense ratio, and is the reported fund return before or after it?

43. **[Ch.4]** What is a 12b-1 fee, how large can it be, and why is it economically like a sales load?

44. **[Ch.4]** What does a back-end load (CDSC) that declines each year do economically, and who receives it?

45. **[Ch.4]** A fund reports an 11% one-year return and an investor paid a 5% front-end load. Does the reported return equal her return?

46. **[Ch.4]** Class A shares charge a front-end load and a lower expense ratio; Class C shares charge no front-end load and a higher expense ratio. Which investor should prefer which class, and why?

47. **[Ch.4]** An ETF trades above the value of its underlying basket. Describe what an authorized participant does and how this restores the ETF price.

48. **[Ch.4]** Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?

49. **[Ch.4]** What does a 2× leveraged ETF deliver, and why can its multi-period return differ widely from twice the index return?

50. **[Ch.4]** Where does the high distribution of a covered-call ETF come from, and what does the fund give up in return?

51. **[Ch.4]** Why must the average actively managed dollar underperform the market after costs? Use the logic of passive and active investors.

52. **[Ch.4]** Of 164 U.S. large-cap funds in the top quartile in 2021, none were still in the top quartile in 2023, while chance alone would give 6.25%. What does this suggest about choosing funds by past returns?

53. **[Ch.4]** A fund charges a 5.75% front-end load and you invest $10,000. Why must the fund earn more than 5.75% just for you to break even?

54. **[Ch.5]** Define the holding-period return (HPR), split it into its two components, and state which price is the denominator.

55. **[Ch.5]** Distinguish the arithmetic average from the geometric average of returns. Which one is the actual compound growth rate, and which is larger?

56. **[Ch.5]** How are the dollar-weighted and the time-weighted returns calculated, and when do they differ? Which one is better for judging a manager?

57. **[Ch.5]** For a given APR, how does the compounding frequency affect the effective annual rate (EAR), and how is a continuously compounded rate related to the EAR?

58. **[Ch.5]** Why does the shortcut real return ≈ nominal − inflation overstate the real return, and what is the exact relation?

59. **[Ch.5]** A bank reports a one-day 99% VaR of ₩40 million. What does the number mean, how often should it be exceeded, and is it the maximum possible loss?

60. **[Ch.5]** Under normally distributed returns, how is a one-day VaR scaled to a T-day horizon, and which common mistake does the rule avoid?

61. **[Ch.5]** As an investor changes the fraction y invested in the risky portfolio P, what happens to the Sharpe ratio of the complete portfolio, and why?

62. **[Ch.5]** How does the nominal interest rate respond to an increase in expected inflation, and what does this preserve?

63. **[Ch.5]** Explain what skew and kurtosis measure, and why they matter when returns are not normal.

64. **[Ch.5]** Compare historical simulation and Monte Carlo simulation as ways to obtain the return distribution for VaR.

65. **[Ch.5]** Why is Expected Shortfall (ES) used alongside VaR, and how did Basel treat the two?

66. **[Ch.5]** What does the Sharpe ratio measure, why is it unchanged when the investor mixes in the risk-free asset, and why is it the wrong tool for a single security?

67. **[Ch.5]** In the utility function $U = E(r) - \tfrac{1}{2}A\sigma^2$, what does A measure, and how do risk-averse, risk-neutral and risk-loving investors differ? What is a certainty equivalent?

68. **[Ch.5]** In $y^* = [E(r_p) - r_f] \div (A\sigma_P^2)$, how does the optimal fraction in the risky portfolio respond to the risk premium, the risk aversion A and the variance?

69. **[Ch.5]** Why does the capital allocation line bend (kink) at the risky portfolio P when the investor can borrow only at a rate above the risk-free rate?

70. **[Ch.5]** What is the capital market line, how is passive investing defended, and what is the Grossman-Stiglitz paradox?

71. **[Ch.6]** Distinguish systematic risk from firm-specific risk. Which can be diversified away, and which earns a risk premium?

72. **[Ch.6]** Suppose every stock has the same standard deviation and every pair has a correlation of 0.30. What happens to an equally weighted portfolio's standard deviation as n grows without limit, and what does the floor represent?

73. **[Ch.6]** In the two-asset portfolio variance formula, which term is the only one that can be negative, and why does it matter for diversification?

74. **[Ch.6]** When can two risky assets be combined into a zero-risk portfolio, and what determines the weights?

75. **[Ch.6]** Starting from 100% in a bond fund with $\sigma_B = 10\%$, when does adding a little of a riskier stock fund with $\sigma_S = 20\%$ reduce portfolio risk?

76. **[Ch.6]** Why is the minimum-variance portfolio generally not the best risky portfolio to combine with T-bills, and which portfolio is chosen instead?

77. **[Ch.6]** What does the separation property imply about the optimal risky portfolio and about the role of risk aversion? When does it break down?

78. **[Ch.6]** When does an investor hold more than 100% of her wealth in the optimal risky portfolio, and what does that mean for her financing?

79. **[Ch.6]** Why is a full Markowitz optimization hard to implement for many securities, and what must be estimated?

80. **[Ch.6]** State the insurance principle. Why is holding mostly your employer's stock in a 401(k) plan the opposite of it?

81. **[Ch.6]** Define covariance and the correlation coefficient. Why is the correlation easier to interpret, and what values can it take?

82. **[Ch.6]** In the 50/50 portfolio of a stock fund and a bond fund (standard deviations 21.73% and 3.54%), in what sense is the benefit of diversification "cost-free"?

83. **[Ch.6]** In which single case is there no benefit from diversification, and what does the opportunity set look like?

84. **[Ch.6]** On the opportunity set of two risky assets, which portfolios are efficient and which are dominated? Why?

85. **[Ch.6]** Define the efficient frontier of risky assets and how it is found. Where do the individual assets lie relative to it?

86. **[Ch.6]** Describe the four-step recipe for building the optimal complete portfolio. Which step depends on who the investor is?

87. **[Ch.6]** What is home bias, why does investing abroad lower portfolio risk, and what is the honest caveat?

Answers

1. [Ch.2] Money market instruments are **short-term** (one year or less), **highly marketable** debt. A **Treasury note** has a maturity of 1 to 10 years, so it belongs to the capital market; negotiable **certificates of deposit**, banker's acceptances and commercial paper are money market instruments.
Keywords: short-term · highly marketable · Treasury note · certificates of deposit

2. [Ch.2] The dealer **sells government securities** to an investor and agrees to **buy them back** the next day at a slightly **higher price**; the difference is the interest. Because the securities serve as **collateral**, a repo is effectively a very safe **short-term** loan to the dealer.
Keywords: sells securities · buy back · higher price · short-term · collateral

3. [Ch.2] Eurodollars are **dollar-denominated** deposits held at **foreign banks or foreign branches of U.S. banks**, outside U.S. bank regulation. The prefix refers to the **location** of the deposit, **not necessarily Europe**: a dollar deposit in a Tokyo bank is a Eurodollar deposit.
Keywords: dollar-denominated · foreign banks · location

4. [Ch.2] A pass-through is an **ownership claim** on a **pool of mortgages**; the homeowners' monthly **principal and interest** are **passed through** to investors net of servicing fees. Securitization shifts **default and prepayment risk** from the originating banks to the investors.
Keywords: ownership claim · pool of mortgages · principal and interest · passed through · default and prepayment

5. [Ch.2] Common stock is a **residual claim**: shareholders are **paid last**, after employees, suppliers and bondholders. It carries **limited liability**: the most a shareholder can lose is the **amount invested**.
Keywords: residual claim · paid last · limited liability · amount invested

6. [Ch.2] Preferred stock is a **hybrid**: like a bond it pays a **fixed dividend** and normally has **no voting rights**, but like equity the dividend is **not a legal obligation**, so skipping it does not trigger default; **cumulative** preferred must clear all arrears before any common dividend.
Keywords: hybrid · fixed dividend · not a legal obligation · no voting · cumulative

7. [Ch.2] An **ADR** is a **dollar-denominated certificate** issued by a **depositary bank** that represents ownership of **foreign shares** and trades on a **U.S. exchange**, so U.S. investors trade and receive dividends in dollars without dealing in a foreign market or currency.
Keywords: certificate · foreign shares · U.S. exchange · dollar-denominated · depositary bank

8. [Ch.2] A split lowers the stock's price, which would mechanically lower the index, so the **divisor** is reset to keep the **index level unchanged**. The split stock's **weight falls permanently**, a known distortion of price weighting.
Keywords: divisor · index level unchanged · weight falls · permanent

9. [Ch.2] It corresponds to investing an **equal dollar** amount in each stock. As prices move the weights drift, so it needs periodic **rebalancing**: **selling outperformers** and **buying underperformers**. This implies **trading costs** that value-weighted indexes do not have.
Keywords: equal dollar · rebalancing · selling outperformers · buying underperformers · trading costs

10. [Ch.2] The holder has the **right to buy** at the **exercise price**, with no obligation. If the asset ends below that price the option **expires worthless**; the **maximum loss** is the **premium** paid, while the gain is open-ended. Unlimited loss describes the call writer, not the holder.
Keywords: right to buy · exercise price · expires worthless · premium · maximum loss

11. [Ch.2] A put gives the **right to sell** at the exercise price, so it is **in the money** when the asset price is **below the exercise price**. The buyer profits only if the price falls far enough to **cover the premium** paid; the writer's gain is capped at that premium.
Keywords: right to sell · in the money · below the exercise price · premium · cover the premium

12. [Ch.2] A long **futures** position is an **obligation** to take delivery (or settle) at the agreed price, so **losses** can be large if the price falls; a long **call** is a **right** that the holder abandons if unprofitable. A call requires a **premium** at purchase, while futures require no premium, only a **margin** deposit that acts as a performance bond.
Keywords: obligation · right · premium · margin · losses

13. [Ch.2] The investor is **indifferent** when the **after-tax yields are equal**, $r_{taxable}(1 - t) = r_{muni}$, which defines a critical **marginal tax rate** t. Investors in **higher brackets** prefer the **tax-exempt** municipal bond; those below that rate prefer the taxable bond.
Keywords: indifferent · after-tax yields are equal · marginal tax rate · higher bracket · tax-exempt

14. [Ch.2] The **bank discount yield** has two flaws: "First, it assumes that the year has only **360 days**. Second, it computes the yield as a fraction of **face value** rather than of the **price the investor paid**." The **bond equivalent yield** corrects both: it uses a **365**-day year and divides by the price paid.
Keywords: bank discount yield · 360 days · face value · price the investor paid · bond equivalent yield · 365

15. [Ch.2] The **bank discount yield** is the **lowest** because it divides by face value instead of the **price paid** and uses a 360-day year. The **bond equivalent yield** corrects both and is higher. The **effective annual yield** is the **highest** because it adds **compounding** of the sub-year return, so BDY < BEY < EAY.
Keywords: bank discount yield · lowest · price paid · bond equivalent yield · effective annual yield · compounding · highest

16. [Ch.2] Money market funds try to hold their **share value at $1** and pass along all earnings as interest. A fund "breaks the buck" when losses force its value per share **below $1**. After **Lehman** Brothers went bankrupt in September 2008, the Reserve Primary Fund, which held Lehman **commercial paper**, fell to $0.97, and the resulting wave of redemptions looked like a **run** on a bank.
Keywords: share value at $1 · below $1 · Lehman · commercial paper · run

17. [Ch.2] The **principal** amount of a TIPS is **adjusted** in proportion to increases in the **Consumer Price Index**, so coupons paid on that principal rise with inflation. TIPS therefore provide a constant stream of income in **real** (inflation-adjusted) dollars.
Keywords: principal · adjusted · Consumer Price Index · real

18. [Ch.2] **General obligation bonds** are backed by the **full faith and credit** (the taxing power) of the issuer, while **revenue bonds** are issued to finance **particular projects** (airports, hospitals, turnpikes) and are backed by the **revenues from that project** or the agency operating it. Revenue bonds are **riskier** in terms of default.
Keywords: general obligation · full faith and credit · revenue bond · particular projects · revenues from that project · riskier

19. [Ch.2] A **price-weighted** index averages share prices, so a **high-priced** stock dominates it. A **value-weighted** index weights each stock by its **market capitalization**, so a **large-cap** firm dominates it. An **equally weighted** index is the simple average of each stock's return.
Keywords: price-weighted · high-priced · value-weighted · large-cap · equally weighted

20. [Ch.3] In the **primary market** a firm sells **newly issued** securities (for example an IPO) and receives the **proceeds**. Later trades between investors take place in the **secondary market** and raise no new capital, but the **liquidity** and prices it provides set the terms on which new issues can be sold.
Keywords: primary market · newly issued · proceeds · secondary market · liquidity

21. [Ch.3] In a **firm commitment** the syndicate **buys the entire issue** at a negotiated price and resells it; the spread is its pay and the **underwriter bears the risk** of a poor sale. In **best efforts** the bank only acts as an **agent** and unsold shares revert to the issuer.
Keywords: firm commitment · buys the entire issue · underwriter bears · best efforts · agent

22. [Ch.3] IPOs are on average **underpriced**: **first-day** returns are large and positive, **money left on the table** by the issuer. Explanations include compensating institutions for revealing demand in **book-building** and the **winner's curse**. Long-run returns reverse, as IPOs tend to **underperform** comparable non-issuing firms.
Keywords: underpriced · first-day · money left on the table · book-building · underperform

23. [Ch.3] The investor **buys at the asked** price ($34.90) and **sells at the bid** ($34.75). The **spread** of $0.15 is the dealer's compensation for **immediacy** and a **round-trip** transaction cost for the investor on top of commissions.
Keywords: buys at the asked · sells at the bid · spread · immediacy · round-trip

24. [Ch.3] A **market order executes immediately** at the **best available price**, so execution is certain but the price is not. A **limit order** names the **worst acceptable price**, so the price is controlled, but it **may never execute** and waits in the limit order book.
Keywords: market order · executes immediately · best available price · limit order · worst acceptable price · may never execute

25. [Ch.3] The order **cannot be filled** now; it sits in the **limit order book** and executes only if the price **falls to $47 or below** (it may expire unfilled). Such standing orders **supply liquidity** to investors who submit market orders.
Keywords: cannot be filled · limit order book · falls to $47 · supply liquidity

26. [Ch.3] The order lies **dormant** until the stock trades **at or below $55**, then becomes a **market order** to sell, which helps **limit further losses**. It does **not guarantee** the price: in a fast-falling or gapping market the sale can be filled well below $55.
Keywords: dormant · market order · at or below $55 · limit further losses · not guarantee

27. [Ch.3] High-frequency trading is **computerized** trading that exploits very **small, fleeting price discrepancies**, with positions held for **milliseconds**. Proponents say it **narrows spreads and adds liquidity**; critics note that the liquidity can be **withdrawn** under stress, as in the 2010 flash crash.
Keywords: computerized · small · milliseconds · narrows spreads · withdrawn

28. [Ch.3] Margin **magnifies both gains and losses** on the investor's equity because the loan is fixed while the collateral value moves. **Interest accrues** whatever the stock does, so the stock must earn more than the **borrowing rate** for leverage to pay. The initial margin requirement is set by the **Federal Reserve**, and brokers may be stricter.
Keywords: magnifies · gains and losses · interest accrues · borrowing rate · Federal Reserve

29. [Ch.3] For a **long** margin purchase the loan is fixed, so a **price fall** cuts equity below the **maintenance margin**; for a **short** position the liability grows when the **price rises**, so the call comes on a rise. In both cases the broker demands **additional funds** or sells shares.
Keywords: long · price fall · short · price rises · maintenance margin · additional funds

30. [Ch.3] The short seller **borrows** shares and **sells** them, then **buys back** later to return them, profiting if the **price falls**. The gain is **capped** because the price cannot fall below zero, while the loss is **unlimited** because the price can rise without bound.
Keywords: borrows · sells · buys back · price falls · capped · unlimited

31. [Ch.3] The sale **proceeds stay in the account as collateral**, and the short seller must post **additional margin**. The lender keeps the economic rights of ownership, so the short seller must **pay the lender any dividends** the stock pays, a **carrying cost** of the trade.
Keywords: collateral · additional margin · dividends · pay the lender · carrying cost

32. [Ch.3] ECNs (electronic communication networks) match buy and sell orders directly with **no dealer** in the middle. **Algorithms** decide the timing, price and size of orders to cut cost and market impact, and ultra-fast high-frequency programs exploit tiny price gaps in **milliseconds**. The benefits are faster execution, **lower trading costs**, greater price transparency, anonymity for large orders and trading beyond exchange hours.
Keywords: ECN · no dealer · algorithm · milliseconds · lower trading costs

33. [Ch.3] The **Securities Acts of 1933** and 1934 require **full disclosure** at issue and created the **SEC** to oversee markets. **Self-regulation (FINRA)** means industry bodies police member conduct and trading practices. **Sarbanes-Oxley** (2002) tightened corporate governance, auditing and financial reporting after major scandals. **Insider-trading** rules prohibit trading on **material non-public information**.
Keywords: 1933 · full disclosure · SEC · FINRA · Sarbanes-Oxley · material non-public information

34. [Ch.4] She owns a **proportional (pro-rata) claim** on the **whole pooled portfolio**, 2% of every asset, and receives 2% of its **income**. She does **not own specific securities**, and no return is **guaranteed**.
Keywords: proportional · whole pooled portfolio · income · specific securities

35. [Ch.4] The four services are **record keeping and administration**, **diversification and divisibility**, **professional management**, and lower **transaction costs** from trading in large blocks. A **guaranteed** return above the market is not among them; whether management beats the market is an empirical question.
Keywords: record keeping · diversification · professional management · transaction costs · guaranteed

36. [Ch.4] NAV = (**market value of assets** − **liabilities**, such as accrued management fees) ÷ **shares outstanding**. New shares are issued **at NAV**, so assets and shares rise in **proportion** and NAV is unchanged, with **no dilution** of existing holders.
Keywords: market value of assets · liabilities · shares outstanding · at NAV · proportion

37. [Ch.4] It is filled at the **NAV struck once a day at the 4:00 p.m. close**, the **same price** for every order that day, and the investor does **not know** it when ordering. Closed-end funds and ETFs trade on an **exchange** at intraday market prices.
Keywords: NAV · 4:00 p.m. · same price · not know · exchange

38. [Ch.4] A closed-end fund has a **fixed number** of shares after its **IPO** and neither issues nor redeems them. Investors trade with each other on an **exchange**, so the price is set by **supply and demand** and can sit at a **premium or discount** to NAV.
Keywords: fixed number · IPO · exchange · supply and demand · premium or discount

39. [Ch.4] A UIT is a **fixed, unmanaged** portfolio deposited in a trust by a **sponsor** and sold as redeemable units; it **terminates on a set date**. With essentially **no trading** after formation, **management fees are very low**.
Keywords: fixed · terminates · sponsor · no trading · very low

40. [Ch.4] Hedge funds are private pools open only to **accredited and institutional** investors and largely **exempt from the Investment Company Act** of 1940. They use **short sales, derivatives and heavy leverage**, charge higher fees (**2 and 20**), and restrict redemptions through **lock-ups**.
Keywords: accredited · Investment Company Act · short sales · 2 and 20 · lock-up

41. [Ch.4] A **balanced fund** holds **stocks and bonds** in a relatively **stable mix**; an **asset allocation fund** actively **shifts** the mix across asset classes as market conditions change.
Keywords: balanced fund · stocks and bonds · stable mix · asset allocation fund · shifts

42. [Ch.4] Operating expenses are **deducted daily from fund assets** before NAV is struck, so the investor is **never billed** separately and the cost is easy to overlook. The reported return is already **net of** the expense ratio.
Keywords: deducted · daily · from fund assets · never billed · net of

43. [Ch.4] A **12b-1 fee** is an **annual** charge of up to **1%** of assets that pays for **marketing and distribution** (including adviser compensation). It is reported inside the **expense ratio**, but economically it is a **load** paid in instalments.
Keywords: annual · marketing and distribution · 1% · expense ratio · load

44. [Ch.4] A **CDSC** is charged on redemption and **declines** each year until it disappears. It makes **early redemption** expensive and so **locks investors in**; it compensates the **selling broker**, not the fund.
Keywords: CDSC · declines · early redemption · locks investors in · selling broker

45. [Ch.4] No: the reported return **overstates** her return. Published returns are computed from **NAV**, which is net of the **expense ratio** but **not of loads**; the load **reduces the amount invested**, so over one year a 5% load can consume almost half of an 11% return.
Keywords: overstates · not of loads · NAV · expense ratio · reduces the amount invested

46. [Ch.4] The load is a **one-time** cost but the higher expense ratio is paid **every year** and compounds. Over a **short** horizon the load dominates, so **Class C** wins; over a **long** horizon the annual fee dominates, so **Class A** wins.
Keywords: Class A · Class C · long · short · one-time · every year

47. [Ch.4] The **authorized participant buys the cheaper basket**, swaps it **in kind** for newly **created** ETF shares, and **sells the ETF shares** on the exchange. The selling pressure pushes the ETF price **back toward the basket value**.
Keywords: authorized participant · buys the cheaper basket · in kind · sells the ETF shares · back toward the basket value

48. [Ch.4] In a **mutual fund**, **redemptions force** the fund to **sell stock**, realizing gains that are distributed to all holders. In an **ETF** the seller finds a buyer on the **exchange**, and creations and redemptions happen **in kind**, not for cash, so the fund does not need to sell.
Keywords: mutual fund · redemptions force · sell stock · ETF · exchange · in kind

49. [Ch.4] It aims for **twice the index's daily return**; the fund **resets** to 2× every day, so gains and losses **compound** on a changing base. In **volatile** markets it falls behind even if the index ends flat, so multi-period returns are not 2×.
Keywords: daily · twice · resets · compound · volatile

50. [Ch.4] The fund **sells call options** on shares it owns and distributes the **premium** received. That premium is the price of the **upside it gives away** (gains are capped above the strike) while the **downside remains open**; it is an option strategy, **not a dividend** and not insurance.
Keywords: sells call options · premium · upside · downside · not a dividend

51. [Ch.4] Passive investors hold the market, so what remains for **active investors as a group** is also the market: their value-weighted average **gross return** equals the market return. After **fees** the average active dollar must **lag** the market. This says nothing about any single fund.

52. [Ch.4] Past top performance is **not a reliable** signal of future top performance: if **skill** drove the rankings, winners would repeat more often than **chance**, not less. The evidence shows no **persistence**; the lasting signal is the **fee**, not past returns.
Keywords: persistence · not a reliable · chance · skill · fee

53. [Ch.4] A front-end **load** is deducted before investing, so a 5.75% load on $10,000 leaves only **$9,425** as the **net** amount invested. The load is 5.75% of the gross, but to get back to $10,000 you must earn **6.10%** on the net, i.e. you need more than the load percentage just to **break even**.
Keywords: load · net · 9,425 · 6.10% · break even

54. [Ch.5] HPR = (ending price − beginning price + dividends) ÷ beginning price, the **capital gain yield** plus the **dividend yield**. The denominator is always the **purchase price**, **never the ending price**.
Keywords: capital gain yield · dividend yield · purchase price · never the ending price

55. [Ch.5] The **arithmetic** average is the simple mean of period returns; the **geometric** average is the constant **compound** growth rate that reproduces the actual ending wealth. The geometric average is **smaller** than the arithmetic one when returns fluctuate, by roughly $\frac{1}{2}\sigma^2$, and the gap grows with **volatility**.
Keywords: arithmetic · geometric · compound · smaller · volatility

56. [Ch.5] The **dollar-weighted** return is the **IRR** of all **cash flows** (purchases, dividends, sales); the time-weighted return is the **geometric** average of **sub-period** returns. They **differ** when money is added or withdrawn at different times, and the time-weighted figure ignores that timing, so it is better for judging the **manager**.
Keywords: IRR · cash flows · geometric · differ · manager

57. [Ch.5] For a given APR, **more frequent** compounding gives a **higher EAR**, $\text{EAR} = (1 + \text{APR}/m)^m - 1$. With **continuous** compounding the APR equals the **natural log** of (1 + EAR).
Keywords: EAR · more frequent · higher · continuous · natural log

58. [Ch.5] The exact relation is the **Fisher** equation, $1 + r = (1 + R) \div (1 + i)$. The shortcut $R - i$ is an **approximation** that **overstates** the real return by the **cross term** $r \times i$, and the error grows when inflation is high.
Keywords: Fisher · approximation · overstates · cross term

59. [Ch.5] VaR is the loss that is **exceeded** with a given **probability** over the horizon; here the loss should exceed ₩40 million on about **1 trading day in 100**. It is **not the maximum** loss: on the worst days losses can be **larger**, and VaR says nothing about how much larger.
Keywords: probability · exceeded · 1 trading day in 100 · not the maximum · larger

60. [Ch.5] Under normal returns $\mathbf{VaR = V_0 \times z \times \sigma \times \sqrt{T}}$: the one-day figure is scaled by the **square root** of time because independent daily variances add, **not multiplied by T**, which would overstate the risk. The **z-value** reflects the **confidence level** (1.645 for 95%, 2.326 for 99%) and σ is the daily **volatility**.
Keywords: square root · not multiplied by T · z-value · volatility

61. [Ch.5] It is **unchanged** for every y: both the **risk premium** and the **standard deviation** of the complete portfolio **scale** with y, so their ratio equals the **slope** of the capital allocation line. This is why the choice of P and the choice of y can be solved separately.
Keywords: unchanged · risk premium · standard deviation · scale · slope

62. [Ch.5] Investors need compensation for expected inflation, so the **nominal** rate **rises** with **expected inflation**: $r_{nom} = r_{real} + E(i)$. The increase in the nominal rate is needed to **preserve** investors' **real** rate of return.
Keywords: expected inflation · nominal · real · rises · preserve

63. [Ch.5] Skew measures the **asymmetry** of the return distribution. **Kurtosis** compares the frequency of **extreme** values with that of the **normal distribution** (whose kurtosis is zero), so it is called a measure of **fat tails**, which signal tail risk that the standard deviation alone understates.
Keywords: skew · asymmetr · kurtosis · fat tails · extreme · normal distribution

64. [Ch.5] Historical simulation sorts the **actual past returns** and reads off the quantile; **no distribution** is assumed so **fat tails** come for free, but it needs a long sample. **Monte Carlo** simulation draws many paths from an **assumed model**, handles **options** and other non-linear payoffs, but carries model risk.
Keywords: historical simulation · actual past returns · no distribution · fat tails · Monte Carlo · assumed model · options

65. [Ch.5] VaR marks the **cut-off** but says nothing about how bad the loss is beyond it. ES (also CVaR) is the **average loss** in the tail **beyond** VaR. ES is **coherent** (subadditive) where VaR is not, so merging books cannot manufacture risk, and **Basel FRTB** replaced the 99% VaR capital charge with **97.5%** expected shortfall.
Keywords: cut-off · average loss · beyond · coherent · Basel · 97.5%

66. [Ch.5] The Sharpe ratio is the **reward per unit** of volatility, $[E(r_P) - r_f] / \sigma_P$. It equals the **slope of the CAL** and is **unchanged** however much of the risk-free asset is mixed in, so it measures the quality of the risky portfolio itself. It is the wrong tool for a **single security** because the denominator includes **diversifiable risk**.
Keywords: reward per unit · slope of the CAL · unchanged · single security · diversifiable risk

67. [Ch.5] A **risk-averse** investor has $A > 0$, a **risk-neutral** investor has $A = 0$ and judges by E(r) alone, and a **risk-loving** investor has $A < 0$. The term $\frac{1}{2}A\sigma^2$ is the **penalty** for risk. The **certainty equivalent** is the rate that a risk-free investment would have to offer to give the same utility as the risky portfolio.
Keywords: risk-averse · risk-neutral · risk-loving · penalty · certainty equivalent

68. [Ch.5] y^* **maximizes utility** and **rises with the risk premium**, **falls with risk aversion** and **falls with variance**: a larger reward per unit of risk raises the risky share, while a larger risk penalty lowers it.
Keywords: maximizes utility · rises with the risk premium · falls with risk aversion · falls with variance

69. [Ch.5] To the left of P the investor **lends** at the risk-free rate. Beyond P she must borrow at a rate above r_f, and because the **borrowing rate is higher** the **slope falls** (the Sharpe ratio of leveraged positions is lower), so the CAL has a **kink** at P.
Keywords: lends · borrowing rate is higher · slope falls · kink

70. [Ch.5] When a **broad market index** is used as the risky portfolio P, the resulting capital allocation line is the **capital market line** (CML). Passive investing is defended by the **free-rider** argument (the analysis of thousands of other investors is already in the price) and by **low cost** (index funds charge about 0.03-0.2% a year against 1% or more for active funds). The open question is **who sets prices** if everyone is passive, the **Grossman-Stiglitz** paradox.
Keywords: capital market line · broad market index · free-rider · low cost · who sets prices · Grossman

71. [Ch.6] Total variance is **systematic** plus **firm-specific** variance. Adding stocks shrinks **firm-specific** (diversifiable) risk toward zero but leaves systematic (market) risk. Prices are set by **diversified investors**, so only the systematic part earns a **risk premium**.
Keywords: systematic · firm-specific · diversified investors · risk premium

72. [Ch.6] As n grows the firm-specific term $(1/n)\sigma^2$ vanishes and σ_P **falls toward a floor** equal to the square root of the **average covariance**. The floor is **market (systematic) risk**, which **cannot be diversified** away.
Keywords: falls · floor · average covariance · market risk · cannot be diversified

73. [Ch.6] The **interaction (covariance) term** $2w_S w_B \cdot \text{Cov}$ is the only one that can be **negative**; the other two terms are **squares** and always positive. When the **correlation** is below +1 it shrinks the variance, and for $\rho < 0$ it pulls σ_P below the **weighted average** of the two standard deviations.
Keywords: interaction term · negative · correlation · weighted average · squares

74. [Ch.6] When the assets are **perfectly negatively correlated** ($\rho = -1$) a **zero-risk** portfolio exists: setting $\sigma_P = |w_S\sigma_S - w_B\sigma_B| = 0$ gives $w_S = \sigma_B \div (\sigma_S + \sigma_B)$, so the weights are set by the **standard deviations**.
Keywords: perfect negative correlation · zero risk · standard deviations

75. [Ch.6] Risk falls when the **correlation** is **below $\sigma_B \div \sigma_S$** (here $10 \div 20 = \mathbf{0.5}$), the point at which the minimum-variance stock weight becomes positive. A negative correlation is **not required**.
Keywords: correlation · below $\sigma_B \div \sigma_S$ · 0.5 · not required

76. [Ch.6] The minimum-variance portfolio only minimizes risk and **ignores expected return**. The optimal risky portfolio is the **tangency** point where the **capital allocation line** is steepest, so it maximizes the **Sharpe ratio**, which is **at least as high** as that of any other mix, including the minimum-variance portfolio.
Keywords: ignores expected return · Sharpe ratio · tangency · capital allocation line · at least as high

77. [Ch.6] One optimal risky portfolio is the same for every investor who shares the same inputs; only the **split** between it and **T-bills** depends on **risk aversion**. The property breaks down with **different inputs, borrowing rates above lending rates, taxes** and human capital.
Keywords: same for every investor · risk aversion · split · T-bills · different inputs

78. [Ch.6] A **risk-tolerant** (low risk aversion) investor gets $y^* = [E(r_P) - r_f] \div (A\sigma_P^2)$ **above 1**, so she **borrows** at the **risk-free rate** and invests more than her wealth in P, a **leveraged** position.
Keywords: low risk aversion · above 1 · borrows · risk-free rate · leveraged

79. [Ch.6] With n securities you must estimate **expected returns** for each, **n variances** and $n(n-1)/2$ **covariances**, a total of $n(n+3)/2$ that **grows with the square of n** (125,750 inputs for n = 500). Every estimate carries **estimation error**.
Keywords: expected returns · n variances · covariances · square of n · estimation error

80. [Ch.6] "Risk reduction by **spreading exposure** across many **independent risk sources** is called the **insurance principle**." An employee who also holds company stock in her 401(k) is **doubling down** on the fortunes of a **single firm**, since her job and her savings depend on the same firm.
Keywords: insurance principle · spreading exposure · independent risk sources · single firm · doubling down

81. [Ch.6] Covariance measures how much two returns move together, but its units are "**percent square**," so its size is hard to interpret. The **correlation coefficient** rescales it by the **product of the standard deviations**, $\rho = \text{Cov} \div (\sigma_S\sigma_B)$, so $\text{Cov} = \rho\sigma_S\sigma_B$, and it lies **between −1 and +1**.
Keywords: covariance · percent square · correlation coefficient · product of the standard deviations · between −1 and +1

82. [Ch.6] The portfolio's expected return (8.00%) is exactly the **weighted average** of the two funds' returns, so you **lost nothing** in return. Yet the portfolio standard deviation (9.57%) is far **below** the weighted average of the two (12.64%). That 3.07-point gap is the **diversification benefit**, and it exists purely because the **correlation** is below +1.
Keywords: weighted average · lost nothing · below · correlation · diversification benefit

83. [Ch.6] Perfect positive correlation ($\rho = +1$) is the **only** case with no benefit from diversification: $\sigma_P = w_B\sigma_B + w_S\sigma_S$ is a **weighted average** of the component standard deviations and the opportunity set is a **straight line** through the two assets. Whenever returns are **less than perfectly** positively correlated, σ_P is below that weighted average.
Keywords: perfect positive correlation · only · weighted average · straight line · less than perfectly

84. [Ch.6] The **upward-sloping** part of the opportunity set, starting at the **minimum-variance portfolio**, is the **efficient** part. Points below it are **dominated**, because a portfolio on the efficient part offers a **higher expected return** at the same standard deviation.
Keywords: efficient · upward-sloping · minimum-variance portfolio · dominated · higher expected return

85. [Ch.6] The **efficient frontier** is the set of portfolios that offer the **highest expected return** for each level of risk. It is found by **minimizing variance** subject to each **target expected return**. Individual assets all lie **inside** the frontier, so every one of them is dominated.
Keywords: efficient frontier · highest expected return · minimizing variance · target expected return · inside

86. [Ch.6] Step 1 collects the **inputs**: expected returns, standard deviations and covariances or correlations. Step 2 traces the **efficient frontier** (upper branch). Step 3 **maximizes the Sharpe ratio** to find the optimal risky portfolio P*. Step 4 sets **y*** from the investor's **risk aversion** and splits wealth between P* and the risk-free asset. Only step 4 depends on the investor; steps 1-3 are identical for everyone, the **separation property**.
Keywords: inputs · efficient frontier · Sharpe ratio · y* · separation property

87. [Ch.6] Investors everywhere **overweight** their home country relative to its share of the world portfolio, a pattern called **home bias**. Adding foreign assets lowers the **covariance** term in the portfolio variance because countries' systematic shocks differ. The caveat is that global **correlations** have risen and **currency** risk adds another layer, yet the benefit is not zero as long as correlations stay **below +1**.
Keywords: home bias · overweight · covariance · correlations · below +1 · currency

Part A: Econometrics 1–3 (Professor's Slides)

Multiple Choice

1. An econometrician wants to learn the average hourly wage of all workers in Korea and collects data on 2,000 randomly selected workers. Which statement is correct?
a. A sample is used only when the population is hypothetical
b. The 2,000 workers are the population, because they are the units actually analyzed
c. The 2,000 workers are the sample, and the estimates are statements about what the population of all Korean workers is like
d. The estimates describe only the 2,000 workers and say nothing about other workers

2. Which statement best defines external validity?
a. Whether the OLS estimator is unbiased within the sample used
b. Whether the sample size is large enough to compute standard errors
c. Whether the R-squared of the regression is high
d. The validity of applying the conclusions of a study outside the context of that study

3. A researcher surveys only students at one private university in Seoul to estimate the average monthly smartphone spending of all Korean adults. What is the main concern?
a. OLS cannot be computed because the population is too large
b. The sample is unlikely to be representative of the population, so the conclusions may have weak external validity
c. The estimates are automatically unbiased for all Korean adults because the students were chosen at random within the university
d. The population becomes hypothetical, so no estimate is possible

4. Consider the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u. How should β_1 be interpreted?
a. The elasticity of price with respect to subsidy
b. The slope from a simple regression of price on subsidy
c. The change in log(price) from a one-unit increase in subsidy, holding log(sunlight) fixed (a partial slope)
d. The total change in price from a one-unit increase in subsidy, including any change in sunlight it causes

5. In the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, which statement about subsidy is correct?
a. Subsidy has a linear relationship with price, but not with log(price)
b. Subsidy has a linear relationship with log(price), but not with price
c. Subsidy has a linear relationship with both price and log(price)
d. Subsidy has a linear relationship with neither price nor log(price)

6. In the model log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, which statement about sunlight is correct?

a. β_2 is the % change in price for a one-unit increase in sunlight

b. β_2 is the change in log(price) for a 1% increase in price

c. log(sunlight) has a linear relationship with log(price), so β_2 is approximately the % change in price for a 1% increase in sunlight, holding subsidy fixed

d. Sunlight has a linear relationship with price, so β_2 is the change in price for one more unit of sunlight

7. Suppose log(price) = β_0 + β_1 battery + β_2 battery^2 + u with $\beta_2 \neq 0$. Which statement is correct?

a. β_1 alone is the constant slope of battery for every value of battery

b. The model violates the assumption of linearity in parameters, so OLS cannot be used

c. Battery has a linear relationship with log(price) because the model is linear in parameters

d. Battery does not have a linear relationship with log(price); the effect of one more unit of battery depends on the current level of battery

8. Suppose log(price) = β_0 + β_1 battery + β_2 sunlight + β_3 battery× sunlight + u with $\beta_3 \neq 0$. Holding sunlight fixed, what is the effect on log(price) of one more unit of battery?

a. β_3· battery

b. β_1

c. β_1 + β_3

d. β_1 + β_3· sunlight

9. Which statement describes random sampling?

a. Every element has an equal probability of being assigned to the treatment or control group

b. The researcher chooses units that look typical of the population

c. Every element in the population has an equal probability of being selected into the sample

d. The sample contains more than half of the population

10. A researcher has a random sample of 500 workers and then flips a coin for each worker to decide who joins a job training program. The coin flip is an example of:

a. Random sampling

b. Perfect collinearity

c. Partialling out

d. Random assignment

11. Consider Model 2: wage = $\beta_0 + \beta_1$ educ + u, where ability is not included. People with better ability tend to get more education. Why can the OLS estimate of β_1 be biased even with a random sample?
a. There is no sample variation in educ
b. Ability goes into u and is correlated with educ, so the zero conditional mean condition E(u|educ) = 0 fails
c. Random sampling guarantees unbiasedness, so the estimate cannot be biased
d. The homoskedasticity assumption fails, which always causes bias

12. In Model 2, wage = $\beta_0 + \beta_1$ educ + u (ability omitted), suppose ability raises wage and is positively correlated with education. The OLS estimate of β_1 tends to:
a. Have a bias whose sign can never be determined
b. Understate the true return to education (downward bias)
c. Be unbiased, because ability is unobserved
d. Overstate the true return to education (upward bias)

13. The slides state: "OLS requires random sampling. It may also require random assignment." When estimating the effect of education on wages with Model 2 (wage = $\beta_0 + \beta_1$ educ + u), why would random assignment help?
a. Random assignment removes the need for any error term
b. Random assignment increases the R-squared of the regression
c. If educ were randomly assigned, it would be unrelated to ability and other factors in u, so the zero conditional mean condition would hold
d. Random assignment makes the sample representative of the population

Part A: Econometrics 1–3 (Professor's Slides) — Answers

1. c
표본은 분석을 위해 선택된 모집단의 부분집합이며, 표본을 연구하는 목적은 모집단이 어떠한지 알아내는 것이다. 추정치는 표본을 바탕으로 한 '모집단에 대한' 판단이다.

2. d
외부 타당성(external validity)은 연구의 결론을 그 연구 맥락 밖에 적용하는 것이 타당한지를 말한다. 표본이 모집단을 대표하는지와 직결된다.

3. b
한 대학 학생들은 한국 성인 전체를 대표하지 못할 가능성이 크다. 표본이 모집단을 대표하는지 따져봐야 하며, 그렇지 않으면 결론을 모집단에 적용하기 어렵다(외부 타당성 약함).

4. c
다중회귀에서 β_1은 단순 '기울기'가 아니라 다른 설명변수(log(sunlight))를 고정한 상태에서의 '부분 기울기(partial slope)'이다. 종속변수가 log(price)이므로 단위도 log(price) 기준이다.

5. b
log(price)는 subsidy의 1차식이므로 선형관계이다. 그러나 price = exp($\beta_0 + \beta_1$ subsidy + ...)이므로 price와 subsidy는 선형관계가 아니다.

6. c
log-log 형태: log(sunlight)와 log(price)는 선형관계이고, β_2는 탄력성(1% 증가 → 약 β_2% 변화)이다. sunlight와 price 자체는 선형관계가 아니다.

7. d
$\Delta \log(\text{price})/\Delta \text{ battery} \approx \beta_1 + 2\beta_2$ battery 로 battery 수준에 따라 달라지므로 선형관계가 아니다. 단, 모델은 '모수에 대해' 선형이므로 OLS로 추정 가능하다.

8. d
battery와 sunlight가 상호작용하므로 battery의 효과는 $\beta_1 + \beta_3 \cdot$ sunlight 로 sunlight 수준에 따라 달라진다. 따라서 β_1만으로는 battery의 기울기가 아니며, battery와 log(price)는 선형관계가 아니다.

9. c
무작위 추출(random sampling): 모집단의 모든 원소가 표본에 뽑힐 확률이 같다. 집단 배정 확률이 같은 것은 무작위 배정(random assignment)이다.

10. d
표본을 뽑는 것은 random sampling, 뽑힌 사람을 동전 던지기로 처치/통제 집단에 나누는 것은 random assignment이다.

11. b
능력이 모델에서 빠지면 오차 u에 들어간다. 능력과 교육이 상관되어 있으면 E(u|educ)=0 (zero conditional mean)이 깨져 random sampling이어도 β_1이 편의된다.

12. d
누락변수 편의 = (능력의 효과 β_2) × (educ와 abil의 상관 방향). 둘 다 양(+)이므로 편의도 양(+), 즉 과대추정된다.

13. c
무작위 배정이면 교육 수준이 능력 등 오차 속 요인과 체계적으로 관련되지 않으므로 E(u|educ)=0이 성립하고 β_1은 불편이 된다. (대표성은 random sampling의 역할)

Part B: Ch.1 The Nature of Econometrics and Economic Data

Multiple Choice

1. A data set records the crime rate and the number of police officers for the same 150 cities in both 2020 and 2023. What type of data is this?
a. Pooled cross sections
b. Cross-sectional data
c. Panel (longitudinal) data
d. Time series data

2. A researcher draws a random sample of house prices in 1993 and a new, independent random sample of house prices in 1995 to evaluate a property tax change. What type of data is this?
a. Panel data
b. A single cross section
c. Time series data
d. Pooled cross sections

3. Which feature is typical of time series data but NOT of a cross section drawn by random sampling?
a. Observations are typically serially correlated, and their ordering conveys important information
b. Observations are more or less independent of each other
c. The ordering of observations does not matter
d. All observations are collected at the same point in time

Part B: Ch.1 The Nature of Econometrics and Economic Data — Answers

1. c
같은 단위(도시)를 시간에 걸쳐 반복 관측하면 패널 데이터이다. 시간불변의 관측되지 않는 특성을 통제할 수 있다.

2. d
서로 독립적으로 뽑은 두 시점의 횡단면을 합친 것이 pooled cross section이다. 같은 집을 추적하지 않으므로 패널이 아니다. 정책 변화 전후 비교에 자주 쓰인다.

3. a
시계열은 관측치 간 계열상관이 있고 순서가 중요한 정보다(추세, 계절성). 무작위 추출 횡단면은 관측치가 대체로 독립이며 순서가 무의미하다.

Part C: Ch.2 The Simple Regression Model

Multiple Choice

1. In the simple regression $y = \beta_0 + \beta_1 x + u$, the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to:
a. Minimize the sum of squared residuals
b. Make the fitted value equal to y for every observation
c. Minimize the sum of residuals
d. Minimize the sum of absolute residuals

2. In a simple regression of y on x, $R^2 = 0.25$. Which statement is correct?
a. 25% of the observations lie exactly on the regression line
b. 25% of the sample variation in y is explained by x
c. x causes 25% of the change in y
d. The estimated slope equals 0.25

3. The fitted regression is log(wage) = 0.584 + 0.083educ. How should 0.083 be interpreted?
a. A 1% increase in education raises the wage by 8.3%
b. One more year of education is associated with an approximately 8.3% higher wage
c. One more year of education raises the wage by 0.083%
d. One more year of education raises the wage by $0.083

4. Real earnings (in $1,000s) are regressed on train (= 1 if the person joined a randomly assigned job training program, 0 otherwise): earn = 4.55 + 1.79train. What does 1.79 measure?
a. Nothing, because OLS cannot be used when x is binary
b. Participants earn about 1.79% more than non-participants
c. The average earnings of participants are $1,790
d. The average treatment effect: participants earn about $1,790 more on average than non-participants

5. Under SLR.1–SLR.5, $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$. Which change makes $\hat{\beta}_1$ more precise (smaller variance)?
a. More sample variation in x
b. Fewer observations
c. A larger error variance σ^2
d. Replacing homoskedasticity with heteroskedasticity

Part C: Ch.2 The Simple Regression Model — Answers

1. a
OLS(최소제곱법)는 잔차 제곱합 $\Sigma \hat{u}_i^2$ 를 최소화한다. 그 결과 잔차의 합은 항상 0이 된다(대수적 성질).

2. b
$R^2 = SSE/SST = 1 - SSR/SST$: y의 표본 변동 중 x로 설명되는 비율이다. 인과관계나 기울기 크기를 뜻하지 않는다.

3. b
log-level(semi-log) 형태: x 1단위 증가 → y 약 $100\beta_1$ % 변화. $0.083 \times 100 = 8.3\%$.

4. d
x가 0/1이면 기울기 = 두 집단 평균 차이. 무작위 배정이면 OLS는 처치효과의 불편추정량이다. 비참가자 평균은 $4,550이며, 1,790은 약 39.3% 증가에 해당한다.

5. a
분산은 오차분산 σ^2이 클수록 커지고, x의 표본변동 SST_x가 클수록 작아진다.

Part D: Ch.3 Multiple Regression Analysis: Estimation

Multiple Choice

1. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the OLS estimate $\hat{\beta}_1$ can also be obtained by:
a. Regressing y on x_2 first, then adding x_1
b. Regressing x_1 on x_2 (with a constant), then regressing y on the residuals from that regression
c. Regressing x_2 on the residuals from a regression of y on x_1
d. Averaging the slopes of the two simple regressions

2. Which regression violates MLR.3 (no perfect collinearity)?
a. log(salary) on log(sales) and ceoten
b. wage on educ and $educ^2$
c. wage on educ and exper, where their correlation is 0.9
d. wage on an intercept, male (= 1 if male) and female (= 1 if female)

3. The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, but x_2 is omitted and y is regressed on x_1 only. In which case is the simple regression slope unbiased for β_1?
a. When the sample size is very large
b. When $\beta_2 = 0$ or when x_1 and x_2 are uncorrelated
c. When the R-squared is high
d. When x_2 is positively correlated with x_1 and $\beta_2 > 0$

4. Using 506 communities, the simple regression gives log(price) = 11.71 − 1.043 log(nox), while adding rooms gives log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms. rooms raises price and is negatively correlated with log(nox). Which statement is correct?
a. The simple regression is biased upward (toward zero), so −1.043 understates the true effect
b. Adding rooms must always make the nox elasticity more negative
c. The simple-regression elasticity −1.043 is biased downward (too negative), as predicted by $\beta_2 > 0$ and Corr(log(nox), rooms) < 0
d. The two estimates differ only because of sampling error

5. High (but not perfect) correlation between x_1 and x_2 in a multiple regression mainly:
a. Violates MLR.3, so OLS cannot be computed
b. Increases the sampling variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ but does not cause bias
c. Causes $\hat{\beta}_1$ and $\hat{\beta}_2$ to be biased
d. Inflates the variance of every coefficient, including an unrelated x_3

6. The Gauss–Markov theorem states that under MLR.1–MLR.5, the OLS estimators are:

a. Exactly normally distributed in any sample

b. The only unbiased estimators of the coefficients

c. The best estimators even when the errors are heteroskedastic

d. BLUE: they have the smallest variance among all linear unbiased estimators

Part D: Ch.3 Multiple Regression Analysis: Estimation — Answers

1. b
partialling out: x_1에서 x_2로 설명되는 부분을 제거한 잔차(x_2와 무상관인 부분)에 y를 회귀하면 그 기울기가 β_1이다.

2. d
절편이 있을 때 male + female = 1 이므로 정확한 선형관계 → 완전공선성. $educ^2$는 선형관계가 아니고, 높은 상관(0.9)은 다중공선성이지 MLR.3 위반이 아니다.

3. b
누락변수 편의 = $\beta_2 \times \delta_1$. 누락변수가 무관하거나 ($\beta_2=0$) x_1과 무상관($\delta_1=0$)이면 편의가 없다.

4. c
누락변수 편의의 부호 = (rooms의 효과 β_2, +) × (log(nox)와 rooms의 상관, −) = 음(−) → 하향편의. 실제로 −1.043이 −0.718보다 더 음수이다. (Wooldridge 연습문제, HPRICE2)

5. b
다중공선성은 관련된 변수의 분산만 키우고(VIF로 진단) 불편성은 유지된다. 엄밀한 의미의 MLR.3 위반이 아니다.

6. d
MLR.1–5 하에서 OLS는 선형 불편추정량 중 분산이 가장 작다(BLUE). 이분산이 있으면 더 나은 추정량이 존재한다. 정규성은 MLR.6이 필요하다.

Part E: Ch.4 Multiple Regression Analysis: Inference

Multiple Choice

1. In a regression, $\hat{\beta}_j = 0.80$ with $se(\hat{\beta}_j) = 0.10$. What is the t statistic for H_0: $\beta_j = 1$?
a. 8.0
b. −2.0
c. 2.0
d. −0.2

2. In a two-sided t test, the p-value is 0.07. Which statement is correct?
a. H_0 is rejected at the 10% level but not at the 5% level
b. H_0 is rejected at the 5% level
c. H_0 is rejected at the 1% level
d. The probability that H_0 is true is 7%

3. The 95% confidence interval for β_1 is [0.02, 0.10]. Which statement is correct?
a. In repeated samples, intervals constructed this way contain the true β_1 95% of the time, and H_0: $\beta_1 = 0$ is rejected at the 5% level (two-sided)
b. There is a 95% probability that $\hat{\beta}_1$ lies in [0.02, 0.10]
c. β_1 equals 0.06 with 95% certainty
d. H_0: $\beta_1 = 0$ cannot be rejected at the 5% level

4. To test H_0: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$ jointly, which statement is correct?
a. The F statistic can be negative when the restrictions help the fit
b. Three separate t tests always give the same conclusion as the F test
c. The restricted model always has a smaller SSR than the unrestricted model
d. Use an F test comparing the SSR of the restricted and unrestricted models; F is large when the SSR rises a lot after imposing the restrictions

5. Which can cause the usual OLS t statistics to be invalid (not t-distributed under H_0)?
a. A sample correlation of 0.95 between two explanatory variables in the model
b. A low R-squared
c. Heteroskedasticity, or omitting an important explanatory variable
d. Having 500 observations instead of 5,000

6. For 32 chemical firms, rdintens = 0.472 + 0.321 log(sales) + 0.050 profmarg, where rdintens is R&D as a percentage of sales. If sales increase by 10%, rdintens changes by about:

a. 3.21 percentage points

b. 0.032 percentage points

c. 32.1 percent

d. 0.321 percentage points

Part E: Ch.4 Multiple Regression Analysis: Inference — Answers

1. b
$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j) = (0.80 - 1)/0.10 = -2.0$. 가설값을 빼고 표준오차로 나눈다.

2. a
p-value는 H_0를 기각할 수 있는 가장 작은 유의수준이다. $p < \alpha$ 일 때만 기각: $0.07 < 0.10$ 이지만 $0.07 > 0.05$.

3. a
신뢰구간의 경계가 확률적이다(모수가 아님). 0이 95% 구간 밖에 있으므로 5% 양측검정에서 $\beta_1 = 0$을 기각한다.

4. d
$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-k-1)]$. 제약을 가하면 SSR은 커지기만 하므로 $F \geq 0$이며, 개별 t검정과 결합검정 결론은 다를 수 있다.

5. c
이분산은 MLR.5, 중요한 변수의 누락은 MLR.4(zero conditional mean) 위반이므로 t분포가 성립하지 않는다. 설명변수 간 높은 상관(0.95)은 CLM 가정 위반이 아니다(완전공선성만 금지). (Wooldridge 연습문제 4.1)

6. b
level-log 형태: $\Delta y \approx (\beta_1/100)(\%\Delta x) = (0.321/100)\times 10 \approx 0.032$. rdintens는 퍼센트 단위이므로 '퍼센트포인트' 변화이다. (Wooldridge 연습문제, RDCHEM)

Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics

Multiple Choice

1. Which statement about the consistency of OLS is correct?
a. Consistency requires the errors to be normally distributed (MLR.6)
b. Consistency requires homoskedasticity (MLR.5)
c. A consistent estimator is unbiased in every finite sample
d. Under MLR.1–MLR.4, the distribution of $\hat{\beta}_j$ collapses to the true β_j as $n \to \infty$, i.e. plim $\hat{\beta}_j = \beta_j$

2. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (pctstck = % of pension in stocks, funds = number of mutual funds offered, risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, the inconsistency in $\tilde{\beta}_1$ is:
a. Zero, because the full model satisfies MLR.1–MLR.4
b. Negative, because risktol is omitted
c. It vanishes as the sample size grows
d. Positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1 > 0$, since $\beta_2 > 0$ and funds and risktol are positively correlated

3. cigs is the average number of cigarettes smoked per day. Can cigs have a normal distribution in the adult population, and what does this mean for OLS?
a. Yes, by the central limit theorem every variable is normal
b. No: cigs is nonnegative, piles up at zero (most adults do not smoke) and is discrete; still, with a large n, OLS t and F tests are approximately valid
c. Yes, because cigs is an average
d. No, so OLS estimates in a cigs equation are biased

4. Under MLR.1–MLR.5 but without normality (MLR.6), with a large sample:
a. t tests become invalid no matter how large n is
b. OLS becomes biased
c. The usual t and F statistics are approximately valid, because $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ is approximately standard normal
d. OLS standard errors cannot be computed

5. Under MLR.1–MLR.5, $\text{se}(\hat{\beta}_j) \approx c_j/\sqrt{\{n\}}$. If the sample size is quadrupled, the standard error is approximately:
a. Quartered
b. Halved
c. Unchanged
d. Doubled

6. Crime model (n = 2,725): narr86 is regressed on pcnv, ptime86, qemp86 (restricted model). Its residuals are regressed on pcnv, avgsen, tottime, ptime86, qemp86, giving $R^2 = 0.0015$. To test H_0: avgsen and tottime have no effect:

a. $LM = nR^2 \approx 4.09$; compared with χ^2_2 (10% critical value 4.61), H_0 is not rejected at the 10% level

b. $LM = 0.0015$, so H_0 is rejected at the 1% level

c. $LM \approx 4.09$; compared with χ^2_5, H_0 is rejected

d. $LM = n(1 - R^2) \approx 2,721$, so H_0 is rejected

Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics — Answers

1. d
일치성: 표본이 커질수록 추정량의 분포가 참값에 몰린다. MLR.1–MLR.4면 충분하고, 더 약한 MLR.4′(E(u)=0, Cov(x_j, u)=0)만으로도 성립한다. 정규성·동분산은 필요 없다.

2. d
비일치성 = $\beta_2 \times \delta_1$, δ_1 = Cov(funds, risktol)/Var(funds). 위험선호가 클수록 주식 비중이 크므로 $\beta_2 > 0$, 상관도 양(+) → 양의 비일치성. 표본이 커져도 사라지지 않는다. (Wooldridge 연습문제 5.2)

3. b
정규분포는 연속·대칭이며 음수도 가능하다. cigs는 0에 질량이 몰려 있어 정규가 아니다. 그러나 정규성(MLR.6)은 불편성과 무관하고, 대표본에서는 점근적 정규성 덕분에 추론이 근사적으로 타당하다. (Wooldridge 연습문제 5.3)

4. c
점근적 정규성(중심극한정리): 정규성 없이도 대표본에서 t·F 검정과 신뢰구간이 근사적으로 타당하다. 단, 동분산(MLR.5)은 여전히 필요하다 — 이분산이면 대표본에서도 일반 표준오차는 틀린다.

5. b
표준오차는 1/√{n} 속도로 줄어든다. √{4} = 2 이므로 약 절반이 된다.

6. a
LM = $n \cdot R^2_u$ = 2,725 × 0.0015 ≈ 4.09, 자유도 = 제약의 개수 q = 2, p-value ≈ 0.13. 제약 모형 잔차를 '모든' 설명변수에 회귀한다. (Wooldridge Example 5.3)

Part G: Ch.6 Multiple Regression Analysis: Further Issues

Multiple Choice

1. The dependent variable is changed from dollars to thousands of dollars (divided by 1,000). What happens?
a. Only the intercept changes
b. t statistics are divided by 1,000
c. R-squared falls
d. All coefficients and standard errors are divided by 1,000; t statistics and R-squared are unchanged

2. For 32 chemical firms, rdintens = 2.613 + 0.00030 sales − 0.0000000070 $sales^2$ (sales in $ millions). With salesbil = sales/1,000, the coefficients on salesbil and $salesbil^2$ become:
a. 0.00000030 and −0.0000000000070
b. 0.30 and −0.0000070
c. 0.30 and −0.0070, with the intercept and R-squared unchanged
d. 0.030 and −0.00070

3. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 $sales^2$ (sales in $ millions), at what level of sales does the marginal effect of sales on rdintens become negative?
a. About 2,143
b. About 21,429, i.e. $21.4 billion
c. About 42,857
d. At sales = 0

4. A housing equation with dependent variable log(price) contains −0.545 rooms + 0.062 $rooms^2$ (other controls included). Moving from 6 to 7 rooms raises price by about:
a. 6.2%
b. 12.4%
c. −54.5%
d. 19.9%

5. In log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms, what is the exact percentage change in price from one more room?
a. Exactly 30.6%
b. 0.306%
c. 3.06%
d. About 35.8%, = 100[exp(0.306) − 1]

6. In a regression where price and all explanatory variables are standardized (z-scores), the coefficient on nox is −0.340. Interpretation:

a. A 1 ppm increase in nox lowers price by 34%

b. A one standard deviation increase in nox lowers price by 0.34 standard deviation, holding the other variables fixed

c. A 1% increase in nox lowers price by 0.34%

d. nox explains 34% of the variation in price

7. log(wage) = 5.65 + 0.047educ + 0.00078educ· pareduc + 0.019exper + 0.010tenure (n = 722), where pareduc is total years of both parents' education. What is the estimated return to education when pareduc = 32?

a. 4.7%

b. 0.078%

c. About 7.2%

d. 2.5%

8. Which statement about adjusted R-squared is correct?

a. It always rises when a variable is added

b. It is always larger than R-squared

c. It can compare a model for y with a model for log(y)

d. It can fall when a variable is added; it rises only if the new variable's |t| exceeds 1

9. College GPA equation (n = 4,137): predicted colgpa = 2.70 at sat = 1200, hsperc = 30, hsize = 5, with se($\hat{y}^0$) = 0.020 and $\hat{\sigma}$ = 0.560. Which is correct?

a. The prediction interval for one student is narrower than the confidence interval for the mean

b. Both 95% intervals are 2.66 to 2.74

c. The prediction interval shrinks to zero width as n grows

d. The 95% CI for expected colgpa is about 2.66–2.74, but the 95% prediction interval for one student is about 1.60–3.80, because it also includes the error variance σ^2

10. The dependent variable is log(y). Using exp(predicted log(y)) as the prediction of y:

a. Systematically underestimates E(y|x); it should be scaled up by $\hat{\alpha}_0$, e.g. exp($\hat{\sigma}^2$/2) under normality or the average of exp($\hat{u}_i$)

b. Gives an unbiased prediction of y

c. Overestimates E(y|x)

d. Is valid only when R-squared is high

Part G: Ch.6 Multiple Regression Analysis: Further Issues — Answers

1. d
y의 단위를 바꾸면 계수와 표준오차가 같은 비율로 바뀐다. 따라서 t통계량, p-value, R^2은 변하지 않는다(통계적 결론 동일).

2. c
salesbil 계수 = 0.00030 × 1,000 = 0.30, $salesbil^2 = sales^2/1{,}000{,}000$ 이므로 계수 = $-0.0000000070 \times 10^6 = -0.0070$. 보고하기 쉬운 형태가 된다. (Wooldridge 연습문제 6.3)

3. b
전환점 $x^* = \beta_1/(2|\beta_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$ ($ millions). 그 이후 한계효과가 음수가 된다. (Wooldridge 연습문제 6.3)

4. d
$\%\Delta \text{price} \approx 100[-0.545 + 2(0.062)(6)] \approx 19.9\%$. 전환점은 $0.545/(2\times0.062) \approx 4.4$ rooms이며, 이보다 작은 지역은 표본의 약 1%뿐이다. (Wooldridge Example 6.2, HPRICE2)

5. d
log-level의 100β 근사는 변화가 작을 때만 정확하다. 정확한 값은 $100[\exp(\beta\Delta x) - 1] = 100(1.358 - 1) \approx 35.8\%$.

6. b
표준화 계수(베타 계수)는 x가 1 표준편차 변할 때 y가 몇 표준편차 변하는지를 나타내므로 단위가 다른 변수의 효과를 비교할 수 있다. (Wooldridge Example 6.1)

7. c
$\Delta \log(wage)/\Delta educ = 0.047 + 0.00078 \times pareduc = 0.047 + 0.025 \approx 0.072$. 부모 학력이 높을수록 교육수익률이 크다. (Wooldridge 연습문제 6.4, WAGE2)

8. d
$\bar{R}^2 = 1 - [SSR/(n-k-1)]/[SST/(n-1)]$: 자유도로 벌점을 준다. R^2은 변수를 추가하면 절대 줄지 않지만 $\bar{R}^2$은 줄 수 있다. 종속변수가 다르면(y vs log y) 비교할 수 없다.

9. d
평균의 신뢰구간: $2.70 \pm 1.96(0.020)$. 개별값 예측구간: $se(\hat{e}^0) = \sqrt{0.020^2 + 0.560^2\} \approx 0.560$, $2.70 \pm 1.96(0.560)$. 오차분산은 n이 커져도 줄지 않는다. (Wooldridge Example 6.5–6.6, GPA2)

10. a
$E(y|x) = \exp(\sigma^2/2)\cdot\exp(x\beta)$ 이므로 $\exp(\log \hat{y})$만 쓰면 과소예측한다. $\hat{\alpha}_0 > 1$을 곱해 보정한다. (Wooldridge 6.4절, Example 6.7)

Econometrics Midterm · Short Answer

1. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is the difference between a population and a sample?

2. **[Part A: Econometrics 1–3 (Professor's Slides)]** Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. **[Part A: Econometrics 1–3 (Professor's Slides)]** Define external validity and state which property of the sample matters for it.

4. **[Part A: Econometrics 1–3 (Professor's Slides)]** In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is β_1 called a partial slope rather than simply "the slope"?

5. **[Part A: Econometrics 1–3 (Professor's Slides)]** In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery}\times\text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is the difference between random sampling and random assignment?

7. **[Part A: Econometrics 1–3 (Professor's Slides)]** For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is a control variable, and why is ability added in Model 1, $\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{abil} + u$?

9. **[Part B: Ch.1 The Nature of Econometrics and Economic Data]** What does "ceteris paribus" mean, and why is it important in econometrics?

10. **[Part B: Ch.1 The Nature of Econometrics and Economic Data]** What is panel (longitudinal) data, and what is one advantage of it?

11. **[Part C: Ch.2 The Simple Regression Model]** What does OLS minimize, and what is a residual?

12. **[Part C: Ch.2 The Simple Regression Model]** What does R-squared measure in a regression?

13. **[Part C: Ch.2 The Simple Regression Model]** What does the homoskedasticity assumption (SLR.5) state?

14. **[Part C: Ch.2 The Simple Regression Model]** What does it mean that the OLS estimator $\hat{\beta}_1$ is unbiased?

15. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** State the Gauss–Markov theorem.

18. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Define the p-value of a hypothesis test.

19. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Interpret a 95% confidence interval for β_j.

21. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta}_1$, and why?

24. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** What is a beta (standardized) coefficient, and why is it useful?

28. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** Why is adjusted R-squared used, and when does adding a variable increase it?

31. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

Answers

1. [Part A: Econometrics 1–3 (Professor's Slides)] A population is the **entire** set of individuals, objects or data points we want to study, while a sample is a **subset** of the population that is **selected** for analysis.
Keywords: entire · subset · selected

2. [Part A: Econometrics 1–3 (Professor's Slides)] Studying the whole population is often too **costly** or **impossible**, so we study a sample; the goal is to find out what the **population is like** from the sample.
Keywords: costly · impossible · population is like

3. [Part A: Econometrics 1–3 (Professor's Slides)] External validity is the validity of applying the **conclusions** of a study **outside** the context of that study; it depends on whether the sample is **representative** of the population.
Keywords: outside · conclusions · representative

4. [Part A: Econometrics 1–3 (Professor's Slides)] β_1 is the change in log(price) from a **one-unit** increase in subsidy while **holding** the **other variables** (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Keywords: holding · other variables · one-unit

5. [Part A: Econometrics 1–3 (Professor's Slides)] Because battery and sunlight **interact**, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which **depends** on the level of sunlight; the slope is **not constant**.
Keywords: interact · depends · not constant

6. [Part A: Econometrics 1–3 (Professor's Slides)] In random sampling every element of the population has an **equal probability** of being **selected**; in random assignment every element has an equal probability of being **assigned** to a group.
Keywords: equal probability · selected · assigned

7. [Part A: Econometrics 1–3 (Professor's Slides)] Ability is omitted, so it goes into the **error** u; because people with better ability get more education, educ is **correlated** with u, so $E(u \mid educ) = 0$ fails and β_1 is estimated with bias.
Keywords: ability · error · correlated

8. [Part A: Econometrics 1–3 (Professor's Slides)] A control variable is included to **hold** another factor **fixed** so it no longer sits in the error; adding ability helps satisfy the **zero conditional mean** condition and removes the **bias** in β_1.
Keywords: hold · zero conditional mean · bias

9. [Part B: Ch.1 The Nature of Econometrics and Economic Data] Ceteris paribus means "**other relevant factors** being **equal**"; most economic questions ask for the **causal** effect of one variable holding all other factors fixed.
Keywords: other factors · equal · causal

10. [Part B: Ch.1 The Nature of Econometrics and Economic Data] Panel data follow the **same** cross-sectional units **over time**; they can account for **time-invariant** unobservables and model lagged responses.
Keywords: same · over time · time-invariant

11. [Part C: Ch.2 The Simple Regression Model] OLS chooses the estimates that minimize the **sum of squared residuals**; a residual is the **actual** value y_i minus the **fitted** value $\hat{y}_i$.
Keywords: sum of squared residuals · actual · fitted

12. [Part C: Ch.2 The Simple Regression Model] $R^2 = SSE/SST$ is the **fraction** of the sample **variation** in y that is **explained** by the explanatory variable(s).
Keywords: fraction · variation · explained

13. [Part C: Ch.2 The Simple Regression Model] Homoskedasticity means the **error** u has the **same** (constant) **variance** for any value of x: $Var(u \mid x) = \sigma^2$.
Keywords: variance · error · same

14. [Part C: Ch.2 The Simple Regression Model] Unbiasedness means $E(\hat{\beta}_1) = \beta_1$: if sampling and estimation were **repeated** many times, the estimates would **on average** equal the **true** population value, though one estimate may differ.
Keywords: on average · repeated · true

15. [Part D: Ch.3 Multiple Regression Analysis: Estimation] There is no omitted variable bias if the omitted variable is **irrelevant** (its coefficient is zero) or **uncorrelated** with x_1.
Keywords: irrelevant · uncorrelated

16. [Part D: Ch.3 Multiple Regression Analysis: Estimation] Multicollinearity **increases** the sampling **variance** of the affected slope estimators, but OLS remains **unbiased**; it can be detected with variance inflation factors.
Keywords: variance · increase · unbiased

17. [Part D: Ch.3 Multiple Regression Analysis: Estimation] Under **MLR.1–MLR.5**, OLS is **BLUE** (best linear unbiased estimator): it has the **smallest variance** among all linear unbiased estimators.
Keywords: BLUE · smallest variance · MLR.1

18. [Part E: Ch.4 Multiple Regression Analysis: Inference] The p-value is the **smallest significance level** at which the null hypothesis would be **rejected**; we reject H_0 when the p-value is below the chosen level.
Keywords: smallest · significance level · reject

19. [Part E: Ch.4 Multiple Regression Analysis: Inference] Not necessarily: statistical significance only shows the effect differs from zero, while **economic** (practical) significance depends on the **magnitude** of the coefficient.
Keywords: not necessarily · magnitude · economic

20. [Part E: Ch.4 Multiple Regression Analysis: Inference] In **repeated** samples, intervals built this way **contain** the **true** population coefficient in 95% of cases; the bounds are random, not the parameter.
Keywords: repeated · contain · true

21. [Part E: Ch.4 Multiple Regression Analysis: Inference] Heteroskedasticity (i) and an **omitted** important variable (iii) violate the CLM assumptions, so the t statistics are invalid; a high **correlation** of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Keywords: heteroskedasticity · omitted · correlation

22. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] An estimator is consistent if, as the **sample size** grows, its distribution **collapses** to the **true** parameter (plim $\tilde{\beta}_j = \beta_j$); OLS is consistent under **MLR.1–MLR.4** (even zero correlation between each x_j and u is enough).
Keywords: sample size · collapse · true · MLR.4

23. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] The inconsistency is **positive**: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher **risktol** raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively **correlated**; it does **not disappear** as n grows.
Keywords: positive · risktol · correlated · not disappear

24. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] No: most adults smoke **zero** cigarettes and cigs is **nonnegative** and discrete, so it is **not normal**; with a **large sample**, asymptotic normality still makes the t and F tests approximately valid.
Keywords: zero · nonnegative · not normal · large sample

25. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] Estimate the **restricted** model, regress its **residuals** on all the explanatory variables, and compute LM = $\mathbf{nR^2}$ from that auxiliary regression; under H_0 it is approximately **chi-square** with q degrees of freedom.
Keywords: restricted · residuals · nR · chi-square

26. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] All coefficients and standard errors are **divided by 1,000**, while the **t statistics** and **R-squared** stay **unchanged**; only the units change, not the statistical conclusions.
Keywords: divided · unchanged · t statistic · R-squared

27. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] A beta coefficient comes from regressing the **standardized** y on standardized x's; it is the change in y, in **standard deviations**, from a one standard deviation increase in x_j, so effects measured in different **units** can be **compared**.
Keywords: standardized · standard deviation · units · compare

28. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] Turning point x^* = $\mathbf{\hat{\beta}_1/(2|\hat{\beta}_2|)}$ = 0.00030/(2 × 0.0000000070) ≈ **21,428.57**, i.e. sales of about **$21.4 billion**; beyond that the estimated marginal effect of sales on rdintens is negative.

29. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] The return is $\beta_1 + \beta_2 \cdot$pareduc, so it **depends** on **parents' education**; the **interaction** coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Keywords: depends · parents · interaction

30. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] R-squared **never falls** when a variable is added, so adjusted R-squared imposes a **penalty** for extra regressors using **degrees of freedom**; adding a variable raises it only if that variable's **t statistic** exceeds one in absolute value.
Keywords: never falls · penalty · degrees of freedom · t statistic

31. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] The confidence interval reflects only the sampling **variance** of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved **error** for an **individual** outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Keywords: error · variance · individual

32. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] exp(log $\hat{y}$) systematically **underestimates** E(y|x); we **multiply** it by $\hat{\alpha}_0 = \exp(\hat{\sigma}^2/2)$ under **normal** errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Keywords: underestimate · multiply · residuals